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Baryonic Matter Organisation on Planetary, Stellar and Galactic Scales: The ΛCDM (M, λ) Parametrization as a Scale-Free Inheritance Law

Jamieson G. Dahl[1]

[1] *Independent Researcher, Safety Bay, Australia*

ABSTRACT

ΛCDM galaxy formation describes a galaxy with two numbers, a mass budget and a spin draw, and three decades of semi-analytic success show that the pair is enough. Nothing in the theory says why. This paper says why: (M, λ) is the inheritance pair of dissipative baryonic organisation, the budget and the collapse-depth record that every parent system hands every child, processed by one universal machine at every scale. The galaxy recipe cosmology found by simulation is the top rung of a law this paper calibrates four rungs down. Jupiter's satellite system closes at 0.9% on the pair inherited from Sol's natal disc at Jupiter's formation slot. The same law, inverted, reconstructs Hyperion, the star whose disc produced Sol. Run forward at full scale, it generates the Milky Way. Dark matter cannot dissipate and lies outside the theory; ΛCDM's cosmological core is imported intact.

The machine. A natal disc's formation seats, its natal slots, are axisymmetric magnetized zonal flows: standing pressure bands that trap drifting solids while the disc lives and catch migrating planets afterwards. The first band forms at the outer boundary's pressure gradient and each successive band at the previous band's shoulder, $r_n = R_{\rm disc} \cdot \rho^n$ with $\rho = 1 - \frac{\sqrt{\ln 2}}{2} \approx 0.584$, a geometric constant with no fitted parameter. Two boundaries hold the disc in a vice: the magnetospheric Alfvén barrier inside, and outside the Davis Dam, the pressure wall against the ambient cloud, at $R_{\rm disc} = 30.07\,(M_\star/{\rm M_{prim}})\,D^{-1/3}$ AU with the nebula density D equal to one for Sol. Each slot's allocation sums rock, snow-line ice, pebble drift, and envelope capture. In primordial solar units Sol sits at exactly $(1, 1, 1)$ with its dam at Neptune; ten clean calibrations and 23 fitted systems follow. One dynamical event, Jupiter's destabilisation of the embryo swarm, joins the Mercury–Vulcan merger to account for every Solar System mass anomaly.

The hierarchy. A daughter system inherits both defining inputs from its seat in the parent's natal disc: the slot allocation budgets her inventory, and the parent's local pressure sets her dam through $D_d/D_p = 0.39\,(r_{\rm site}/R_{\rm disc})^{-2.12}$, an exponent that matches the disc's own density profile. Three generations check one another. The

jamiesongrahamdahl@gmail.com

Galilean disc prefers Jupiter's formation site over its present orbit by a factor of 35; all four moons sit in situ in their observed order, Callisto on the dam, the inner trio displaced outward in the Laplace tidal pattern. Inverted, the recursion calibrates Sol's A-class progenitor and predicts its remnant: a white dwarf of $0.69 \pm 0.04\,\mathrm{M}_{\odot}$, cooling age near 3.4 Gyr, on a thin-disc orbit consistent with co-natal dispersal, travelling with a main-sequence sibling that carries Sol's isotopic fingerprint.

The factory. The dam is also a production line. The T Tauri firehose drove $660\,\mathrm{M}_{\oplus}$ of solids through it, and the Kuiper belt is the residue that stuck. The factory is a moving shoreline whose deposits are dated cohorts, and the trans-Neptunian census confirms that the dam's exterior carries structure in time, not in space: the resonant populations are the earliest cohort swept into Neptune's comb, the cold classicals are the mid-firehose deposition band, and the Kuiper cliff is the radius where production shut down. The boundary still operates. Its stocked shelf is observed as the heliopause hydrogen wall, its counterparts as the astrospheres of other stars, and its galactic instance is already named in the literature: the galactopause, where thermal instability condenses cloud-mass products that rain back inward. A dense-cloud transit 2–3 Myr ago, independently proposed to explain the deep-sea ^{60}Fe layers, predicts young volatile-fresh bodies at 40–150 AU carrying live radionuclides.

The galaxy. A galaxy forms the way a planetary system does, one rung up. A seed condenses in a collapsing protogalactic cloud, and its outflow against the ambient medium sets a Davis Dam, the pressure boundary where the factory runs; here its products are stars. The vice also bounds the seed: collapse runs only deep enough to form it, its outflow erects the two dams, and subsequent infall is processed into stars rather than accreted. The dam is the system's outer edge at every epoch. While the cloud is still collapsing, ambient pressure is at its peak and pins the dam to a small radius, so the factory makes its stars close in, building the bulge and feeding the seed together, which is why black-hole mass tracks classical-bulge mass ever after. The seed only templates: the factory's output is limited by the fuel arriving at the boundary, so the budget, not the seed, sets the final mass, and the finished structure outweighs its founder ten thousand to one. As ambient pressure falls the dam advances, and the disc is its deposition track: every annulus of stars is minted at the dam as it passes, so the disc ages from the centre outward, the inside-out growth observers measure, at constant mass per radius, which is the Mestel profile (Mestel 1963) and the origin of the flat rotation curve. Each stellar cohort the factory mints lands at a golden-angle offset of 137.5° from the last, the division of the circle at which successive batches least overlap. Cohorts one lattice step apart trace a logarithmic spiral of pitch 12.65°, two steps apart 24.2°, both angles fixed by ρ and the golden angle alone; the measured pitch distribution across 4,378 disk galaxies is bimodal at $12.0° \pm 3.4$ and $23° \pm 4.3$. Once the deposited mass outweighs the seed, dynamics pass from Keplerian to self-gravitating, and the templating epoch survives as fossils: the founder coupling, the deposition profile, the pitch lattice, and a bulge fraction set by collapse depth. A generative run of this sequence with one tuned normalisation reproduces the Milky Way's eight gross statistics: stellar mass, star count, white-dwarf census, star-formation rate, half-mass epoch, old-mass fraction, bulge fraction, and the inside-out age gradient. Slot-determined stellar metallicity dissolves the Sun's metal-rich anomaly along the way.

The paper closes with a dated predictions register: every mechanism is listed with the observation that would falsify it.

Keywords: planet formation, protoplanetary disks, Solar System formation, Kuiper belt, Oort cloud, satellite systems, spiral galaxies, Galaxy structure, stellar abundances

1. INTRODUCTION

ΛCDM galaxy formation rests on a remarkably economical recipe. To first order a galaxy is two numbers, its halo mass and its spin parameter λ (Peebles 1969), plus an assembly history. Three decades of semi-analytic modelling attest that the pair suffices, yet within ΛCDM the pair is phenomenology; nothing in the theory says why two numbers, or why these two. This paper proposes the why, with evidence from far below the galactic scale. Our thesis: (M, λ) is the universal inheritance pair of dissipative baryonic organisation, and λ is the effect, not the cause. The cause is compression: the collapsing cloud squeezes the system's outer pressure boundary down against the seed's magnetospheric truncation radius, and spin is the record of that compression ($\Omega = D^{2/3}$; §2). Every bounded baryonic system is determined by the mass budget and the collapse-depth record handed down by its parent, processed by one universal machine. ΛCDM's galaxy recipe is the top rung, operating in the self-gravitating regime, discovered top-down by simulation at the one scale where the parent is the environment itself. The lower rungs are calibrated in this paper at the sub-percent level (Table 1).

The machine operates in two dynamical regimes. In the Keplerian regime, where the seed dominates its natal disc, it is fully calibrated: a ladder of natal slots at a fixed geometric ratio, held between two pressure boundaries, with a closed-form mass allocation. In the self-gravitating regime the disc has come to outweigh the seed that organised it. The Keplerian formulas for period structure and resonant locking do not transfer there, and they are not needed: the templating epoch writes its record into the structure itself, and Section 9 reads that record. The boundary grammar crosses the transition whole. Dissipative matter settles into flat ordered components, dissipationless matter occupies spheroidal holding reservoirs, and boundaries between media operate as factories at every scale we examine. Where dissipation itself fails the theory ends: dark matter is the permanently unorganisable remainder, and ΛCDM's cosmological core enters this paper as infrastructure, imported rather than contested.

Table 1. The inheritance pair across the hierarchy. Each system is determined by a mass budget and a collapse-depth (spin) draw handed down by its parent. The machine processing the pair is universal; its dynamical regime changes once, at the top, where the disc outweighs its seed. Throughout this paper the dissipative disc from which a bound system's members condense is called the *natal disc*, and its discrete formation seats are the *natal slots*. Protogalactic, protoplanetary, circumplanetary, and protolunar discs are scale-specific instances of one object, the realisation of the inheritance pair.

System	Parent	Budget (M)	Spin draw (λ)	Status
Galaxy	environment	$f_b\, M_{\rm halo}$	$\lambda_{\rm halo}$ (tidal torques)	generative facsimile (§9)
Star + disc	cloud / parent disc	core / slot allocation	recorded $\Omega = D^{2/3}$ (jaw-lock)	10 clean calibrations (§3)
Sol	A-class progenitor	slot-1 allocation	$D = 1$ recorded at Sol's seat	reconstruction (§7)
Giant-planet sub-cascade	host star's disc	slot allocation $\sigma_{\rm AAF}$	$D_d = 0.39\, x^{-2.12} D_p$	measured at 0.9% (§7)

The machine tier is developed first and carries the paper's calibrations; its strongest instance is the Solar System. The architecture of the Solar System is unusually well constrained by both dynamical and cosmochemical evidence, yet no single theoretical framework simultaneously accounts for: (i) the exact masses of the eight planets, (ii) Mars's anomalously small mass relative to extrapolated rocky surface density, (iii) the Moon-forming impactor's existence, and (iv) the near-5:2 mean-motion resonance between Saturn and Jupiter (their orbital periods sitting close to a 5:2 integer ratio).

Existing frameworks address subsets. The Grand Tack hypothesis (Walsh et al. 2011) postulates that Jupiter migrated inward to ~ 1.5 AU and then outward, dragging Saturn behind it; this scenario explains Mars's mass deficit but is in tension with isotopic evidence (Kruijer et al. 2017) indicating that Jupiter's core formed in < 1 Myr and separated the non-carbonaceous and carbonaceous reservoirs without subsequent excursions. The Nice Model (Tsiganis et al. 2005) postulates a large-scale post-formation outward migration of Saturn through Neptune. This paper requires no such reorganization: all four giants form at or within an AU of their present orbits, and Neptune sits in situ at the dam to four significant figures, where it anchors the entire calibration. Core accretion (Pollack et al. 1996) and pebble accretion (Lambrechts & Johansen 2012; Bitsch et al. 2015) provide mechanisms for individual planet growth but do not produce a closed-form mass-allocation rule across the entire disc.

We propose a complementary approach: a *local-capacity* rule that allocates planetary mass to each orbital radius r on the basis of the viscous-disc surface density profile, snow-line ice partition, pebble drift, and gas-envelope accretion timescales. The rule is closed-form and admits direct comparison against observed planetary masses. The principal advantage is parsimony: three system-level parameters plus per-planet formation radius and time account for ten Solar System bodies (including the proposed impactors Theia and Vulcan); the two anchor planets fit exactly, and every other deviation from cascade allocation traces to one of two dynamical events, chiefly Jupiter's scattering of the embryo swarm at the snow-line pile-up (slots 4–5, 2–3.5 AU).

Evidence structure.—The law is calibrated where the hierarchy can be resolved and predictive where it cannot yet be. Sections 2 through 7 carry the calibrations: ten clean system fits, the Solar System at exactly $(1, 1, 1)$ in primordial units, and the three-generation closure at 0.9%. Sections 8 and 9 carry the same law into territory where its tests are still running, and state for each prediction the observation that would kill it; the register in Section 10 consolidates these with dates. Two matters lie outside the theory by its own structure and are said so once: dark matter, which cannot dissipate and therefore cannot be organised by any dissipative machine, and late-stage system evolution.

Scope of this paper.—The paper proceeds as follows. Section 2 introduces the local-capacity machine: the geometric cascade, the two-jawed vice (Alfvén boundary and Davis Dam), the nebula-density variable, and Tanigawa-Ikoma gas-envelope physics. Section 3 presents the Solar System calibration, including the Vulcan-Mercury merger. Section 4 establishes Venus and Neptune as the ISU primordial-plane reference and quantifies impact signatures via inclination and eccentricity excesses. Section 5 demonstrates Saturn and Jupiter in-situ at slots 2 and 3 respectively, with the slot 4-5 embryo-swarm dispersal mechanic as the framework's single dynamical event. Section 6 demonstrates scale-invariance of the cascade at multi-stellar and cluster scales. Section 7 formalises the hierarchical recursion by which each generation budgets and confines the next, and carries its three-generation calibration. Section 8 extends the boundary physics outward, deriving the Kuiper belt and Oort cloud as reservoirs at the system's magnetic and gravitational outer boundaries, with the dam as a moving production shoreline. Section 9 presents the galactic tier: the regime transition, the inheritance pair under its ΛCDM names, the signatures of the geometry surviving self-gravity, and the boundary-factory programme. Section 10 enumerates testable predictions. Section 11 discusses novelty, the empirical $f_{\rm disc}$ stochasticity finding, model limitations, and relation to prior frameworks. We conclude in Section 12. A browser-based interactive implementation of the model, with all calibrated systems pre-loaded, is available at https://jamesgdahl.github.io/HYDROS-Planet-Formation-Model/.

2. THE LOCAL-CAPACITY ALLOCATION MODEL

2.1. *Physical foundation*

The model rests on three established disc-physics results:

Viscous disc surface density.—For a steady-state viscous accretion disc with viscosity $\nu \propto r$, self-similar solutions of the Lynden-Bell–Pringle (hereafter LBP) equations yield surface density

$$\Sigma(r) \propto r^{-1}, \tag{1}$$

giving constant mass-per-annulus $2\pi r\Sigma(r)\,\mathrm{d}r = \mathrm{const}$ (Lynden-Bell & Pringle 1974; Pringle 1981; Shakura & Sunyaev 1973). This is observationally confirmed for discs around T Tauri stars (young pre-main-sequence stars still accreting from their discs), with measured power-law indices $\gamma \approx 0.9$–1.1 (Hartmann et al. 1998; Andrews et al. 2009, 2010; Williams & Cieza 2011; Andrews 2020; Long et al. 2018; Trapman et al. 2019). The Hayashi minimum-mass solar nebula (MMSN) (Hayashi 1981) adopts a steeper $\Sigma \propto r^{-1.5}$ for solid material, but modern observations with the Atacama Large Millimeter/submillimeter Array (ALMA) favor the LBP scaling. We adopt $\Sigma \propto r^{-1}$ here.

The Davis Dam (outer-edge pressure barrier).—We name the outer boundary for Leverett Davis Jr., who identified its present-day descendant before the solar wind itself was accepted: Davis (1955) proposed that the Sun's corpuscular emission carves a cavity in the interstellar medium whose edge is set by *pressure balance against the interstellar field*, the heliopause as a confinement surface, three years before Parker (1958) supplied the outflow that inflates it. The two boundaries form a *vice* with distinct owners: the inner jaw is stellar (spin and field, through disc-locking) while the outer jaw is environmental, the ambient pressure differential of the birth site. The single spin parameter reads both jaws at once because they correlate through their common driver, the birth-environment density; the extreme effective spins of the inverted regime are the outer jaw's signature, not stellar rotation. The dam's local masonry is magnetic: through the bulk of the disc the magnetorotational instability (MRI) , the magnetic stirring that makes disc gas effectively viscous, is *suppressed* by non-ideal effects (ambipolar diffusion, low ionization; Bai & Stone 2013; Turner et al. 2014); only where ionization rises does the midplane re-couple to the magnetic field. At the outer disc boundary, cosmic-ray and stellar ionization penetrate the thinning column and revive the MRI (Gammie 1996; Flock et al. 2015), producing a sharp rise in effective viscosity: a step, because the bulk on one side is magnetically dead and the edge on the other is alive. Solid material drifting inward from the outer disc pools against this viscosity step, and the associated pressure gradient halts further drift (Pinilla et al. 2012, 2016; Lyra et al. 2009; Regály et al. 2012). Observed disc outer radii behave like such transport-set boundaries: gas-disc sizes track angular-momentum transport rather than free viscous spreading (Najita & Bergin 2018), and measured dust radii do not grow with age over 1–10 Myr (Hendler et al. 2020), edges are boundary-controlled, not spreading fronts. We name this boundary the *Davis Dam*, $R_{\bar{A}}$, in parallel to the inner *Alfvén Dam*, R_A (next paragraph); both are viscous barriers in a flowing fluid, not gravitational traps: material accumulates because the downstream channel narrows. The two dams bracket the planet-forming annulus $[R_A, R_{\bar{A}}]$, each backed by a reservoir that fills to a baseline mass set by the balance of inward drift flux against the dam's holding capacity: the inner reservoir is the model's intercept a (Eq. 8); the outer reservoir is the edge pile-up of Eq. 12. (The "dead zone" of the literature is this quiescent interior; the dynamically important structure is the outer boundary where material pools.) In the model we identify $R_{\bar{A}} \equiv R_{\mathrm{disc}}$.

The inner-reservoir mass a (Eq. 8) corresponds physically to what Chatterjee & Tan (2014) term the "Inside-Out Planet Formation reservoir": solids drift inward via aerodynamic drag and pool against the inner edge, supplying the close-in planet zone (Boley & Ford 2013; Hansen & Murray 2013; Liu, Ormel & Lin 2017; Romanova & Lovelace 2006). Indeed, in-situ assembly of close-in super-Earths from a smooth disc fails without exactly such an inner solid pile-up (Ogihara et al. 2015). Our intercept a is the steady-state baseline fill of that reservoir; the scaling $a \propto M_\star \cdot \Omega^{2/7}$ reflects the dam-pressure dependence on stellar magnetic moment and spin via $R_A \propto \Omega^{4/7}$.

The Alfvén Dam (magnetospheric inner barrier).—Stellar magnetic-field pressure halts disc gas at the Alfvén radius,

$$R_A = \left(\frac{\mu^4}{2GM_\star \dot{M}^2} \right)^{1/7}, \tag{2}$$

where μ is the stellar magnetic moment and $\dot{M}$ is the accretion rate (Ghosh & Lamb 1979; Königl 1991; Shu et al. 1994; Long et al. 2005; Romanova et al. 2008); the underlying magnetic-pressure concept traces to Alfvén (1942). The truncation formula itself carries no explicit spin dependence: the $R_A \propto \Omega^{4/7}$ scaling we use enters through the disc-locking equilibrium, in which the star's rotation is regulated to corotation with the truncation radius throughout the accretion phase (Königl 1991; Bouvier et al. 2007), dynamically coupling spin to the inner boundary. We refer to this boundary as the *Alfvén Dam*: the magnetic field obstructs the inward-flowing disc, and solids pool against it, filling the inner reservoir a described above. For solar-mass stars during the protoplanetary phase, $R_A \sim 0.05$–0.2 AU, scaling with stellar mass and spin.

Snow line and ice condensation.—The snow line r_{snow} is the orbital radius beyond which the disc is cold enough for water to condense onto grains as ice, enhancing the local solid surface density by a factor ~ 3–4 (Hayashi 1981; Pollack & Bodenheimer 1989; Mulders et al. 2017). Its position is set by the disc's thermal structure, which depends on dust opacity and therefore on the size distribution of the suspended dust (Min et al. 2011; Mulders, Ciesla & Min 2015; Bitsch et al. 2015). In a viscously heated disc, opacity-rich ISM-like (small-grain-dominated) dust blankets the midplane, keeping it warm and pushing the snow line outward; grain growth lowers the opacity and draws the snow line up to a factor ~ 2 closer to the star (Mulders, Ciesla & Min 2015). We parameterize this through the dimensionless *grain-opacity parameter* $g \in [0, 1]$ (Eq. 6), interpolating between the grain-grown, opacity-poor limit $r_{\mathrm{snow}}(g{=}0) = 1.6$ AU and the ISM-like, opacity-rich limit $r_{\mathrm{snow}}(g{=}1) = 3.3$ AU at solar mass and $f_{\mathrm{disc}} = 0.01$, matching the Mulders-Ciesla viscous-disc range.

Pebbles drifting inward across the snow line additionally create a local pile-up: they lose their volatiles at the ice-sublimation boundary and accumulate there as a pressure-trap concentration of refractory and residual-ice material (Mulders et al. 2017; Schoonenberg & Ormel 2017; Drążkowska & Alibert 2017). In the two-population picture of Birnstiel et al. (2012), the small grains that carry the opacity (and so set $r_{\rm snow}$) coexist with the grown pebbles that carry the inward mass flux (and so feed the trap). In the model the trap's amplitude and position both follow $r_{\rm snow}$ (Eq. 11), giving g a dual role: it sets where the snow-line bump sits and how much mass it concentrates.

What a slot is.—The slot is geometric in origin, magnetic in operation, and fossil in survival. Its position is fixed by the projection constant ρ, with no field in the derivation; its effects are delivered entirely through the magnetized gas. Every component of this embodiment is published. Long-lived axisymmetric pressure bumps arise spontaneously in magnetorotational turbulence as zonal flows, generated by magnetic-pressure variations (Johansen et al. 2009), driven by magnetic flux concentration (Bai & Stone 2014); solids drift into them and form planetesimals there (Dittrich et al. 2013), and dust trapping in such rings is now directly observed (Dullemond et al. 2018). Migration torque reverses across density structure, so the same bumps trap passing planets (Masset et al. 2006), with the magnetospheric cavity rim as the family's innermost member (Lin et al. 1996). One structure therefore both builds occupants and catches them. What this paper adds is the organisation. The first band forms at the steepest pressure gradient in the system, the dam itself; no second band can organise within the half-amplitude point of the first's profile taken at the 45° projection, so the next band forms at the previous band's shoulder, $r_{n+1} = \rho\, r_n$, and the lattice descends shoulder by shoulder until the geometry ends at the Alfvén Dam. The bands precede their occupants. The lattice is directly observed: HL Tau's rings at an age younger than any planet-formation timescale (ALMA Partnership 2015), and the ringed-disc census of the DSHARP survey (Huang et al. 2018). The mainstream reading of those rings runs the causal arrow the other way, as gaps carved by already-formed planets; HL Tau's youth is this paper's witness, and two registered discriminators follow (§10). First, geometry: lattice occupancy predicts adjacent-ring ratio peaks at 1.31 and 1.71 and their multiples across the DSHARP catalogue; planet carving predicts none. Second, occupancy: carving requires a planet in every gap, while the lattice predicts bands at *every* slot, occupied or not – most rings should be planet-free, and seats whose material is lost to neighbours remain permanently vacant. The deep direct-imaging searches of DSHARP gaps, which have yielded confirmed planets in essentially one system (PDS 70), are a standing tension for carving and the expected result here. When the gas disperses the addresses persist as fossils, fittable gigayears later.

2.2. *Fundamental inputs*

The model has three fundamental system-level inputs, one per-planet input, and a conditional perturber block:

$M_\star$, **primordial stellar mass:** (in $M_\odot$, before pre-main-sequence mass loss).

D, **nebula density:** dimensionless, with $D = 1$ defined as the Sun's birth-environment value: the causally primary outer jaw, setting the dam radius, the factory surface density, the quantity the inheritance law propagates, and (through the locked spin) the XUV retention dose. The primordial spin is its jaw-locked record, $\Omega = D^{2/3}$; the two are interchangeable readings of one dial, and the catalogue records both.

$f_{\rm disc}$, **disc-to-star mass ratio:** $\in [0.001, 0.3]$, the integrated gas-disc mass divided by the stellar mass.

r, **per-planet orbital radius:** (in AU).

($M_{\rm pert}$, q, $a_{\rm bin}$), **perturber block:** only for encounter-edited or companion-bound systems (§8.10): perturber mass and encounter periapsis (supplied or bisected), with $a_{\rm bin}$ present when the perturber is bound (standing tide) rather than a one-shot flyby.

Per-planet formation time $t_{\rm form}$ is derived from the cascade rule (§2.5) rather than fit independently. Numerically the model works in *primordial-solar units* (stellar mass, spin, and nebula density all equal 1 for Sol, with one mass unit $\equiv$ 1.14 present-day $M_\odot$) so Sol is the exact baseline $(M_\star, \Omega, D) = (1, 1, 1)$ and the calibration constants carry no residual 1.14 factors. Catalogue host masses, quoted in present-day solar masses, are adopted *directly* as primordial-unit values: under uniform fractional early mass loss (each star, like Sol, retaining 1/1.14 of its birth mass through the T Tauri wind era) the 1.14 cancels in the ratio. Young hosts that have not yet shed their T Tauri mass are the exception, β Pic ($\sim$ 20 Myr) and HR 8799 ($\sim$ 30–40 Myr) enter at literature mass /1.14. Spectral classes follow the same convention as primordial identities: a slot of X units is the star the literature quotes at $X\,M_\odot$ after uniform loss, so letter-class thresholds scale by 1.14 while ignition thresholds (deuterium, hydrogen) remain physical at birth mass. Sol is born, lives, and is classified G, and the convention validates against the sky: Alpha Centauri B's 0.80-unit slot classifies K, matching its observed K1V.

The jaw-lock.—The lock between the two readings is physical: the formation cloud's ambient confinement establishes the Davis Dam at

$$R_{\rm disc} = 30.07 \cdot \frac{M_\star}{1.14\,M_\odot} \cdot D^{-1/3}\ \text{AU}, \qquad \Omega = D^{2/3}, \tag{3}$$

so that $D = 1$ places Sol's dam at 30.07 AU, Neptune's exact slot, with spin the angular-momentum *record* of the same squeeze (the jaw-lock: both jaws driven by birth-site density). The calibration machinery that anchors $R_{\rm disc}$ on the outermost planet is therefore a bisection of D expressed through the lock, and fitted spins double as density readouts. Inverting Sol's $D = 1$ through the collapse centrifugal radius (at the measured rotational-energy fraction $\beta \approx 0.02$; Burkert & Bodenheimer 2000) yields a birth core of ~ 0.015 pc at $n(\mathrm{H_2}) \approx 10^6$ cm^{-3}: a *clustered* prestellar core, independently consistent with the meteoritic birth-cluster evidence and the cluster-era boundaries of §8. In the inverted regime the fitted $D_{\rm eff}$ exceeds any static cloud (TRAPPIST-1: $\sim 5 \times 10^4$), there the outer jaw involves more than birth pressure, and the lock reads as effective compression rather than literal cloud density.

The grain-opacity constant g.—A fourth quantity, the dimensionless grain-opacity parameter $g \in [0, 1]$ (from fully grain-grown, opacity-poor dust ($g = 0$) to ISM-like, small-grain-dominated dust ($g = 1$)) sets the snow-line position (Eq. 6). It is treated as a *universal constant*, $g = 0.82$ (Table 2), not a per-system input: calibrated once on Sol, the same value serves every calibrated system, since the mass-position data that drive the calibrations constrain $M_\star$, Ω, and $f_{\rm disc}$ but give little independent leverage on g, whose snow-line effect is partially degenerate with both (Eq. 6). The value sits close to the ISM-like, opacity-rich limit, consistent with the opacity-bearing small-grain population being inherited from the interstellar medium's grain-size distribution (Mathis et al. 1977) with only modest processing by the epoch of planet formation. We note the calibration sample is confined almost entirely to the solar neighbourhood, so the apparent constancy of g may be a property of our local galactic region rather than truly universal; per-system constraints await compositional or resolved-snow-line observations (§11).

Disc compression and its spin record.—Ambient confinement sets the disc's outer radius, and the star's primordial rotation records it: a deeply compressed system is a fast rotator by angular-momentum conservation ($\Omega \propto J/MR^2$ at fixed J, so $R \propto \Omega^{-1/2}$, Eq. 4, with spin communicated to the star by disc-locking; Königl 1991; Bouvier et al. 2007). Modern ALMA surveys reveal a wide spread in protoplanetary disc sizes, from ~ 3 AU to $\gtrsim 200$ AU (Andrews et al. 2010; Long et al. 2018; Trapman et al. 2019; Andrews 2020; Manara et al. 2023; Hendler et al. 2020; Najita & Bergin 2018), which under the lock is a spread in birth-site compression: compact super-Earth chains (TRAPPIST-1, Kepler-90) resolve to recorded spins $\Omega \gtrsim 5$–20 (deeply squeezed births), while wide architectures like Sol sit near $\Omega = 1$. The natal draw of specific angular momentum (Burkert & Bodenheimer 2000) sets the family constant (30 AU at solar values), and its measured spread, propagated through $R \propto \Omega^{-1/2}$, predicts the *width* of the disc-size distribution at fixed stellar mass, directly comparable to the ALMA spread cited above.

2.3. *Derived disc properties*

$$R_{\rm disc} = 30.07 \cdot \frac{M_\star}{M_{\odot,\rm prim}} \cdot \Omega^{-1/2}\,\mathrm{AU}, \tag{4}$$

$$R_A = 0.20 \cdot \frac{M_\star}{M_{\odot,\rm prim}} \cdot \Omega^{4/7}\,\mathrm{AU}, \tag{5}$$

$$r_{\rm snow} = \left[1.6 + 1.7\,g^{2.2}\right] \cdot \left(\frac{M_\star}{M_{\odot,\rm prim}}\right)^2 \sqrt{\frac{f_{\rm disc}}{0.01}}\,\mathrm{AU}, \tag{6}$$

$$\sigma \equiv \frac{M_\star \cdot Z \cdot f_{\rm rock} \cdot f_{\rm disc} \cdot \eta_{\rm rock}}{R_{\rm disc}} \quad [M_\oplus/\mathrm{AU}], \tag{7}$$

$$a = 0.596 \cdot \frac{M_\star}{M_{\odot,\rm prim}} \cdot \Omega^{2/7} \quad [M_\oplus], \tag{8}$$

where $M_{\odot,\rm prim} = 1.14\,M_\odot$ (Sol calibration anchor) and the universal physical constants are listed in Table 2. The $M_\star^2$ exponent in Eq. 6 is a model choice calibrated within the framework; accretional snow-line treatments give shallower stellar-mass scalings (Kennedy & Kenyon 2008; Mulders, Ciesla & Min 2015), and the difference is testable across the M-dwarf catalogue systems. The quantity σ (Eq. 7) we name the *Annulus Allocation Factor*: it is the intensive rate at which rocky mass is allocated per unit radius along the disc. For an LBP $\Sigma \propto r^{-1}$ profile, $\sigma = \mathrm{d}M/\mathrm{d}r = 2\pi r\Sigma(r)$ is a radial constant, which is what makes the linear allocation rule (Eq. 10) well-defined as a function of r alone. The 1970s disc-dynamics literature (Lynden-Bell & Pringle 1974; Pringle 1981) arrived at this constant implicitly via the self-similar viscous solution but did not name it.

Disc compression regime.—The dimensionless compression $C \equiv R_A/R_{\rm disc}$ classifies the disc: $C < 1$ is the normal regime; $C \geq 1$ is the *inverted* regime, where rocky allocation concentrates at the magnetospheric inner edge (relevant for compact M-dwarf systems such as TRAPPIST-1).

Half-step occupancy in the inverted regime.—Inversion changes how viscous matter behaves, and thereby which rungs of the ladder are fed. With the entire disc inside the magnetosphere, magnetic stresses stiffen the gas toward corotation (reducing the Keplerian shear that cross-feeds adjacent annuli) and the compressed reservoir's gas density crushes the random velocities of solids. In the normal regime a rung body's feeding zone runs at its geometric (Gaussian-HWHM) width: adjacent zones at full step touch at half amplitude, so the midpoint material is consumed during consolidation and full-step occupancy is enforced by starvation, not exclusion. In the inverted regime, low-mass bodies' zones become *drag-limited*, narrower than geometric: midpoint material survives, and interstitial bodies consolidate at the half-step sites $r_n\sqrt{\rho}$ wherever the resulting half-step lattice is Hill-stable for the occupant

Table 2. Universal physical constants.

Symbol	Value	Meaning
g	0.82	Grain opacity (0 grain-grown $\rightarrow$ 1 ISM-like)
Z	0.014	Solar metallicity
$f_{\rm rock}$	0.22	Rocky fraction of condensables
$\eta_{\rm rock}$	0.78	Rock retention factor
$\eta_{\rm ice/rock}$	3.5	Ice/rock ratio past snow line
$\epsilon_{\rm pebble}$	0.40	Pebble capture efficiency
$M_{\rm thresh}$	3.0 $M_\oplus$	Gas-accretion threshold
A_0	4.39 $M_\oplus^{-1}$	Tanigawa-Ikoma H/He accretion coefficient
$k_{\rm H/He}$	0.691 Myr^{-1}	Envelope decay rate
$t_{\rm disc}$	5 Myr	Disc dispersal time
$M_\odot/M_\oplus$	332946	Solar mass in $M_\oplus$

(gap/$R_{H,\rm mut} \geq 7$). We name this half-step occupancy state the *Luger lattice*, after the work that exhibited its complete form (the TRAPPIST-1 resonant chain, whose member h was predicted from the occupied rungs before detection (Luger et al. 2017)) and we note the eponym is earned twice over: the lattice's natural period ratio $\rho^{-3/4} = 1.497$ sits within 0.2% of the 3:2 commensurability that dominates that chain. Giants are Hill-limited rather than drag-limited, their reach spans the midpoint regardless of damping, so the gate excludes them naturally. The two occupancy harmonics map onto the two observed resonant-chain families: full-ladder chains (adjacent period ratio $\rho^{-3/2} = 2.24$) relax into 2:1 locks (GJ 876, HR 8799) while Luger-lattice chains are born essentially *on* 3:2: TRAPPIST-1's seven planets occupy seven consecutive lattice sites in the calibrated fit (§11.2), with its outermost member h anchored directly on the dam (slot 0 = $R_{\rm disc}$), and compact Kepler chains follow the same pattern. Mixed occupancy follows the same gate: in Kepler-90 the two giants hold full rungs while the six small inner planets occupy the lattice. The inverted ladder's inward projection is terminated not by geometry but by the star itself: during the disc era the host sits on its Hayashi track, bloated to $R_{\rm HT} \approx 2.3\,R_\odot(M_\star/M_\odot)^{2/3}(t/{\rm Myr})^{-1/3}$, and rungs interior to that photosphere are *consumed*: their allocation is real, its destination is the star, and the predicted planet mass at such sites is zero (TRAPPIST-1's slots 6–10; 55 Cancri's slots 9–10). The inner edge of a compact system thereby tracks the host's protostellar radius: 55 Cancri e sits at the first rung outside its formation-era photosphere.

2.4. *Mass-allocation function*

The total planet mass at r is

$$M(r) = M_{\rm rock}(r) + M_{\rm ice}(r) + M_{\rm peb}(r) + M_{\rm H/He}(r) + \delta M, \tag{9}$$

where δM accommodates per-planet post-formation modifications such as giant-impact mass loss.

Rock allocation.—For the normal regime ($C < 1$),

$$M_{\rm rock}(r) = \begin{cases} \sigma \cdot \left[(r + \frac{5}{6} R_A) - R_A \right] \cdot f_{\rm trunc}(r) & R_A \le r < 2R_A \\ [a + \sigma \cdot r] \cdot f_{\rm trunc}(r) & 2R_A \le r \le R_{\rm disc} \\ 0 & \text{otherwise} \end{cases} \tag{10}$$

plus a snow-line pile-up bump (active when $r \le r_{\rm snow}$):

$$M_{\rm bump}(r) = \tfrac{1}{2} \sigma r_{\rm snow} \exp\left[-\frac{(r - r_{\rm snow})^2}{2(0.15 r_{\rm snow})^2} \right] \tag{11}$$

and an outer-edge contribution:

$$M_{\rm outer}(r) = \sigma R_{\rm disc}\, C \, \exp\left[-\frac{(r - R_{\rm disc})^2}{2(0.3 R_{\rm disc})^2} \right]. \tag{12}$$

Edge truncation of the feeding zone: the hexagonal void.—A planet near a disc edge loses part of its feeding zone to the void beyond the dam, and the fraction lost follows from packing geometry alone. In two dimensions the densest arrangement of equal bodies is hexagonal close-packing, in which every interior body has exactly six nearest neighbours spaced at 60° intervals: six being the two-dimensional "kissing number" (Thue 1910; Conway & Sloane 1999). A growing embryo embedded in a drifting pebble field competes for material along the six sectors separating it from those six neighbours, so its feeding zone (in the sense of Safronov 1972) divides naturally into six equal shares. For a body in the disc interior all six sectors are populated with solids. For a body sitting on the outer dam ($r = R_{\rm disc}$), the outward-facing sector lies beyond the dam, in the void where no solid material exists: the neighbour that would have occupied that sector is absent, and the share of material that sector would have supplied is simply never delivered. The body therefore accretes from five of its six sectors (retention $= 5/6 \approx 0.83$, loss $= 1/6 \approx 0.17$) a geometric constant with no fitted parameter. Equivalently, the feeding zone can be written as an asymmetric radial interval with inward range L and outward range S in the ratio $L{:}S = 5{:}1$; smoothing that interval across the edge gives

$$f_{\rm trunc}(r) = \max\left[0,\ 1 - \frac{(r + S) - R_{\rm disc}}{L + S} \right], \qquad S = 0.025\, R_{\rm disc},\ L = 5S, \tag{13}$$

equal to 1 well inside the disc (all six sectors populated), exactly $5/6$ at $r = R_{\rm disc}$, and 0 beyond $1.125\, R_{\rm disc}$. Neptune, sitting on Sol's outer dam, carries precisely this one-sixth ($\sim 17\%$) reduction in its rock and ice allocation. The same one-sixth loss recurs at the inner edge, where the missing sector faces the magnetospheric void inside R_A; it enters Eq. 10 through the ramp offset $\frac{5}{6} R_A$. The outer-reservoir Gaussian (Eq. 12) is the dam pile-up itself, not a feeding zone, and is not truncated.

Ice allocation.—Full condensation past the snow line gives an ice-to-rock mass ratio of $\eta_{\rm ice/rock} \approx 3.5$, set by solar condensable-element abundances (Lodders 2003; Asplund et al. 2009), which yield a cosmochemical ice/rock partition of ~ 3–4 depending on whether organics and CO/CO_2 ices are included (Pollack et al. 1994; Ciesla & Cuzzi 2006; Oka, Nakamoto & Ida 2011).

Only a fraction of this ice is retained as planet-building material, and the fraction falls with radius: slower accretion at large r allows more ice to drift inward and sublimate (Ciesla & Cuzzi 2006; Garaud & Lin 2007; Kennedy & Kenyon 2008); outer-disc solids are more easily lost to scattering and photoevaporation (Owen 2017; Picogna et al. 2019); and late-forming bodies lose ice as the disc photosphere recedes (Krijt et al. 2016, 2018). We capture these combined effects through an empirical exponential ice-retention factor:

$$\eta_{\rm ice}(r) = \exp\left[-\frac{r - r_{\rm snow}}{0.8 R_{\rm disc}}\right], \tag{14}$$

$$M_{\rm ice}(r) = \sigma\,(r - r_{\rm snow}) \cdot \eta_{\rm ice/rock} \cdot \eta_{\rm ice}(r) \cdot f_{\rm trunc}(r) + M_{\rm bump}(r). \tag{15}$$

The decay length $0.8\,R_{\rm disc}$ is calibrated to Uranus's and Neptune's ice fractions and is consistent with the depletion scaling expected from Ciesla & Cuzzi (2006)'s viscous transport modeling. The snow-line bump $M_{\rm bump}$ (Eq. 11) is predominantly ice past $r_{\rm snow}$, since the pressure trap concentrates condensable material at the ice-condensation front (Schoonenberg & Ormel 2017; Drążkowska & Alibert 2017).

Pebble flux allocation.—Drifting solids in a protoplanetary disc are aerodynamically transported inward as a disc-wide *pebble flux* (Birnstiel et al. 2012; Lambrechts & Johansen 2012, 2014; Bitsch et al. 2015). Gas-eligible cores along the drift path capture a share of this flux via pebble accretion. We allocate the total budget across eligible cores according to per-planet weights set by their radial distance from the snow line (the dominant pebble source region). The total pebble-flux budget is

$$M_{\rm peb,total} = f_{\rm disc} \cdot Z \cdot (1 - f_{\rm rock}) \cdot M_\star \cdot \epsilon_{\rm pebble}, \tag{16}$$

distributed among gas-eligible planets according to weights $w_i = (r_i - r_{\rm snow})^{-1/2}$ (zero if $r_i \leq r_{\rm snow}$):

$$M_{\rm peb,i} = M_{\rm peb,total} \cdot \frac{w_i}{\sum_{j\in{\rm eligible}} w_j}. \tag{17}$$

Gas-accretion threshold.—A planet is *gas-eligible* if its core mass (rock + ice) exceeds $M_{\rm thresh} = 3\,M_\oplus$ *and* its formation time $t_{\rm form} < t_{\rm disc} = 5$ Myr. The threshold reflects the critical core mass at which the envelope's self-gravity overwhelms its thermal pressure support, triggering runaway gas accretion (Mizuno 1980; Pollack et al. 1996; Ikoma, Nakazawa & Emori 2000; Rafikov 2006; Piso & Youdin 2014; Lee & Chiang 2015). The classical Mizuno value ($\sim 10\,M_\oplus$) assumed solar-metallicity opacity-rich envelopes; modern revisions accounting for grain-size evolution and reduced opacity in young protoplanetary discs place the critical core mass closer to 3–5 $M_\oplus$ (Ikoma, Nakazawa & Emori 2000; Hori & Ikoma 2010; Piso & Youdin 2014). We adopt 3 $M_\oplus$ consistent with the low-opacity end of this range, appropriate for the high-accretion-rate protoplanetary phase. The disc-dispersal timescale of 5 Myr is the observationally-determined median gas-disc lifetime for solar-mass stars (Haisch et al. 2001; Hartmann et al. 1998; Mamajek 2009; Williams & Cieza 2011).

Hydrogen-helium envelope.—For gas-eligible cores, the H/He envelope grows by Kelvin-Helmholtz contraction (the envelope radiates away heat, shrinks, and admits more gas) until either the core reaches its runaway-accretion mass or the gas disc disperses (Pollack et al. 1996; Ida & Lin 2004; Mordasini 2014; Lissauer et al. 2009; Tanigawa & Ikoma 2007). The Tanigawa-Ikoma 2007 formulation gives a quadratic dependence of gas-envelope mass on core mass during the runaway-accretion phase, integrated over the gas-disc dispersal window:

$$A(t_{\rm form}) = A_0 \exp[-k_{\rm H/He}\, t_{\rm form}], \tag{18}$$

$$w_{\rm wind}(r) = \frac{1}{1 + (\Omega/30)(0.5/r)^2}, \tag{19}$$

$$M_{\rm H/He} = A_0 \cdot M_{\rm core}^2 \cdot \exp[-k_{\rm H/He}\, t_{\rm form}] \cdot w_{\rm wind}(r), \tag{20}$$

where $M_{\rm core} = M_{\rm rock} + M_{\rm ice} + M_{\rm peb}$, and $t_{\rm form}$ is the formation time inferred from the Annulus Allocation Factor via $t_{\rm form} = 0.10 \cdot r/\sigma$ (Eq. 22, §2.5).

The accretion coefficient $A_0 = 4.39\, M_\oplus^{-1}$ is the most empirically-tuned parameter in the model. Its physical origin is the integrated runaway gas accretion rate (Tanigawa-Ikoma 2007), which scales as

$$A_0 \sim \frac{\Sigma_{\rm gas,0} \cdot \Omega_K \cdot \tau_{\rm disc}}{M_\star}, \tag{21}$$

where $\Sigma_{\rm gas,0}$ is the local gas surface density at formation, Ω_K is the Keplerian angular velocity, and $\tau_{\rm disc}$ is the gas-dispersal timescale. For Sol's minimum-mass solar nebula values at Jupiter's slot-3 position ($\Sigma_{\rm gas,0} \approx 8$ g/cm^2 at 5.2 AU, $\Omega_K \approx 0.53$ rad/yr, $\tau_{\rm disc} \approx 3$ Myr), substituting into Eq. 21 reproduces $A_0 \approx$ 4–6 in the correct order of magnitude. The framework currently treats A_0 as a calibration constant tuned to Jupiter's bulk composition; a first-principles derivation from disc evolution models (Pollack-Bitsch-Mordasini class) is identified as future work.

The exponential decay rate $k_{\rm H/He} = 0.691$ Myr^{-1} (half-life ~ 1 Myr) reflects the rapid loss of available gas as the disc disperses: protoplanetary disc gas surface density declines exponentially over $\sim$ 1–3 Myr (Haisch et al. 2001; Hartmann et al. 1998; Mamajek 2009), suppressing envelope accretion for late-forming cores.

Inner gas-envelope boundary (solar-wind suppression).—The wind-suppression factor $w_{\rm wind}(r)$ encodes the well-established extreme-ultraviolet (XUV) photoevaporation of close-in gas envelopes by stellar irradiation (Murray-Clay et al. 2009; Lopez, Fortney & Miller 2012; Owen & Wu 2013; Jin et al. 2014; Lopez & Fortney 2013). Young stars exhibit XUV luminosities $\sim 10^3$ times the modern Sun's value (Tu et al. 2015; Ribas et al. 2005), and this flux drives hydrodynamic escape from H/He envelopes at small orbital radii. The Ω dependence in w_{wind} captures the correlation between primordial stellar rotation and XUV luminosity: rapidly-rotating young stars emit substantially higher XUV flux (Wright et al. 2011; Tu et al. 2015), stripping envelopes more efficiently. The r^{-2} scaling is the canonical inverse-square dependence of incident flux. Together, $w_{\text{wind}}(r)$ establishes an effective *inner boundary* for gas-envelope retention: close-in planets in fast-rotating systems retain little to no H/He, consistent with the observed bimodal radius distribution of Kepler super-Earths and sub-Neptunes (Fulton et al. 2017; Owen & Wu 2017).

2.5. *Formation-time cascade*

The formation time t_{form} for a planet at radius r is inferred directly from the Annulus Allocation Factor σ (Eq. 7) via a linear pebble-drift-timescale relation:

$$t_{\text{form}}(r) = 0.10 \cdot \frac{r}{\sigma} \quad [\text{Myr}]. \tag{22}$$

This is the only place in the model where t_{form} enters as an independent quantity, and it feeds directly into the H/He envelope accretion (Eq. 18) and the gas-window eligibility ($t_{\text{form}} < t_{\text{disc}}$, §2 paragraph on gas-accretion threshold). The formula's physical content is:

Scaling with σ (the Annulus Allocation Factor).—Higher σ means more local solid mass per AU available to a growing core. For pebble accretion, the core-assembly rate scales linearly with the local solid surface density and therefore with σ: doubling σ roughly halves the time to reach the critical core mass. The $1/\sigma$ dependence in Eq. 22 captures this directly. Across systems with very different disc masses (e.g., HR 8799's $f_{\text{disc}} \sim 30\%$ versus TRAPPIST-1's $f_{\text{disc}} \sim 0.5\%$), this single scaling reproduces the observed formation-timescale spread without per-system tuning.

Scaling with r.—At fixed σ, formation time grows linearly with orbital radius. The dominant physical effect is the radial pebble-flux integration length: cores at larger r must wait longer for inward-drifting pebbles to traverse the disc and reach them (Ormel & Klahr 2010; Lambrechts & Johansen 2012, 2014; Bitsch et al. 2015, 2018; Johansen & Lambrechts 2017). The linear r dependence (as opposed to the steeper $r^{5/2}$ that pure local-density arguments would suggest for a $\Sigma \propto r^{-1}$ LBP profile) reflects the gas-drag-mediated accretion regime in which core growth is rate-limited by the pebble-flux supply rather than by the local solid reservoir alone, a result robust across the Ormel & Klahr (2010), Lambrechts & Johansen (2012), and Bitsch et al. (2015) pebble-accretion frameworks.

Cascade structure and calibration.—Together, these two scalings produce an outward-sweeping formation cascade in which inner planets form first and outer planets form later, in a single disc. The proportionality constant 0.10 is calibrated so that Sol's Jupiter (at $r = 5.20$ AU, $\sigma = 0.304\, M_\oplus/\mathrm{AU}$) lands at $t_\mathrm{form} = 1.71$ Myr, consistent with Kruijer et al. (2017)'s isotopic constraint of < 1 Myr for Jupiter's core formation followed by envelope growth. Once calibrated, the same constant reproduces all eight Solar System bodies' formation timings (Table 3) and generalizes across the 22 further catalogued systems.

Sensitivity.—For cores below $M_\mathrm{thresh} = 3\, M_\oplus$ the envelope physics is gated off and t_form has no further effect. For gas-eligible cores, t_form sets the envelope mass through Eq. 18; across the calibrated systems the observed envelope masses correspond to formation times within a factor of ~ 2 of the cascade value, indicating that the r/σ scaling captures the dominant physics.

Mass-driven vs. radius-driven cascades.—In Sol, mass and radius correlate (heavier planets are further out), so the r/σ cascade produces a clean monotonic t_form-by-radius sequence. In compact systems such as 55 Cnc or Kepler-90, where inner gas giants exist at $r < 1$ AU and the system has been compressed by primordial rotation (Eq. 23), t_form is driven primarily by core mass (through the envelope term's exponential sensitivity to t_form) rather than by formation radius. This dual behavior, natural radius-cascade in spread architectures, mass-cascade in compact architectures, is an emergent property of the model, not an imposed feature.

2.6. *Boundary conditions*

The allocation vanishes outside the two viscous dams. At the inner edge the ramp in Eq. 10 clamps to zero (for $r < R_A/6$, inside the magnetospheric void); at the outer edge f_trunc (Eq. 13) fades the allocation smoothly to zero by $1.125\, R_\mathrm{disc}$. The linear rule $a + \sigma r$ is never extrapolated beyond the disc. In the present model $R_{\bar{A}} \equiv R_\mathrm{disc}$.

2.7. *Anchoring the cascade to the outermost planet*

The outermost observed planet anchors the cascade: slot 0 sits at the Davis Dam, so the outermost planet's orbital radius marks R_disc itself (or an inner slot of it, $r_\mathrm{out} = \rho^k R_\mathrm{disc}$, when the outermost surviving body formed at slot $k > 0$). The primordial spin then follows from Eq. 4:

$$\Omega = \left(\frac{30\, M_\star / M_{\odot,\mathrm{prim}}}{R_\mathrm{disc}} \right)^2 . \tag{23}$$

Compact systems (e.g., TRAPPIST-1, Kepler-90) thereby resolve to fast primordial rotators with strongly compressed discs, while wide architectures like Sol sit near $\Omega = 1$. In rare cases the outermost planet lies beyond the dam entirely: a gas giant that formed at slot 0 and was destabilized outward (HD 219134's outermost giant is the catalogue's clearest example).

Table 3. Solar System fit. Inputs: $M_\star = 1.14\,M_\odot$, $\Omega = 0.995$, $f_{\rm disc} = 0.0104$, calibrated on the ISU anchors Venus and Neptune. Planets form on their slots exactly: $r_{\rm slot}$ is the formation position, and every difference from $r_{\rm obs}$ is post-formation displacement signed by a documented event. Jupiter and Saturn sit inside their slots because they paid the swarm's ejection bill; Uranus sits outside because it rose on the hand-down exchange; Neptune, seated on the dam trap, never moved (Table 6 and the displacement ledger in the text).

Body	Slot	$r_{\rm slot}$ (AU)	$r_{\rm obs}$ (AU)	$t_{\rm form}$ (Myr)	$M_{\rm pred}$ ($M_\oplus$)	$M_{\rm obs}$ ($M_\oplus$)
Mercury	8	0.41	0.387	0.13	0.724	0.055†
Venus	7	0.69	0.723	0.22	0.815	0.815
Earth	6	1.19	1.000	0.38	0.971	1.000‡
Mars	5	2.04	1.524	0.65	1.336	0.107*
Jupiter	3	5.98	5.203	1.90	317.83	317.83
Saturn	2	10.25	9.537	3.25	145.9	95.16***
Uranus	1	17.55	19.189	5.57	35.8	14.54††
Neptune	0	30.07	30.070	10.90	17.15	17.15
Inferred bodies (not observed)						
Vulcan (merger partner)	9	0.24	···	0.08	0.064	merged → Mercury
Slot-4 planet**	4	3.49	···	1.11	2.57	struck Saturn at t^*
Slot-5 planet**	5	2.04	···	···	1.23	struck Uranus at t^*

† Mercury's mass is the iron-enriched merger remnant of slot 8 + slot 9 Vulcan ($\eta_{\rm retain} = 0.070$ at $\Delta v/v_{\rm esc} \approx 1.49$; §3.1).
‡ Earth's +3% excess attributed to inner-scattered slot-4 debris (the Theia delivery; Moon-forming event; §5.6).
* Mars's −92% deficit attributed to Jupiter-driven slot 4–5 dispersal: the surviving inner-scattered embryo settles in the band swept between Jupiter's formation and final positions (§5.6, §4.6).
** The slot 4 and slot 5 primaries are the identified impactors of Saturn and Uranus respectively, struck at the common epoch $t^* = 4.61$ Myr; their swarms dispersed inward and outward (§4.6).
*** Saturn's −35% deficit is accretion *truncation*: the slot-4 primary's impact at $t^* = 4.61$ Myr cut off runaway gas accretion with 61% of the predicted gas delivered. The deficit is gas that never arrived, not gas removed (§4.6).
†† Uranus's −59% deficit: its formation clock ($t_{\rm form} = 5.57$ Myr) postdates t^*, so it *never gassed* – the observed mass is the bare core (predicted 15.0, observed 14.54, −2.9%). Its +1.64 AU outward displacement is the erosive-impact recoil of the slot-5 hit (the polarity law; §4.6).

3. CALIBRATION TO THE SOLAR SYSTEM

We apply the model to the Solar System using inputs $M_\star = 1.14\,M_\odot$, $\Omega = 0.995$ (from Eq. 23 with $R_{\rm disc}$ at Neptune's orbit), and $f_{\rm disc} = 0.0104$. These three parameters, with the universal constants of Table 2, fix the disc properties ($R_{\rm disc} = 30.07$ AU, $R_A = 0.20$ AU, $r_{\rm snow} = 2.70$ AU, $\sigma = 0.304\,M_\oplus$/AU, $a = 0.596\,M_\oplus$).

Table 3 reports the resulting fit for the ten bodies of the Solar System, including the inferred Theia swarm at the snow line and Vulcan as Mercury's merger partner. The ISU anchors Venus and Neptune fit exactly by construction; every other body's deviation is attributed, with independent observational support, to one of the two dynamical events (table footnotes).

The deviations between predicted and observed masses are all attributable to the framework's two dynamical events: the Mercury–Vulcan adjacent-slot merger (§3.1) and Jupiter's scattering of the slot 4–5 embryo swarm (§5.6), whose asymmetric inner/outer split accounts for the Theia impact on Earth, the Mars deficit, the Saturn and Uranus gas truncations and tilts, and the near-empty asteroid belt.

3.1. *The Vulcan merger and Mercury's iron-enriched composition*

Mercury's anomalously high core mass fraction ($\sim 70\%$ Fe by mass) (Nittler et al. 2011; Hauck et al. 2013) is the most prominent compositional anomaly in the inner Solar System. The mainstream interpretation has been that Mercury was originally a chondritic-composition body that subsequently lost a substantial fraction of its silicate mantle. Three mechanism classes have been proposed: single giant impact (Benz et al. 1988, 2007), hit-and-run sequences (Asphaug & Reufer 2014; Chau et al. 2018), and high-temperature nebular processing (Cameron 1985; Fegley & Cameron 1987; Ebel & Alexander 2011). Each has tensions with observations (Peplowski et al. 2011), and none has emerged as a consensus mechanism. The sharpest discriminator is volatile abundance: high-temperature nebular processing and vaporizing giant impacts predict strong depletion of moderately volatile elements, yet MESSENGER found Mercury's surface volatile-*rich* (high S and K; Peplowski et al. 2011; Nittler et al. 2011): falsifying the high-temperature class outright, while remaining fully consistent with a *cold* merger of two icy-rocky inner-cascade bodies whose energy went into mantle ejection rather than global vaporization.

Our framework reverses the direction of inference. Mercury is not a stripped chondritic body; it is the *merger remnant* of two adjacent inner-cascade slot bodies, with the iron enrichment arising naturally from the energetics of the merger event itself.

The local-capacity cascade allocates planet-forming material to geometrically-spaced slots inward from the outer Davis Dam at $R_{\rm disc}$. For Sol, the geometric ratio $\rho \approx 0.584$ (the half-amplitude-at-45° projection of the Gaussian accretion-zone HWHM; see §2) generates slot positions $r_n = R_{\rm disc}\rho^n$. Slot 8 sits at $r_8 \approx 0.41$ AU (Mercury's current position), and slot 9 sits at $r_9 \approx 0.24$ AU. The framework's slot 9 body, which we name *Vulcan* after Le Verrier's 19th-century hypothesis of a sub-Mercury planet (Le Verrier 1859), grew during the protoplanetary phase to $\sim 0.06\,M_\oplus$ in primordial composition (rocky core).

Hill spacing.—Orbital separations in this paper are expressed in Hill radii. The Hill radius of a body of mass m orbiting at semi-major axis a,

$$R_H = a\left(\frac{m}{3M_\star}\right)^{1/3}, \tag{24}$$

is the radius of the region in which the body's gravity dominates over the star's tidal field (Hill 1878; Murray & Dermott 1999). For a planet pair the *mutual* Hill radius, $R_{H,\rm mut} = \frac{1}{2}(a_1+a_2)\,[(m_1+m_2)/3M_\star]^{1/3}$, is the natural unit of orbital spacing: numerical experiments show that similar-mass pairs separated by less than $\sim 10\,R_{H,\rm mut}$ are dynamically unstable, encountering each other on timescales short compared to the system age (Gladman 1993; Chambers et al. 1996), with ~ 7 mutual Hill radii the empirical stability boundary for adjacent bodies in multi-planet chains (Smith & Lissauer 2009).

The Mercury–Vulcan merger.—The two bodies are dynamically packed at ~ 4 mutual Hill radii of separation: well below the stability threshold. They are guaranteed to encounter each other gravitationally; the only question is the encounter outcome. The slot mechanism shapes the approach: while the gas lived, the persisting zonal flows damped the destabilised partner's excursions toward the adjacent occupied band (§2), funnelling the pair into a low-relative- inclination, in-plane encounter rather than a hyperbolic one.

The relevant encounter energy is set by the orbital velocity difference between the two slots:

$$\Delta v_{\rm orb} = |v_{\rm orb}(r_9) - v_{\rm orb}(r_8)| \approx 15.4 \text{ km/s}, \tag{25}$$

where $v_{\rm orb}(r) = 29.785\sqrt{M_\star/r}$ km/s at Sol. The escape velocity of the combined merger body ($M_{\rm combined} = 0.789\, M_\oplus$ at the slot allocations) is

$$v_{\rm esc} = 11.186\, M_{\rm combined}^{1/3} \approx 10.3 \text{ km/s}. \tag{26}$$

The energy ratio $\Delta v_{\rm orb}/v_{\rm esc} \approx 1.5$ places the encounter in the *erosive-stripping regime*: the kinetic energy delivered substantially exceeds the gravitational binding energy of the merged body's silicate mantle. Standard impact-erosion physics (Asphaug 2010; Leinhardt & Stewart 2012) predicts that under these conditions, the silicate mantle of both bodies is preferentially ejected while the dense iron cores merge and are retained. The controlling parameter $\Delta v/v_{\rm esc}$ is classical: it is the inverse of the Safronov number that has governed accrete-versus-erode outcomes since Safronov (1972), and the quantitative accretion-efficiency curves of Leinhardt & Stewart (2012) and the hit-and-run mapping of Asphaug & Reufer (2014) supply the impact-physics basis for the retention law adopted below: mechanically, our adjacent-slot merger is a similar-mass, modest-Δv collision of exactly the class those works parameterize.

We adopt the empirical retention function

$$\eta_{\rm retain} = \max[0.05,\ 0.969 - 0.605\,(\Delta v_{\rm orb}/v_{\rm esc})] \tag{27}$$

with both velocities evaluated at the participants' *slot* radii and the combined mass taken at the *slot-radius allocations* – the on-slot doctrine applied to the collision itself. Two anchors fix the two constants: (a) the Mercury–Vulcan merger (Sol slots 8/9, combined primordial $0.789\, M_\oplus$; $\Delta v \approx 15.4$ km/s against $v_{\rm esc} \approx 10.3$ km/s gives $\Delta v/v_{\rm esc} \approx 1.49$ and $\eta_{\rm retain} = 0.070$) and (b) the milder Tau Ceti e merger ($\Delta v_{\rm orb}/v_{\rm esc} \approx 0.275$, $\eta_{\rm retain} = 0.803$). The intercept sits below unity because even the gentlest merger sheds percent-level ejecta; the 0.05 floor is the bound-core residual of a hit at several escape velocities. An earlier calibration of this law (floor 0.3, slope 0.37) carried the observed-radius conflation – it priced the collision at where the participants are seen today rather than at the seats where they formed – and consequently underestimated the primordial masses entering every merger budget in the catalogue.

Composition prediction.—Combined primordial Vulcan + Mercury rocky mass at the slot allocations is $0.789\,M_\oplus$. At $\eta_{\rm retain} = 0.070$, the residual post-merger body has mass $\sim 0.055\,M_\oplus$, matching Mercury's observed $0.055\,M_\oplus$. The retained composition is iron-dominant because (i) iron is centrally concentrated in each pre-merger body's differentiated core, (ii) the combined post-encounter core sinks inward as denser material, and (iii) silicate mantle is preferentially ejected as outer-layer material. Mercury's $\sim 70\%$ Fe bulk composition is thus the natural consequence of the merger geometry, not the result of post-formation impact stripping of an originally chondritic body.

This resolves the Mercury anomaly without requiring (i) a separate giant impactor of specific kinematics, (ii) extreme nebular processing, or (iii) hit-and-run with larger embryos. The Mercury–Vulcan pair were always packed at unstable mutual Hill spacing; their merger was energetically constrained to produce an iron-enriched remnant.

Comparison to other merger systems.—The retention function's mild end is anchored by the Tau Ceti e partial-merger interpretation: at $\Delta v_{\rm orb}/v_{\rm esc} \approx 0.275$ the formula gives $\eta_{\rm retain} \approx 0.80$, a low-energy merger retaining most of the combined mass. The two anchors thus calibrate opposite ends of the merger-energy spectrum. We predict that mergers at other heliocentric distances and stellar masses follow the same scaling, with retention set by local orbital and escape velocities rather than ad hoc impactor kinematics.

The same model has been applied to the 22 further catalogued systems (Table 9), spanning spectral classes A–M and architectures from compact super-Earth chains to wide-orbit super-Jupiters and multi-star systems; the calibrated $f_{\rm disc}$ distribution across that catalogue is analysed in §11.2.

4. ORBITAL SIGNATURES OF IMPACT EVENTS: VENUS AND NEPTUNE AS THE PRIMORDIAL-PLANE REFERENCE

The standard practice of using Earth's orbit (the ecliptic) as the reference plane for Solar System inclinations is dynamically biased. Earth's orbital plane was perturbed by the Theia impact (Canup & Asphaug 2001; Canup 2012); using it as zero-reference contaminates the inclination of every other planet by a contribution that is itself part of the impact-event signature we seek to identify. We resolve this by using *Venus and Neptune as the primordial-plane reference.* This choice is supported by four independent lines of evidence presented in §4.1.

4.1. *Venus and Neptune as ISU: a four-line proof*

The framework's identification of Venus and Neptune as the Solar System's only in-situ unchanged (ISU) bodies is not an arbitrary or convenience-based assignment. Four observationally independent lines of evidence converge on this same identification.

Line 1: Venus and Neptune are the most circular orbits.—The eccentricities of the eight Solar System planets, ordered from most to least circular:

Venus	$e = 0.0068$
Neptune	$e = 0.0086$
Earth	$e = 0.0167$
Uranus	$e = 0.0457$
Jupiter	$e = 0.0489$
Saturn	$e = 0.0565$
Mars	$e = 0.0934$
Mercury	$e = 0.2056$

Venus and Neptune occupy the top two positions with a clean factor-of-two gap separating them from Earth and all other planets. No other planetary pairing in the Solar System shows this kind of clean ranking separation. We note that osculating eccentricities oscillate over secular cycles (Earth's, for instance, will fall to ~ 0.003–0.005 over the next ~ 30 kyr), so the snapshot ranking is suggestive rather than decisive on its own; the durable content of this diagnostic is carried by the free amplitudes and angular-momentum-deficit argument of Line 2.

Line 2: Eccentricity is a one-way integral of post-formation dynamical history.—Gas-disc dynamical friction and gas-drag damping during the disc-active phase drive orbits toward a near-circular *floor* ($e \lesssim 0.01$): damping is efficient but finite, leaving a small residual rather than $e = 0$ (Goldreich & Tremaine 1980; Ida 1990; Kominami & Ida 2002). Once the gas disc disperses ($\sim$ 5–10 Myr post-formation), no efficient eccentricity-damping mechanism exists for planetary-mass bodies, and Kominami & Ida (2002) show explicitly that post-gas dynamics pump eccentricity *up* from that floor. Subsequent dynamical events, migration (Walsh et al. 2011; Pierens & Raymond 2011), scattering (Tsiganis et al. 2005; Morbidelli et al. 2007), mean-motion resonance crossings, giant-impact events (Canup & Asphaug 2001), can only *add* eccentricity, not remove it: a body observed today *at* the gas-era floor has experienced none of them. Individual eccentricities do oscillate reversibly through secular angular-momentum exchange between the planets, but the system-level budget of that exchange, the angular momentum deficit (AMD; Laskar 1997), can only grow once the gas is gone, so a planet whose free eccentricity amplitude is near zero can never have been significantly excited.

Venus's and Neptune's $e \approx 0.008$ is therefore not just small, it is small in an irreversible sense. Whatever has happened to these two bodies since gas-disc dispersal ~ 4.5 Gyr ago, it has not been substantial enough to perturb them from their primordial near-circular state. The same statement cannot be made for any other planet in the system: Earth has been perturbed by Theia, Mars by the Borealis impactor, Mercury by Vulcan, the gas giants by mutual resonance evolution. This reading has support independent of the present framework: N-body reconstructions of the pre-excitation terrestrial system require Venus's primordial free eccentricity amplitude to have been near zero, with the terrestrial orbits dynamically colder than today (Brasser et al. 2013).

Line 3: ISU diagnostics are the inverse of established gravitational-capture diagnostics.—The orbital signatures used to identify gravitational capture are well established in the satellite-dynamics literature (Jewitt & Haghighipour 2007; Nesvorný et al. 2007; Sheppard & Jewitt 2003):

- High orbital inclination relative to the host's reference plane (Neptune's Triton at $i \approx 157^\circ$; Saturn's Phoebe at $i \approx 175^\circ$);
- High eccentricity (most Jupiter and Saturn irregular satellites have $e \gtrsim 0.1$);
- Retrograde orbital motion (the strongest single capture indicator);
- Position far from the host's natural formation/cascade zone.

In-situ formation is, by definition, the absence of these signatures. A body that formed and remained in its current orbit shows: prograde orbit, low inclination relative to the system's formation plane, low eccentricity, and position consistent with the local formation cascade. Applying the established capture-diagnostic methodology in the inverse direction identifies in-situ bodies by their lack of capture signatures.

For Venus: $e = 0.0068$, $i = 3.39^\circ$ (ecliptic-referenced; only 1.99° from the VN baseline), prograde orbit, position consistent with slot 7 of the cascade.

For Neptune: $e = 0.0086$, $i = 1.77^\circ$ (ecliptic-referenced; only 1.99° from the VN baseline), prograde orbit, position consistent with slot 0 of the cascade.

Both bodies score positive on every in-situ indicator and negative on every capture indicator. By the same methodology that identifies Triton as captured, Venus and Neptune are identified as in-situ.

Line 4: Mass-measurement precision.—Independent of orbital considerations, Venus and Neptune have masses known to part-per-million precision from spacecraft tracking (Magellan, Cassini-Huygens, Voyager 2). Their cascade-allocated mass predictions ($M_{\rm pred}^{\rm Venus} = 0.815\,M_\oplus$ from the slot 7 rocky allocation; $M_{\rm pred}^{\rm Neptune} = 17.15\,M_\oplus$ from the slot 0 cascade) match observed values to better than $0.001\,M_\oplus$. No other Solar System body achieves this combination of precision measurement *and* exact cascade match. The other rocky planets either have observed masses substantially different from cascade predictions (Mercury −92%, Mars −92%, both attributable to the framework's two dynamical events) or have δM contributions from late delivery (Earth +3% from inner-scattered slot-4 debris). The non-anchor gas giants show impact-truncated gas inventories: Saturn −35%, its runaway cut short by the slot-4 primary at $t^* = 4.61$ Myr; Uranus −59%, its gas era cancelled outright ($t_{\rm form} > t^*$), with the slot-5 impact also producing the 98° axial tilt (§4.6).

Convergence.—The four lines, most circular orbit (Line 1), one-way eccentricity preservation (Line 2), inverse-of-capture diagnostics (Line 3), exact mass-cascade match (Line 4), converge on the same identification. Venus and Neptune are uniquely ISU among Solar System bodies. The designation is not a convenience choice but a physically grounded observational conclusion.

Relation to prior art.—For Venus, the equivalent concept has precedent under different nomenclature: Jacobson et al. (2017) attribute Venus's absent magnetic dynamo to a primordially stratified core that was never mechanically mixed (that is, to Venus never suffering a late energetic giant impact) and Brasser et al. (2013) require its primordial free eccentricity to be near zero. The Jacobson et al. (2017) mechanism constrains more than the absence of one late impact: a stratified, never-stirred core requires Venus's *entire* accretion to have proceeded through many small, low-Δv impactors, a geophysical witness, independent of the orbital evidence, that Venus sits on the quiet branch of the impact-energetics ledger this framework assigns it. For Neptune, the mainstream holds the opposite: the resonant Kuiper belt populations (Pluto and the Plutinos in Neptune's 3:2 mean-motion resonance, with pumped eccentricities) are interpreted as bodies swept up and captured during an outward migration of Neptune by several AU (Malhotra 1993, 1995). The framework reads the same populations differently. In the cascade picture the Kuiper belt is not cascade material at all: it forms *outside* the Davis Dam ($R_{\rm disc} = 30.07$ AU, Neptune's slot), by slow accretion in the exterior reservoir that pools against the dam from outside (§2), a different formation mechanism from the slot cascade. Neptune's exterior mean-motion resonances (3:2 at 39.4 AU, 2:1 at 47.7 AU) all lie within that exterior region, and bodies drifting inward toward the dam are captured into the resonances of a *stationary* Neptune by drag-induced convergent drift, their eccentricities pumped to equilibrium during continued drift: the classic exterior-resonance trapping mechanism of Weidenschilling & Davis (1985). Adiabatic resonance capture requires only convergent relative drift: the standard model moves the planet, the framework moves the particles.

Drag trapping cannot hold dwarf-planet-mass bodies (drag acceleration falls with size), but those do not need it. Bodies that condense at the dam itself are born sharing Neptune's orbital radius (the dam is two-sided, producing Neptune from the interior cascade allocation and Pluto-class bodies from the exterior pool) and a body born co-orbital with a $17\,M_\oplus$ giant has no non-resonant survival path: non-resonant Neptune-crossing orbits are cleared on 10–100 Myr timescales, so after 4.5 Gyr the only survivors are those in protecting mean-motion resonances, whose libration geometry forbids close approaches. Pluto's eccentricity is then the scar of the scattering that preceded capture, not the pump of a migrating resonance. Its deep libration and Kozai state, historically read as fingerprints of slow adiabatic capture, are equally fingerprints of *survival*: erosion preferentially removes large-amplitude librators (Tiscareno & Malhotra 2009), so any initial resonant population converges to deep, Kozai-rich microstates, and resonance sticking continually feeds the islands from the scattered population (Lykawka & Mukai 2007). Two further observations favour this channel: the distant 5:2 resonance is anomalously well-populated, a standing puzzle for smooth migration (Gladman et al. 2012) but the natural outcome of sticking from an eccentricity-pumped scattered source, and Triton (retrograde, Pluto-like) is the expected prize of binary-exchange capture (Agnor & Hamilton 2006) acting on the factory's natively-binary output. The two claims interlock: binary-exchange capture *requires* a substantially binary source population, and gravitational collapse at the dam natively produces exactly the wide, near-equal-mass binary fraction observed in the cold classicals (Nesvorný et al. 2010): the factory supplies the prerequisite that makes Triton's capture mechanism available. The competing channels predict different joint distributions of resonant occupancy, eccentricity, and libration amplitude (§10), making the resonant Kuiper belt a sharp observational discriminator between the framework and migration-based models rather than a refutation of Neptune's ISU status.

4.2. *The Venus-Neptune mean plane*

Each planet's orbital plane is specified by its inclination i (tilt from a reference plane) and longitude of ascending node Ω. The orbital plane normal vector (in ecliptic-referenced Cartesian coordinates with $\hat{z}$ along the ecliptic pole) is

$$\hat{n}(i, \Omega) = (\sin i \sin \Omega,\ -\sin i \cos \Omega,\ \cos i). \tag{28}$$

For Venus ($i_V = 3.39°$, $\Omega_V = 76.68°$) (Williams 2024a; Standish 1998):

$$\hat{n}_V = (0.0575,\ -0.0136,\ 0.9982). \tag{29}$$

For Neptune ($i_N = 1.77°$, $\Omega_N = 131.78°$) (Williams 2024b; Standish 1998):

$$\hat{n}_N = (0.0230,\ 0.0206,\ 0.9995). \tag{30}$$

The Venus-Neptune mean plane normal is the average:

$$\hat{n}_{VN} = \frac{\hat{n}_V + \hat{n}_N}{|\hat{n}_V + \hat{n}_N|} \approx (0.0403,\ 0.0035,\ 0.9989). \tag{31}$$

The VN plane is offset from the ecliptic by $i_{VN} \approx 2.71°$ at ascending node $\Omega_{VN} \approx 95°$. By construction, both Venus and Neptune sit at 1.99° from this plane (in opposite directions).

The angle of any planet's orbital plane from the VN plane is

$$i_{\mathrm{VN}} = \arccos(\hat{n}_{\mathrm{planet}} \cdot \hat{n}_{VN}). \tag{32}$$

Table 4. Solar System inclinations relative to the Venus-Neptune primordial plane. Negative values indicate orbits below the VN plane (toward the invariable plane).

Planet	$i_{\rm VN}$ (°)	$\Delta i_{\rm VN}$ (°)	Status
Saturn	1.62	-0.37	slot-4 impact, gas truncated
Jupiter	1.81	-0.18	anchor body
Venus	1.99	$\equiv 0$	**ISU**
Neptune	1.99	$\equiv 0$	**ISU**
Mars	2.14	$+0.15$	Borealis impact
Uranus	2.13	$+0.14$	slot-5 impact, gas era cancelled
Earth	2.71	$+0.72$	Theia impact
Mercury	5.94	$+3.95$	Vulcan merger

4.3. *Inclinations relative to the VN baseline*

Computing Eq. 32 for all eight planets, then subtracting the 1.99° baseline so that Venus and Neptune sit at zero:

The pattern is consistent with the framework's predicted impact history. Bodies identified as impact-modified (Mercury via Vulcan, Earth via inner-scattered swarm embryo, Mars via swarm dispersal, Uranus via outer-scattered swarm impactor) all show positive inclination excess relative to the VN baseline. Bodies identified as in-situ formation products (Venus, Neptune) sit at zero by construction. Jupiter and Saturn show small negative offsets, reflecting the current invariable plane (dominated by their mass-weighted angular momentum) sitting slightly below the VN primordial plane. The $\sim 0.3°$ offset between the current invariable plane and the VN primordial plane is the cumulative signature of impactor-induced perturbations on Jupiter and Saturn during the slot 4-5 swarm dispersal event, not of any large-scale migration.

4.4. *Eccentricity excess*

Having established Venus and Neptune as the ISU reference (§4.1), we use their near-circular orbits ($e_V = 0.0068$, $e_N = 0.0086$) as the baseline ($\bar{e}_{VN} \approx 0.008$) and compute the eccentricity excess of each remaining planet relative to this baseline:

Table 5. Eccentricity excess relative to the Venus-Neptune baseline ($\bar{e}_{VN} = 0.008$).

Planet	e	Δe	$e/\bar{e}_{VN}$
Venus	0.0068	-0.001	~ 1
Neptune	0.0086	$+0.001$	~ 1
Earth	0.0167	$+0.009$	2
Uranus	0.0457	$+0.038$	6
Jupiter	0.0489	$+0.041$	6
Saturn	0.0565	$+0.049$	7
Mars	0.0934	$+0.085$	12
Mercury	0.2056	$+0.198$	26

Mercury's eccentricity ($e \approx 0.21$) is dramatically elevated, a factor ~ 26 above the VN baseline. This is consistent with a high-$\Delta v/v_{\rm esc}$ merger event delivering substantial orbital momentum exchange in addition to the catastrophic mass-loss signature. Mars's $e \approx 0.09$ ($12\times$ baseline) is similarly consistent with its scattering history and the Borealis impact. Saturn and Uranus show $\sim$ 6–7$\times$ baseline eccentricity, consistent with the swarm bombardment overlaid on mutual gas-giant secular evolution; Jupiter's comparable excess reflects its role as the scattering source.

4.5. *Coupled signatures: four mass-change events, four orbital records*

The combination of inclination excess and eccentricity excess provides a coupled diagnostic for the four mass-changed planets. For each event, three or four independent signatures should appear: mass anomaly (§3), inclination excess (§4.3), eccentricity excess (§4.4), and (for some) axial tilt:

Table 6. Coupled orbital signatures of the framework's four mass-change events. Each attribution is supported by *multiple* independent orbital observables on its target body, including the orbital displacement $\Delta a = r_{\rm obs} - r_{\rm slot}$, whose signs close as one angular-momentum ledger (see text).

Target	Mass anomaly	Δa (AU)	$\Delta i_{\rm VN}$	Δe	Tilt / spin
Mercury (Vulcan merger)	70% Fe	−0.02	+3.95°	+0.198	0° tilt (locked)
Mars (swarm + Borealis)	$-1.22\ M_\oplus$	−0.52	+0.15°	+0.085	25.2° tilt
Saturn (slot-4 impact)	$-50.7\ M_\oplus$	−0.71	−0.37°	+0.049	26.7° tilt
Uranus (slot-5 impact)	$-21.3\ M_\oplus$	+1.64	+0.14°	+0.038	98° tilt, gas era cancelled

Mercury (Vulcan merger).—Four independent signatures from the same high-$\Delta v/v_{\rm esc}$ encounter event: composition, mass, inclination, and eccentricity all extreme. The Vulcan body (slot 9 sub-Mercury) merger explains all four through the impact retention formula calibrated on this event.

Mars (inner-scattered swarm survivor).—Hemispheric dichotomy (Borealis basin), severe mass deficit (−92%), mild inclination perturbation, substantial eccentricity excess. Mars is one of $\sim$ 1–2 inner-scattered embryos from the slot 4-5 swarm that survived at $r < r_5$ (below Jupiter's chaotic zone); the Borealis-basin impact was a smaller sibling-embryo arriving at high velocity.

Saturn (slot-4 primary plus swarm debris).—−35% gas deficit (truncation; §4.6), 26.7° axial tilt, $7\times$ baseline eccentricity, and a small negative $\Delta i_{\rm VN}$ (the invariable-plane offset of §4.7): intermediate tilt with distributed, rather than single-event, orbital perturbation.

Uranus (outer-scattered swarm impactor).—98° axial tilt, a −59% gas deficit (a cancelled gas era; §4.6), small inclination perturbation, moderate eccentricity excess. The small $\Delta i_{\rm VN}$ despite massive impactor energy indicates an *in-plane impact trajectory*, consistent with the impactor having been scattered through the Jupiter-Saturn ecliptic plane to reach Uranus's orbit. The impact transferred angular momentum to spin (tilting the planet) without significantly displacing the orbital plane.

The displacement ledger.—The Δa column closes as one angular-momentum account, and its signs are mass-ordered. Orbital angular momentum scales as $m\sqrt{r}$. Uranus's outward displacement costs $14.5\,(\sqrt{19.19} - \sqrt{17.55}) \approx +2.8$ in units of $M_\oplus\sqrt{\mathrm{AU}}$; the $\sim 3.5\,M_\oplus$ of outward-scattered swarm debris that Uranus relayed inward and down surrendered $\sim 3.5\,(\sqrt{18} - \sqrt{5.2}) \approx 6.9$ of the same currency, paying that bill at 40% efficiency. Jupiter, which ejected what Uranus handed it, paid $318\,(\sqrt{5.20} - \sqrt{5.98}) \approx -52$ and sank inside its slot; Saturn, the junior ejector, paid -11 and sank less; Neptune, seated on the dam trap, did not move at all. Mercury's small inward offset is the merger barycentre, weighted toward the heavier slot 8 partner, and Mars's is the remnant's re-circularised orbit below Jupiter's chaotic zone. The account has two terms with distinguishable signatures. The hand-down exchange is smooth and cumulative: it moves a while leaving eccentricity nearly untouched. The impacts are impulsive, and debris reaching a giant's radius arrives near aphelion moving slower than the planet, so every hit is a retrograde brake, an inward Δv, and an impulse is precisely what pumps Δe and, delivered off-centre, the tilt. Saturn's two terms therefore share a sign, ejection payment plus impact braking, both inward, with the impacts countersigning through Δe and the 26.7° tilt. Uranus's two terms oppose, and the exchange won because Uranus relayed several times the debris mass it absorbed; the impulse receipts, $\Delta e = +0.038$ and the 98° tilt, are the minority term's signature on the same event. Bodies too small to eject relay debris and rise; bodies massive enough to eject pay and sink; the dam-seated body holds. This mass-ordered division is the planetesimal-exchange hierarchy of Fernández & Ip (1984), operating on the framework's own dispersed swarm rather than a conjectured residual disc; the giants' positional residuals are thereby accounted quantities, and every planet formed on its slot.

4.6. *Impact forensics: the truncation clock and the polarity law*

The dispersal of slots 4–5 leaves two primaries unaccounted for: the slot-4 planet ($2.57\,M_\oplus$ at the snow-line pile-up) and the slot-5 planet ($1.23\,M_\oplus$). This subsection identifies where each one went, and when, using nothing but masses and orbits – no axial tilts. The restriction is deliberate: tilt is unobservable for exoplanets, so a forensic method that needs it cannot travel. One that does not becomes a prediction class for every system in the catalogue.

Truncation, not blow-off.—A giant struck mid-runaway does not have its envelope blown off; the energy budget forbids it. Even granting the impactor a 10× extended-radius cross-section, its kinetic energy covers less than a quarter of the binding energy of Saturn's missing gas. What an impact can do is *truncate*: disrupt the accretion flow and end the gas era early. The deficit is gas that never arrived, not gas removed. Truncation turns the deficit into a clock. Runaway capture delivers gas as $1 - e^{-k(t-t_{\rm form})}$ with $k = 0.691$ Myr^{-1} (the catalogue's Tanigawa–Ikoma calibration), so a body holding a fraction κ of its predicted gas was interrupted at

$$t^* = t_{\rm form} - \frac{\ln(1-\kappa)}{k}. \tag{33}$$

Saturn kept 61% of its predicted gas with $t_{\rm form} = 3.25$ Myr, giving $t^* = 4.61$ Myr – squarely in the window after Jupiter completes ($t_{\rm form} = 1.90$ Myr) and destabilises slots 4–5.

The cross-prediction.—The same epoch must govern every victim, and Uranus's formation clock reads $t_{\rm form} = 5.57$ Myr: it postdates t^*. Uranus therefore *never gassed at all*, and its observed mass should equal its bare allocated core: predicted $15.0\,M_\oplus$, observed $14.54\,M_\oplus$, an error of -2.9% with no parameter adjusted. The clock dated on Saturn's deficit predicts Uranus's entire mass. This is also, in passing, an answer to a question the framework did not set out to ask: Uranus is an ice giant rather than a gas giant because its gas era was cancelled before it began.

The polarity law.—Which primary hit which giant? The retention line (Eq. 27) supplies the discriminant. A high-retention impact is *accretive*: the impactor arrives near aphelion moving slower than its target, so absorbing it is a retrograde brake and the target steps inward ($\Delta a < 0$). A low-retention impact is *erosive*: most of the impactor mass leaves again as backward-directed ejecta, and the recoil boosts the target outward ($\Delta a > 0$). Retention sets the polarity, and the assignment must reproduce each struck body's observed displacement – sign first, then magnitude. Testing both routings: slot 4 $\to$ Saturn with slot 5 $\to$ Uranus reproduces both signs (Saturn: $\Delta v/v_{\rm esc} = 0.28$, retention 0.80, accretive brake against an observed $\Delta a = -0.71$ AU; Uranus: $\Delta v/v_{\rm esc} = 0.66$, retention 0.57, erosive boost against an observed $\Delta a = +1.64$ AU) at a combined residual of 1.81 AU; the swapped routing scores 2.23 AU and mispolarises. The verdict – slot-4 primary struck Saturn, slot-5 primary struck Uranus, both at $t^* = 4.61$ Myr – uses no tilt. Note the inversion the polarity law explains: Uranus received the most powerful impact relative to its mass and carries the largest single-planet displacement in the system, yet that displacement is *outward*. An accretive story cannot produce it; the erosive recoil does.

The inward ledger closes.—Each dispersed zone also scatters 5–10% of its allocation inward. The slot-5 band, 0.067–$0.134\,M_\oplus$, closes on a single settled survivor of $0.107\,M_\oplus$: Mars. The survivor's seat is itself predicted, because the scattering authority moved – Jupiter formed at slot 3 (5.98 AU) and settled at 5.20 AU, so its chaotic-zone edge swept a band, and the survivor settles in the strip between the edge's initial and final positions, 1.46–1.68 AU, bracketing Mars's observed 1.524 AU. The slot-4 band, 0.129–$0.257\,M_\oplus$, pays Earth's $+3\%$ excess as a retained $0.029\,M_\oplus$

from a $\sim 0.09\,M_\oplus$ impactor arriving at 12.2 km/s with retention 0.31 – the Moon-forming Theia delivery, here an entry in a mass ledger rather than a hypothesis – and leaves 0.04–0.16 $M_\oplus$ unaccounted, for which the predicted sinks are the corridor debris field (the asteroid belt; §8.2), a hit-and-run on a mass-pinned body (an orientation receipt with no mass signature – Venus's retrograde spin is the candidate), a star-grazer, or ejecta lost past the dam.

The same ledger prices the water. Slot-4 material is 34% ice (it formed beyond the 2.75 AU snow line), so every inward delivery carries a water line in ocean units ($2.3 \times 10^{-4}\,M_\oplus$ each): Earth's receipt is 21–43 oceans *delivered*, against one surface ocean. The gap cannot have escaped: hydrodynamic escape fractionates hydrogen over deuterium, and Earth's ocean D/H sits at the carbonaceous-chondrite value – the unfractionated signature of water that arrived and stayed (Venus's 150× enrichment is what losing an ocean looks like). What arrived and did not escape is still here, so the ledger *requires* a deep interior reservoir of tens of ocean equivalents – a delivery-side derivation of the conclusion the retention side has been converging on independently: hydrous ringwoodite in superdeep diamonds puts $\sim$ 1–3 oceans in the transition zone alone (Pearson et al. 2014), and high-pressure partitioning experiments put tens of ocean-equivalents of hydrogen in the core (Tagawa et al. 2021). The majority of Earth's water was never on the surface. The unaccounted residual crossed every interior orbit and bounds the others: Mars at up to a few tens of oceans delivered against a geological record of $\lesssim 1$ retained (a body too small to hold what arrived), and Venus at up to ~ 8 – consistent with the $\sim 150\times$ deuterium enrichment that records an ocean Venus had and lost. Earth's oceans, Mars's valley networks, and Venus's D/H anomaly are one delivery event read through three different retention histories.

Every number above comes from the fitted slots and the observed masses and orbits; nothing is Sol-specific. The same forensics runs unmodified on every catalogue system, so for any fitted exoplanet system the framework now states which seats were dispersed, who absorbed them, at what epoch, and with what displacement – falsifiable wherever a mass or a period is eventually measured to higher precision.

4.7. *The original disc plane and the invariable-plane offset*

The Venus-Neptune plane's 1.99° offset from both Venus and Neptune reflects the natural inclination dispersion of bodies forming in a protoplanetary disc of finite vertical scale height ($H/r \approx 0.03$–0.05 at typical disc temperatures and stellar masses; Armitage 2018). Bodies forming within such a disc inherit small random inclinations on the order of $\arctan(H/r) \approx 2$–$3°$. Venus and Neptune happen to sit on opposite sides of the disc midplane at characteristic disc-thickness offsets.

The *current* invariable plane of the Solar System, defined by the total angular momentum vector (dominated by Jupiter and Saturn), sits at $i \approx 1.58°$ relative to the ecliptic (Souami & Souchay 2012), or equivalently $\approx 1.6°$ from the VN primordial plane. This offset is the **cumulative impactor-perturbation signature**: Jupiter's and Saturn's orbital planes shifted by $\sim 0.3°$ under the swarm-impactor bombardment, dragging the system's angular-momentum-weighted invariable plane below the primordial disc plane.

4.8. *Refined ISU criterion*

The orbital analysis suggests a refined criterion for assigning the ISU flag in the framework:

ISU-eligible: when a planet has *both*:

- Precise, well-characterized observed mass and orbit
- Orbital state (i, e) consistent with undisturbed formation (low Δi, low Δe relative to a system-mean baseline)

Not ISU-eligible: when orbital signatures indicate post-formation perturbation, even if mass is precisely measured. Such planets show distinct Δi or Δe excursions whose origin is identifiable as an impact, scattering, or migration event.

For Sol, only Venus and Neptune satisfy both conditions, making them the natural ISU anchors: they are the only bodies whose mass and position can be used as absolute constraints on the disc parameters without contamination from post-formation events.

Operational diagnostic: minimum-eccentricity selection.—In practice, the planets with the lowest observed eccentricities are the strong ISU candidates, with the number selected set by how cleanly an eccentricity gap separates near-circular bodies from the rest of the system. For Sol, the factor-of-two gap below Earth's $e = 0.0167$ selects exactly Venus and Neptune. Because eccentricity integrates post-formation dynamical history one way (Line 2 above), preserved near-circularity directly identifies undisturbed bodies. Systems with no clean gap may have no undisturbed bodies; the disc parameters are then inferred jointly from all observed planets rather than pinned to ISU anchors, Beta Pictoris's directly-imaged super-Jupiters, whose orbital states are poorly constrained, are handled this way.

5. SATURN AND JUPITER: IN-SITU OUTER ARCHITECTURE

The geometric cascade $\rho = 1 - \sqrt{\ln 2}/2 \approx 0.584$ places slot 2 at $r_2 = 10.24$ AU and slot 3 at $r_3 = 5.98$ AU for Sol. Saturn (observed 9.54 AU) and Jupiter (observed 5.20 AU) sit ~ 0.7 AU inside their cascade positions, a minor inward drift, not a 4-AU or 8-AU migration. **The Solar System's outer architecture is essentially in-situ.** The framework reproduces Saturn's and Jupiter's positions and masses with the standard cascade allocation plus the slot-4 impact that truncated Saturn's gas accretion (§4.6); no Grand Tack or Nice Model is invoked.

5.1. *Jupiter as the in-situ anchor*

Jupiter at slot 3 ($r_3 = 5.98$ AU) with cascade-default $t_{\rm form} = 1.90$ Myr predicts $M_{\rm pred} = 317.83\, M_\oplus$, matching the observed mass. The Tanigawa-Ikoma $A_0 = 4.39$ $(1/M_\oplus)$ and $k = 0.691$ Myr^{-1} calibration is anchored on Jupiter's mass-position match, and the same constants reproduce Neptune at slot 0. Jupiter's ~ 0.78 AU inward drift from slot 3 to its observed position is consistent with gas-disc torque damping during the gas-active phase, and is a factor of five smaller than the Grand Tack's postulated ~ 4 AU inner excursion.

Jupiter's stable position is consistent with the isotopic constraint of Kruijer et al. (2017): Jupiter formed early and remained stably between the NC and CC reservoirs without major inner excursions. The constraint is in fact stronger than a formation clock: the non-carbonaceous and carbonaceous meteorite reservoirs remained isotopically *separated* from ~ 1 to ~ 3–4 Myr, which requires a sustained transport barrier between inner and outer disc for the entire interval: a barrier a stationary slot-3 Jupiter atop the framework's snow-line pile-up supplies naturally, and that any large radial excursion would have breached. The same cosmochemistry thereby witnesses both of this section's claims at once: the giant stayed put, and the disc was partitioned by standing dams.

5.2. *Saturn in-situ at slot 2*

Saturn at slot 2 ($r_2 = 10.25$ AU) with default $t_{\rm form} = 3.25$ Myr predicts $M_{\rm pred} = 145.9\, M_\oplus$. Saturn's observed mass is $95.16\, M_\oplus$, a $-50.7\, M_\oplus$ (-35%) deficit relative to the cascade prediction. The deficit is accretion *truncation*: the slot-4 primary's impact at $t^* = 4.61$ Myr ended Saturn's runaway with 61% of the predicted gas delivered (§4.6). The missing mass is gas that never arrived – the energy budget rules out stripping it after the fact – and the same truncation epoch, applied to Uranus's later formation clock, predicts Uranus's observed mass as a bare core to -2.9%.

Saturn's observed position at 9.54 AU sits 0.70 AU inside slot 2, consistent with combined disc-gas torque damping and cumulative gravitational interactions during impactor transits.

5.3. *Saturn's axial tilt as an impact signature*

Saturn's 26.7° axial tilt is intermediate between Jupiter's 3.1° (anchor, no impacts) and Uranus's 97.8° (extreme direct impact). The progressive tilt magnitudes (Jupiter $\rightarrow$ Saturn $\rightarrow$ Uranus) reflect cumulative angular momentum delivery from the slot 4–5 impactor population: Jupiter is the scattering source and receives no swarm impacts; Saturn absorbs the slot-4 primary plus swarm debris, tilting its axis while the impact truncates its gas era at 61% delivered (§4.6); Uranus takes the highest-velocity direct hit, producing its extreme axial tilt while the same epoch cancels its gas accretion outright. The progression is a natural prediction of the framework's swarm-dispersal mechanism, with impact severity increasing with heliocentric distance from the Jupiter scattering source as the swarm acquires successively higher orbital eccentricities through repeated Jupiter encounters.

The standard explanation for Saturn's obliquity is instead a secular spin-orbit resonance with Neptune, entered as Kuiper-belt mass depletion slowed the precession of Neptune's orbit plane (Ward & Hamilton 2004). The two accounts are not exclusive here: the framework's trans-Neptunian evolution (§8) supplies the required belt-mass depletion without planetary migration, leaving the resonance pathway open. Saturn's gas-mass deficit, however, requires an impact in either case.

5.4. *Contrast with conventional outer-system narratives*

***Grand Tack* (Walsh et al. 2011):** requires Jupiter to migrate inward to ~ 1.5 AU and back, dragging Saturn behind it. Our model has Jupiter at 5.20 AU, essentially in-situ at slot 3, with only ~ 0.8 AU inward drift.

***Nice Model* (Tsiganis et al. 2005):** requires the outer giants to form in a compact configuration ($\sim$ 15–30 AU) and reorganize via specific mean-motion-resonance crossings. Our model has all four outer giants within ~ 1.6 AU of their geometric cascade slot positions; no post-formation reorganization is invoked. We note a convergence rather than only a conflict: the compact, near-resonant initial chain the Nice family must *assume* (Morbidelli et al. 2007) is precisely the configuration the cascade *produces* (adjacent slots at period ratio 2.24, just wide of 2:1), the frameworks disagree on what happened next, not on how the giants started. Removing the instability does leave the Nice Model's auxiliary successes (the putative Late Heavy Bombardment spike; Trojan capture) in need of separate accounts; irregular-satellite capture, at least, ties naturally to the framework's own scattering encounters (§8.9; Nesvorný et al. 2007).

The near-5:2 Jupiter-Saturn period ratio, $P_S/P_J = (9.54/5.20)^{3/2} \approx 2.48$, emerges as a natural consequence of the cascade's geometric structure: adjacent slots with ratio $\rho \approx 0.584$ have a period ratio $\rho^{-3/2} \approx 2.24$, within 10% of the observed 2.48. Both planets occupy positions close to their geometric slots without requiring ad-hoc resonance trapping or migration sequences.

5.5. *The framework's parsimony claim*

Sol's outer-giant architecture (Jupiter, Saturn, Uranus, Neptune all within ~ 1.6 AU of cascade slots), the Jupiter-Saturn 5:2 mean-motion resonance, Saturn's -35% truncated gas mass, Saturn's modest axial tilt, Uranus's extreme axial tilt and cancelled gas era, the asteroid belt's near-empty mass, and Earth's Moon are all reproduced from: (i) the geometric cascade allocation; (ii) the Mercury-Vulcan adjacent-slot merger event (§3.1); (iii) Jupiter's standard gravitational scattering of the slot 4-5 embryo swarm (§5.6).

The Solar System's observed architecture is the outcome of cascade formation plus one dynamical event in the inner-mid region; migration sequences, resonance crossings, and reorganization events are not required once the cascade structure is identified.

5.6. *Embryo-swarm dispersal as a general mechanic*

A predictive criterion follows: for any cascade slot N with mass M_N, if there exists a perturber slot P with $M_P \gg M_N$ within $\sim$ 5–7 mutual Hill radii of N's position, then N's pre-consolidation embryo swarm is dispersed by Jupiter-style asymmetric scattering. The dispersal produces:

- **Inner fraction** ($\sim 5\%$): scattered to lower AU, manifests as either an embryo impact on inner planets or a small surviving body at $r < r_N$.
- **Outer fraction** ($\sim 95\%$): scattered outward via multi-pass perturber acceleration, manifests as cumulative impacts on outer-slot bodies.

The scattering mechanism itself is established: N-body simulations show that a giant's runaway gas accretion destabilizes and scatters nearby planetesimals in all directions, with a fraction deposited inward (Raymond & Izidoro 2017). The framework differs in elevating this from one ingredient among migration scenarios to the Solar System's *single* dynamical event, and in tying the outward-scattered majority to the Saturn and Uranus impact record.

For Sol, Jupiter at slot 3 has Hill radius $R_{H,J} = 0.34$ AU (Eq. 24). First-order mean-motion-resonance overlap (Wisdom 1980) makes orbits formally chaotic within $\Delta a/a \approx 1.3\,(M_J/M_\odot)^{2/7} \approx 0.18$ of Jupiter (~ 0.9 AU on each side), and repeated distant encounters destabilize loosely-bound pre-consolidation embryo swarms out to the framework's $\sim 10\,R_H$ dispersal criterion (~ 3.4 AU). Slots 4 and 5 of the cascade ($r_4 = 3.49$ AU, $r_5 = 2.04$ AU) both lie within this reach. The framework predicts:

Table 7. Predicted dispersal of cascade slots under Jupiter's gravitational influence for Sol.

Slot	Predicted ($M_\oplus$)	Status	Outcome
4	2.5	Fully dispersed	Theia swarm
5	1.3	Partially dispersed + survivor	Mars (0.107) + outer scatter
6	0.9	In-situ + Theia chunk	Earth + Moon
7–8	rocky	In-situ	Venus, Mercury

This dispersal mechanic is the framework's single dynamical event beyond cascade formation. It is generalizable to other systems: any cascade slot within scattering range of a sufficiently massive perturber is predicted to be either unobservable (fully dispersed) or displaced to a position *below and outside* the perturber's chaotic zone with surviving mass at $\sim$ 5–10% of the slot's primordial allocation. The settling radius is the inner stability edge of the chaotic zone, $r_{\rm survivor} \approx r_{\rm perturber} - 11\,R_H$. For Sol this predicts a survivor at $5.20 - 11 \times 0.34 \approx 1.46$ AU carrying 0.07–0.13 $M_\oplus$; Mars is observed at 1.524 AU with $0.107\,M_\oplus$ (8.1% of slot 5's predicted $1.323\,M_\oplus$): a position match within 5%. The position-plus-mass-fraction signature is the characteristic outcome of partial dispersal.

5.7. *Migration is gravitational, not gas-torque*

Conventional gas-disc migration mechanisms (Type-I, gas torque on sub-thermal-mass cores; Type-II, viscous transport of gap-opening giants) are not invoked in this framework. Planets in the calibrated catalogue certainly migrate, and some migrate far: Mu Arae c steps off its dam outward, the wrecking pairs of HD 60532 and HD 142 fall from their birth seats to the magnetospheric rim, and 55 Cancri b transits its entire inner system. But the driver in every case is gravitational: angular-momentum exchange with more massive bodies. A body that surrenders angular momentum to a heavier neighbour drops to a lower orbit; one that extracts angular momentum rises to a higher one. Luna's tidal recession is the same bookkeeping with Earth's spin as the reservoir.

The same interaction obeys a mass-ratio law that splits its outcomes. When the participants are comparable, or the migrator is itself large, the body transports intact and arrives whole at its destination. When the mass ratio is extreme, the smaller body does not migrate; it is taken apart, and the catalogue documents the spectrum: swarm dispersal with the asymmetric inner/outer split (§5.6), Martian-type remnants at 5–10% of the parent, mantle-grazed survivors, and Roche deliveries ground into rings: large bodies migrate intact, small bodies at extreme ratios are disassembled.

The gas disc still matters, but its role is passive and architectural: it traps rather than torques. The slot lattice is the zonal-flow ladder of §2, pressure bands at which migration torque cancels, so a gravitationally driven migrator in a living disc is caught at the next vacant seat, and the magnetospheric cavity rim is the terminal trap (Lin et al. 1996); the wrecking pairs park bracketing their fitted Alfvén radii, and HD 142 c sits on the exact seat its companion vacated. Parking position therefore dates the move: a migrator seated on the lattice moved while the gas lived; one parked off-lattice moved after the gas was gone, as Jupiter's small post-swarm recoil and Luna's tidal march both illustrate.

This is a substantive claim that extends beyond the framework's calibration: the observational evidence for Type-I and Type-II migration as continuous gas-torque processes may be substantially weaker than commonly attributed. What has been called migration is better described as gravitational scattering settling into trap-stabilised orbits. The distinguishing tests are the asymmetric inner/outer mass fraction of scattering (the ∼5%/95% split) against continuous migration's smoothly varying distributions, and the parking statistics: gas-torque migration predicts no preference for fossil seat positions among migrated planets, while trap-terminated gravitational migration predicts that wet-era migrators sit on them.

5.8. *Brown-dwarf vs gas-giant mutual eviction in other systems*

Eviction is the migration grammar of §5.7 carried past its boundaries. The same gravitational angular-momentum exchange that lifts a body to a higher orbit, given enough exchange, lifts it past escape velocity and out of the system entirely; the same exchange that drops a body to a lower orbit, continued, drops it through the rim and into the star. The catalogue holds both endpoints: Proxima left Alpha Centauri with an energy surplus of barely ten percent, the marginal escape of a body nudged just past the threshold, while the protostellar-consumption channel records the opposite exit, and Sol's own eviction from Hyperion sits in between as the gentlest recorded departure. Migration, ejection, and consumption are one process read at three depths of the exchange; shredding (§5.7) is what the same process does to a body too small to survive it. Which depth each participant reaches depends on the encounter. The pair must balance its own books only when no third party pays: in a full relaxation the destroyed slots and consumed bodies carry part of the angular-momentum bill, so the menu includes one partner up and one down, both partners lifted together, one partner held fast while the other goes (Alpha Centauri's relaxation ejected Proxima from slot 1 while the dam-keeper B never moved, the bill paid by the obliterated interior slots; Upsilon Andromedae's ratcheted the keeper itself off the dam, paid in sunk factory products), mutual ejection when both clear escape, and ejection paired with a plunge into the star.

Sol's outer giants do not undergo system ejection: encounters between $\sim 100\,M_\oplus$-scale bodies do not deliver hyperbolic excess velocity (speed beyond what is needed to escape the system), so mass is redistributed within the bound system rather than ejected. Genuine mutual eviction requires bodies above roughly the deuterium-burning threshold ($13\,M_J \approx 4000\,M_\oplus$, the mass at which an object briefly fuses deuterium and is classed a brown dwarf; Spiegel et al. 2011), where mutual encounter velocities can exceed the system escape velocity. The exit assignments then follow the encounter, not a rule: one ejected and one consumed, both ejected, or both displaced outward with the deficit charged to the rest of the cascade. The framework flags empty adjacent slots whose predicted masses both exceed this threshold as mutually evicted. The distinction matters for Sol: the literature's "ejected fifth giant" hypotheses (Nesvorný 2011; Nesvorný & Morbidelli 2012) demonstrate that Jupiter encounters *can* eject an ice-giant-mass body, their five-planet simulations succeed at it, but the framework's point is parsimony, not impossibility: no fifth body is needed here, because the outer architecture is reproduced by Jupiter's scattering of the slot 4–5 swarm alone, and the cascade allocates no slot for one.

Jupiter's recoil from swarm scattering.—Jupiter's ~ 0.8 AU inward drift from slot 3 ($r_3 = 5.98$ AU) to its observed position (5.20 AU) is consistent with momentum-conservation recoil from scattering $\sim 3.9\,M_\oplus$ of the slot 4–5 swarm material outward. For typical scattering velocity boost $\Delta v_{\rm swarm} \approx 30$ km/s, momentum conservation gives $\Delta v_J = -(M_{\rm swarm}/M_J)\,\Delta v_{\rm swarm} \approx -0.37$ km/s, producing an inward drift $\Delta r_J \approx$ 0.4 AU. This accounts for roughly half of Jupiter's observed inward drift from slot 3; the remainder is consistent with disc-gas dynamical friction during the gas-active phase. No large-scale Type-II migration is required.

6. CASCADE FRAMEWORK AT STELLAR SCALES

The cascade framework's geometric ratio $\rho = 0.5837$ and mass-allocation formula are dimensionless. Their natural extension to multi-star systems and stellar nurseries follows once the host mass and primordial $f_{\rm disc}$ are scaled appropriately. This section presents three demonstrations: (i) Alpha Centauri as binary/triple formation in a single protoplanetary disc; (ii) the Crab Nebula's supernova progenitor as a scaled-up version of the same physics producing a multi-star cluster; (iii) the predicted compositional bimodality between cascade bodies forming inside vs outside their parent's snow line.

6.1. *Alpha Centauri as in-disc multi-star formation*

Alpha Centauri is fit the way every system is now fit: each body enters at its *observed* position, and the configuration below is the unaided winner of the scan under the framework's fitting laws, in particular the overlord rule (no hypothesis may conjure an unobserved body exceeding every observed member) and the requirement that every observed body justify its existence at both its formation seat and its present position. With $M_\star = 1.08\,M_\odot$ and the fitted spin 1.90 ($D = 2.62$), the cascade sets the dam at $R_{\rm disc} = 23.52$ AU and produces a nine-seat system:

- Slot 0, the dam seat (23.52 AU): **Alpha Cen B** ($0.91\,M_\odot$ current = 302,770 $M_\oplus$), *in situ*: its observed orbit is its formation seat, and the seat's allocation, dam pile-up included, sources its mass at 0.0006% (predicted 302,772). The dam-keeper never moved.
- Slot 1 (13.73 AU): **Proxima Centauri** ($0.122\,M_\odot$ current = 40,625 $M_\oplus$; predicted 40,625.3, 0.0008%), the interior sibling, evicted in the primordial relaxation to its marginally bound $\sim 10^4$-AU orbit (Kervella et al. 2017).
- Slots 2–3 (8.0 and 4.7 AU): brown dwarfs ($\sim$ 15,900 and $\sim$ 5,600 $M_\oplus$); sunk and dispersed in the relaxation.
- Slots 4–7 (2.7–0.5 AU): gas giants of decreasing mass ($\sim$ 2,000 down to $\sim$ 60 $M_\oplus$); all purged.
- Slot 8 (0.32 AU): rocky; purged.

Both stellar masses close at four digits with B seated and Proxima's seat identified from its mass alone, and the purge of the entire interior cascade is itself testable: A's planetlessness through decades of searches is consistent with a fully scoured ladder.

B and Proxima formed 9.8 AU apart with overlapping Hill spheres (packing index 0.8; §8.9): the relaxation was compulsory, not contingent.

The exchange differs from Upsilon Andromedae's (§5.8) in exactly one respect: there the dam-keeper itself was ratcheted off its seat; here the keeper held, and the interior sibling alone took the ejection. Lifting Proxima from its 13.7 AU seat to its present orbit requires a specific energy of $\sim 0.036\, GM_A$ AU^{-1}; the exit pass at the fitted periapsis $q = 19.7$ AU (§8.10) supplies $\sim GM_B/q \approx 0.040\, GM_A$ AU^{-1}, almost exactly the requirement. The geometry corroborates: $q = 19.7$ AU lies between Proxima's seat (13.7) and B's dam (23.5), precisely the corridor an eviction from slot 1 past the dam-keeper must climb through. The sunk brown-dwarf and giant seats ($\sim 24{,}000\, M_\oplus$) released the binding energy that scoured the interior and ejected the lightest member. Proxima did not quite unbind, stalling on the marginally bound, moderately eccentric ($e \approx 0.5$), prograde, plane-aligned orbit that is observed: the signature of an internal scattering, not an external capture.

The same exit edited Proxima's own planetary system, and the edits remain legible in its fit residuals – under a disc geometry that is no longer fitted at all. Because Proxima formed at A's slot 1, the hierarchical inheritance law applies at $x = \rho$ exactly: $D = 0.39\, \rho^{-2.12} D_A = 3.20$, spin $\Omega = 2.17$ – Proxima's disc is a *prediction of A's fit*, with no per-system disc parameter bisected (only $f_{\rm disc}$, measured from the planets, and the encounter's own periapsis remain free). A stellar flyby truncates a protoplanetary disc at $r_t \approx 0.28\, q\, (M_{\rm pert}/M_*)^{-0.32}$ (Breslau et al. 2014); the re-bisected encounter ($q = 19.7$ AU, §8.10) gives $r_t = 2.9$ AU, beheading the disc's outer rungs (the would-be slot-0 gas giant at 2.5 AU, in the stirred zone, lost 43% of its inventory and is unobserved) while every surviving planet closes essentially exactly under the *inherited* geometry: c at -0.01% via a merger absorbing slot 2 (the same mechanism as 55 Cancri's c), b at -0.00% and d at $+0.6\%$ with late delivery from the inward debris. The encounter energetics forbid $q \gtrsim 45$ AU and the truncation requires the pass to reach the planet region; the re-bisected $q = 19.7$ AU sits inside the window, in the same eviction corridor the energy ledger requires. This is also why Proxima's system resists unperturbed-cascade calibration: no anchor family closes its per-planet masses, because the allocation curve was overwritten by the encounter. The framework's one resistant system is the one it independently identifies as an ejected, disc-truncated star – and the third star of the triple now closes at sub-percent with its disc inherited from the first, the strongest single test of the inheritance law to date. The closure also over-resolves the catalogue: the registered masses are $b = 1.2700$, $c = 7.000$, $d = 0.2615\, M_\oplus$, with d's third digit (+0.6% over the catalogued 0.26) the live discriminator at the next radial-velocity refinement.

Slot-determined metallicity, tested on the triple.—Because B and Proxima are slot children of A's natal disc, their compositions are A's cloud plus their seats' solid enrichment. Under the resolved seating B holds the dam seat, where the pile-up concentrates solids, so the qualitative ordering A < Proxima < B survives the seating revision, and its sharpest entry remains confirmed: differential abundance analyses consistently measure B more metal-rich than A by +0.02 to +0.05 dex (Morel 2018), a difference co-natal formation does not explain and the seat structure requires. Proxima's measured metallicity carries M-dwarf systematics of ± 0.1 dex and does not yet discriminate. The snow line supplies a second-order tag, and it must be stated at the level of the solid seed: both stars are gas-dominated and ingested A's full volatile inventory as vapor, so neither is bulk-dry. What differs is the flavour of the trace enrichment. B's seat lay outside A's ice line (23.5 against 15.7 AU), so its dam pile-up delivered ice-bearing solids; Proxima's slot-1 seed (13.7 AU, inside the line) was refractory rock. The prediction is therefore an abundance-*ratio* signature riding on the metallicity offsets: B's metal excess tilted toward volatile elements (elevated [O/Fe], [C/Fe]), Proxima's toward refractories.

6.2. *The predicted system of Alpha Centauri B*

B was sourced on A's dam seat, so the hierarchical inheritance law (§7, measured at Jupiter to 0.9%) hands its natal disc the collapse record at $x = 1$: $D_B = 0.39 \times 2.62 = 1.02$, spin 1.02. The notional disc radius, 23.8 AU, exceeds B's orbit around A, so the standing binary tide sets the real boundary: the recurring periapsis of the AB orbit ($q_{AB} = 11.3$ AU, every 80 years, permanently) truncates B's disc at $r_t = 2.87$ AU, and the dam forms there. Proxima's departure pass sheared at 5.3 AU, entirely outside the tide dam: its truncation of B's system was feather-light, mild stirring of the outer seats only. The cascade then predicts, with no remaining freedom beyond the scale of $f_{\rm disc}$, six seats: 2.76, 1.61, 0.94, 0.55, 0.32, and 0.19 AU, with masses 0.05–2.4 $M_\oplus$ at $f_{\rm disc} = 0.01$; the outer two sit in the tide-stirred zone (the engine's stirring transform erodes them by 56% and 7% respectively), the inner four are quiet. The snow line at 1.7 AU makes the inner four dry rock and the outer two icy: a miniature Sol.

Three features make this forecast unusually consequential. First, every seat sits below the detection limits of the decades of radial-velocity monitoring this star has received ($K \sim 10$–60 cm s^{-1}): the famous null results are consistent with the prediction, while the $f_{\rm disc} = 0.03$ branch, which would place a $\sim 67\, M_\oplus$ body at 2.76 AU at $K \sim 5$ m s^{-1}, is already excluded. The nulls measure $f_{\rm disc} \lesssim 0.02$. Second, B's habitable zone maps the seats onto Sol compressed. We bound the zone empirically and asymmetrically: the inner edge at the recent-Venus limit (Kopparapu et al. 2013) – a delivered Venus is hot but habitable, and the framework's discriminator is the delivery receipt, not the thermostat – and the outer edge at the first-CO_2-condensation limit of 1.37 AU-equivalent (Kasting et al. 1993), inside Mars's orbit. Mars's Noachian wetness does not widen this bound, because it is an epoch fact. The framework's luminosity history has three phases: dim during the embedded disc era (as the formation-era snow line at 2.75 AU requires), bright through the Hayashi descent once the star is assembled, and – the framework's own extension – elevated for as long as the T Tauri mass shed remains incomplete, since a 1.14-unit Sun runs at

$\sim 1.7\times$ the present luminosity ($L \propto M^4$) and the shed follows the firehose-then-trickle profile rather than a step. Noachian Mars sat inside that early habitable zone (outer edge ~ 1.8 AU-equivalent at full primordial mass); under present luminosity its orbit lies beyond the CO_2-condensation edge. The conjunction dates Mars's drying: the zone's outer edge crosses Mars's orbit when L falls below $1.24\,L_\odot$ – between 12% and 61% of the mass shed complete, depending on whether the standard main-sequence age factor underlies the M^4 mass term – so the Noachian–Hesperian transition is a measurement of the trickle curve, datable independently from the spin-down record, since wind braking and wind mass loss are one outflow. Either reading keeps the Sun above its present luminosity throughout the first gigayear, which carries an Earth-side receipt: warm-to-hot Archean surface oceans, as the long-disputed chert isotope palaeothermometry reads (50–80°C; Knauth & Lowe 2003) – data contested largely because they conflict with a faint young Sun the framework does not have. The zircon record pins the curve's bright early phases as well. A magma ocean under a steam atmosphere cannot cool while absorbed insolation exceeds the atmosphere's radiation ceiling (Hamano et al. 2013); isolated-body solidification takes < 1 Myr, yet Martian zircons date crust formation at ~ 20 Myr (Bouvier et al. 2018) and Earth's at 160–200 Myr (Wilde et al. 2001), its ocean persisting through the Theia impact. The solidification dates therefore measure when the declining $L(t)$ released each planet below its stall threshold, and the observed ordering – the *farther* planet solidifying *first*, with Venus (stalled longest) retaining no ancient crust at all – is the signature of declining external irradiation, which no internal heat source produces. Three independent records (Martian and terrestrial zircons, Martian climate stratigraphy) thereby pin $L(t)$ at ~ 20, ~ 200, and ~ 900 Myr along the Hayashi-plus-shed profile, cross-checkable against the spin-down record throughout. The early-Mars limit is an epoch statement, not a present-day bound. At $L = 0.5\,L_\odot$ the zone is 0.51–0.97 AU. The 0.55 AU seat is the Venus analogue just inside the hot edge, the 1.61 AU seat the Mars analogue well beyond the cold edge, and the 0.94 AU seat, $0.77\,M_\oplus$ of quiet dry rock at an insolation equivalent to 1.33 AU at Sol, is an Earth analogue on the zone's cool side, with $K \approx 7.5$ cm s^{-1} and $P \approx 349$ d: just below current instrument floors and inside next-generation design envelopes. Third, the volatile audit. A's tide is a standing conveyor, applied every periapsis for six gigayears, and its permanence settles stability first: the tide-dam ($r_t = 2.87$ AU) is cross-checked by the independent N-body stability boundary for S-type planets (Holman & Wiegert 1999), $a_{\rm crit} = 2.41$ AU at ($\mu = 0.57, e = 0.52, a = 23.5$) – tidal truncation and resonance-overlap integrations bracketing the same wall. Seats interior to 2.4 AU are gigayear-stable; seats beyond 2.9 AU never assembled. The icy 2.76 AU seat lies between the two radii, Breslau-stable per pass but Holman–Wiegert-unstable secularly, and is therefore feedstock: $\sim 90\%$ processed over the system age ($\sim 2.1\,M_\oplus$ of icy material), of which the Sol-calibrated 5–10% inward share (0.11–$0.21\,M_\oplus$) crosses the inner system. Retention routes the water: arrival Δv rises inward and the retention line (Eq. 27) floors at 5%, so the inner seats stay dry while the outermost safe seat – the habitable-zone seat – retains at 0.24: 0.009–0.037 $M_\oplus$, i.e. 40–160 Earth oceans ($2.3 \times 10^{-4}\,M_\oplus$ each). Unsequestered, that is a 120–500 km ocean with an ice-VI/VII floor; but Earth's own ledger (21–43 oceans delivered, one kept at the surface; §4.6) shows interior sequestration dominates, and applying Earth's surface fraction (2–5%) gives 1.5–5 surface oceans: a 5–17 km rock-floored global ocean (floor pressure 0.04–0.15 GPa, an order of magnitude below the ice-VI line) with hydrothermal contact and a photic zone, and no continents. The sealed ice floor requires sequestration to fail tenfold; the bracket spans drowned to soaked, never dry, and percent-level water mass fractions measurably lower the bulk density, so a transit radius discriminates the endmembers. The 0.55 AU Venus-analogue seat carries its own receipt (8–32 oceans at the retention floor): two delivered seats in one habitable zone. The general statement: habitability outcomes (dry, Earth-like, drowned) are set by the delivery channel's duty cycle – Earth's channel was episodic, this seat's is standing – so the framework predicts volatile *delivery histories*, not only habitable-zone occupancy. The live discriminator is the 2.76 AU seat's present state: a warm debris signature at ~ 2.5 AU.

6.3. *Thought experiment: the Crab Nebula's primordial cluster*

This subsection is a thought experiment, not a claim or proof: it depends on assumed (not measured) primordial parameters and illustrates how the framework extrapolates to massive-star scales.

The Crab Pulsar (PSR B0531+21, $\sim$1.4 $M_\odot$ neutron star, formed in SN 1054 AD) had a B-class progenitor of approximately 10 $M_\odot$. Applied to this progenitor with $M_\star = 10\ M_\odot$, spin = 2.58 (typical B-class primordial fast rotation), and $f_{\rm disc} = 0.25$ (binary-fragmentation regime, identical to Alpha Centauri), the framework allocates an 8-slot cascade with $R_{\rm disc} = 164$ AU:

- Slot 0 (164 AU): B-class star ($\sim$3.4 $M_\odot$)
- Slot 1 (96 AU): B-class star ($\sim$3.1 $M_\odot$)
- Slot 2 (56 AU): A-class star ($\sim$1.5 $M_\odot$)
- Slot 3 (33 AU): K-class star ($\sim$0.65 $M_\odot$)
- Slot 4 (19 AU): M-class star ($\sim$0.26 $M_\odot$)
- Slot 5 (11 AU): M-class star ($\sim$0.10 $M_\odot$)
- Slot 6 (6.5 AU): brown dwarf ($\sim$0.04 $M_\odot$)
- Slot 7 (3.8 AU): gas giant ($\sim$15 $M_\oplus$)

This is a septuple star system: one 10 $M_\odot$ central host plus six cascade-allocated stellar companions, with one brown dwarf and one gas giant. The total cascade mass is approximately 9 $M_\odot$, producing a primordial system mass of $\sim$19 $M_\odot$ in stars alone. The architecture is comparable to compact young clusters such as θ^1 Orionis (Trapezium core, 4–5 confirmed massive stars within $\sim$0.02 pc) or AR Cassiopeiae and ν Scorpii (both confirmed septuple systems).

The framework's mutual-eviction detector flags every cascade member as gravitationally unstable. Within the $\sim$30 Myr main-sequence lifetime of the central B-class star, mutual scattering would have dispersed most cascade members. By the time of supernova in 1054 AD, only 1–2 surviving close companions likely remained; SN 1054 then destroyed any remaining envelope and inner companions while leaving the surviving cluster siblings moving outward at $\sim$10–30 km/s peculiar velocities.

6.3.1. *Stellar nurseries as cascade fragmentation events*

The Crab progenitor's primordial 7-star configuration is identical in structure to observed compact young clusters: θ^1 Orionis, the Arches cluster, R136 in 30 Doradus, Westerlund 1. Each has at its dynamical core a high-mass primary plus a small number of massive companions in tight orbits. The cascade framework identifies these as fragmentation patterns of massive primordial discs, not stochastic Jeans-instability events. Disc fragmentation is an established binary-formation channel: gravitational instability in massive discs forms stellar and brown-dwarf companions beyond $\sim$ 50 AU (Kratter & Lodato 2016); the cascade adds the prescription of *where* (the slot positions) and *how much* (the slot allocations).

The key parameters distinguishing single-star vs multi-star formation outcomes are: host mass $M_\star$, primordial $f_{\rm disc}$, and cascade slot mass at slot 0 relative to the stellar formation threshold, the hydrogen-burning minimum mass of $0.08\,M_\odot \approx 25{,}400\,M_\oplus$ (Chabrier & Baraffe 2000). For low-mass primaries ($M_\star < 1.4\,M_\odot$, G/K/M class), slot 0 mass falls below the stellar threshold unless $f_{\rm disc} > 0.3$; the cascade produces planets only. For intermediate-mass primaries (A/B class), slot 0 readily crosses the stellar threshold at $f_{\rm disc} \gtrsim 0.05$; cascade fragmentation yields binary, triple, or higher-multiplicity systems. For massive primaries (O class), the slot 0 stellar threshold is crossed at any reasonable $f_{\rm disc}$; cascade fragmentation inevitably produces multi-star clusters.

This is consistent with observation: the stellar multiplicity fraction increases monotonically with primary mass, from $\sim$25% for M-dwarfs to $\sim$80% for B-class to $\sim$100% for O-class (Duchêne & Kraus 2013). The cascade framework predicts this scaling directly from slot 0 mass allocation versus the deuterium-burning and hydrogen-burning thresholds.

6.4. *Compositional bimodality across the parent snow line*

A direct compositional consequence of the cascade framework at stellar scales: cascade bodies forming inside their parent's snow line have qualitatively different ice budgets from those forming beyond it. The framework's snow line is a function of parent luminosity (and hence parent mass), scaling roughly as $r_{\rm snow} \propto L_\star^{1/2}$. For a

$M_\star = 2.5\,M_\odot$ A-class parent, the snow line falls at ∼40 AU; for a $M_\star = 10\,M_\odot$ B-class parent, the snow line is at ∼650 AU, well outside the entire cascade.

Cascade bodies forming beyond the parent's snow line incorporate condensed ice in their solid allocation; the cascade rock/ice/pebble formula gives explicit ice fractions. Bodies forming inside the snow line acquire no condensed ice in their solid seed. For stellar-mass occupants the distinction must be drawn carefully: the gas, which dominates their mass, carries the disc's volatile elements as vapour on both sides of the line, so the snow line partitions the *solid seed's* composition, not the bulk volatile inventory. The tag is therefore strongest for solid-dominated bodies (planets, dwarf-planet cohorts) and survives in stars only as trace abundance and isotopic ratios riding on the metal seed.

This compositional bimodality affects multiple observables:

- **D/H ratio** (the deuterium-to-hydrogen isotope ratio): ice-condensed water at low temperatures is enhanced in deuterium relative to gas-phase water (Tielens 1983). Stars formed beyond the parent's snow line should have slightly elevated D/H, though the signature is diluted by H/He envelope mass.
- **^{18}O/^{16}O ratio**: cold gas-grain chemistry selectively enriches ice in ^{18}O. Outer-slot stars carry this enrichment in their composition.
- **Condensed-ice seed mass**: outer-slot stars ingest ∼10–500 $M_\oplus$ of cascade-condensed ice with their solid seed; inner-slot stars effectively none. The differential appears in abundance ratios ([O/Fe], [C/Fe] tilts on the metallicity offset), not in bulk volatility.
- **Isotopic heredity in the sub-cascade**: a daughter disc inherits the parent's gas with volatiles intact and re-condenses ices beyond its *own* snow line, so planet water budgets are gated locally, not by the parent seat. What the parent seat does transmit is the ratio set: D/H, ^{18}O/^{16}O, and C/O tilts imprinted on everything the daughter disc builds.

For the Crab progenitor cluster (slots 1–7 all inside the 650 AU snow line), every cascade companion carried a refractory solid seed. By contrast, the Alpha Centauri pair splits across A's disc ice line (15.7 AU): B's dam seat at 23.5 AU accreted ice-enriched solids, while Proxima's slot-1 seed at 13.7 AU, inside the line, was refractory – a solid-seed abundance-ratio tag, not a bulk volatile difference, since both stars took A's volatiles in the gas phase (§6.1).

Observationally, this bimodality should appear in differential abundance surveys as two families of metal-excess flavour: volatile-tilted outer-slot stars and refractory-tilted inner-slot stars. The refractory-tilted inner-slot population should be statistically associated with cascade companions to high-mass primaries; the volatile-tilted outer-slot population should be statistically isolated or associated with low-mass primaries (where the snow line falls within the cascade range).

A direct prediction for the Sun follows from the calibrated reconstruction (§7.6): Sol's seat lay beyond Hyperion's snow line, so its $\sim 411\,M_\oplus$ of seat solids were 45% ice, water that contributed a few percent of Sol's primordial oxygen budget, with the remainder of the heavy inventory arriving as envelope-dissolved disc gas.

7. THE HIERARCHICAL RECURSION: EACH GENERATION BUDGETS THE NEXT

A daughter system's two defining inputs are both read off the parent's structure at the daughter's seat; nothing else about the parent enters. The budget is the slot allocation: the daughter's inventory is the mass the parent's natal disc assigned to her slot (§2). The boundary is the parent's local pressure. A daughter forming at radius $r_{\rm site}$ inside her parent's disc inherits the nebula density

$$\frac{D_d}{D_p} = 0.39 \left(\frac{r_{\rm site}}{R_{\rm disc}}\right)^{-2.12}, \tag{34}$$

and the exponent is not free in practice. It reproduces the disc's own density profile ($\rho \propto \Sigma/H \propto r^{-2.1}$), so the law says simply: record the ambient compression at your seat. The daughter's dam follows from her pair through $R_{\rm disc} = 30.07\,M\,D^{-1/3}$, her cascade runs on the universal machinery, and her slots budget granddaughters. Two numbers enter at any rung and a complete subtree comes out. This is the (M, λ) inheritance pair of Table 1, with λ realised at these scales as the recorded compression D through the jaw-lock (§2). Two daughters of two different parents fix the constants of Eq. 34: Proxima against Alpha Centauri A's disc, and Jupiter against Sol's. That the resulting exponent independently matches the disc profile is the law's first self-consistency check.

7.1. *The measured rung: Jupiter's satellite system*

The Galilean system supplies the clean test, because the daughter's pair can be obtained two independent ways. Fitted from the moons alone, Jupiter's disc requires $D_J = 11.87$. Predicted from Jupiter's seat in Sol's disc via Eq. 34, the answer depends on which radius is used. Jupiter's formation slot ($r_3 = 5.98$ AU) gives $D = 11.97$, agreement to 0.9%. Jupiter's current orbit (5.20 AU) gives 16.1 and misses by 35%. The satellite disc remembers where Jupiter formed, not where it is. A disc 2,400 times smaller than the orbit it remembers thereby confirms, independently, the ~ 0.8 AU inward recoil that the interior fit attributes to the embryo-swarm scattering event (§5.6).

With the inherited pair imposed, all four Galileans sit in situ in their observed order. Callisto is the dam resident, at slot 0 (0.012586 AU against an observed 0.012585), and is fittingly the one undifferentiated, geologically primitive Galilean. The position residuals of the inner three decline outward in exactly the Laplace-resonance tidal pattern: Io +13.0%, Europa +4.9%, Ganymede −2.3%, Callisto 0.0%.

The framework's geometry plus four gigayears of known tidal drift accounts for the system without any orbital rearrangement.

The Roche parameter completes Jupiter's interior census. Slots 5 and 6 fall inside the fluid Roche limit (1.36×10^5 km): their allocations could never consolidate and persist as the ring system, whose main edge sits on the line. Slot 4, at 1.6 times the limit, consolidated and was dragged inward across the line by gas drag, the loss channel satellite-formation theory has long required (Canup & Ward 2006); the machine itemises one instance. Its bulk fed Jupiter. Its highest shards survive as the Amalthea group, stranded from the seat down to the line: Thebe on slot 4 to 1.5%, Amalthea between, at a measured density of 0.86 $\mathrm{g\,cm^{-3}}$ a rubble pile sitting inside the Roche limit for its own density and held together by friction, and Metis and Adrastea on the line feeding the main ring. Each fragment trails its own gossamer ring: the debris field of the torn slot-4 body is directly observed.

7.2. *The diagnostic rung: Saturn*

Saturn's seat (slot 2, 10.25 AU) fixes its inherited pair, and the pair refuses to fit the present mid-sized moons at any input. Mimas through Rhea total ~ 0.7 milli-$M_\oplus$, two orders of magnitude below any cascade allocation for a disc of Saturn's budget; the refusal is itself the diagnostic. Cassini's gravity and ring-pollution measurements date the rings to $\sim 10^7$–10^8 yr (Iess et al. 2019), and dynamical arguments date the mid-sized moons comparably (Ćuk et al. 2016). The recursion reads the same story from the masses alone. Saturn's interior satellite cascade was destroyed recently; the present mid-moons are re-accreted debris; the rings are the destruction ledger, visible because fresh. The Roche parameter sharpens the ring reading into a law. Saturn's forbidden zone (ice-Roche near 1.37×10^5 km) swallows slot 4 and everything below, and the ring system spans that zone edge to edge: the zone is primordial, a permanent trap in which no slot could ever consolidate, while the ring material is fresh, the latest refill from the recent wreck. This accounts at once for the rings' existence and their youth, and generalises: every giant's ring system sits inside its primary's Roche zone, the standing allocation of forbidden slots, refilled by each destruction epoch. Slot 3, marginal at 1.05 times the line, repeats Jupiter's slot-4 fate one seat deeper: the body just above the line is the first casualty of inward drag. Titan and Iapetus formed beyond the dam at the exterior boundary (the factory of §8 at sub-cascade scale) and were unaffected. They are the system's Kuiper analog; Titan dominates because the dam's outer face is everywhere the best-fed seat in a system.

7.3. *The adopted rung: Neptune*

Neptune adds a third reading of the same machinery. Its inherited pair puts its dam near 3.2×10^5 km: the inner regulars, Naiad through Proteus, sit inside as interior cascade products, and Triton sits just beyond the wall. Triton's retrograde orbit rules out condensation in place and marks it as adopted, and the framework names the source. Neptune is Sol's slot-0 resident, so Neptune's trans-dam territory and Sol's dam-exterior factory floor are the same place. Triton is compositionally Pluto's twin, and Pluto is a Sol-dam cohort object: Triton came off the same production line and was bound by the planet parked beside it, during the firehose era when the newborn cohort was dense and binary-exchange capture easy. The adoption is also the system's destruction event, the third instance of the wrecked-interior pattern after HD 69830 and Saturn: the original regulars were destroyed by the arriving heavyweight, the present inner moons are re-accreted debris, Neptune's faint rings are the remaining ledger, and Nereid rides its $e = 0.75$ orbit as the scattering relic. The trans-dam adoptee outweighs the interior survivors five hundred to one: the exterior dominance Titan shows (§7.2). The test is isotopic: Triton and Pluto should match in D/H and nitrogen isotopes as production-line siblings, and the re-accreted inner moons should show young surfaces.

7.4. *The rebuilt rung: Uranus*

Uranus completes the four-giant gallery. Its inherited pair ($D = 1.22$ from Sol's slot-1 seat) puts its dam near 1.6×10^5 km, and the census splits cleanly across the wall: the ring system, the small-moon swarm, and Miranda inside; Ariel, Umbriel, Titania, and Oberon outside, prograde and equatorial, native trans-dam products outweighing the interior ~ 125 to one. The architecture is the receipt for a catastrophe this paper has already charged to Uranus. The Solar System fit attributes Uranus's 98° tilt and cancelled gas era to the slot-5 primary's erosive impact (§4.6), and the satellite system records the same event: the major moons orbit in the tilted equatorial plane, so they formed in a disc rebuilt after the knockdown; the rings and inner swarm are the wreck ledger; Miranda, sitting near the interior half-step site, is the re-accreted body, and its patchwork coronae geology has read as shattering and reassembly since Voyager; and the dam seat itself is empty, its original resident a casualty. Across the four giants one variable now organises the trans-dam yield: Jupiter completed its gas capture and drank its exterior pool dry (its only trans-dam natives are the small prograde Himalia-group candidates), while Saturn, Uranus, and Neptune left pools that condensed into Titan, the Uranian big four, and the cohort Neptune adopted Triton from. The emptiest larder belongs to the biggest eater.

7.5. *The kinetic rung: Earth and Luna*

The recursion's smallest rung begins with a prohibition. Run Earth's inherited pair from Sol's slot-6 seat and the gas-era dam lands at 1,700 km, inside the planet: the nebula's ambient pressure at 1.2 AU crushes any terrestrial circumplanetary disc to nothing. Terrestrial planets cannot run the machine during the gas era, which is why none of them carries a primordial satellite system. Earth's pair arrived another way. The Theia impact, the inner share of the same slot-4–5 swarm event this paper charges for Mars, Saturn, and Uranus, delivered it kinetically: spin (the $\sim$ 5-hour post-impact day is the λ-record, written by collision rather than collapse), a stirred core and dynamo whose field is the Alfvén boundary, a debris disc as the natal disc, and a Davis Dam where the new magnetosphere meets the post-gas solar wind. With the impact-era field and wind, that dam stands at 5–7 $R_\oplus$, and the cascade under it has exactly two seats outside the Roche limit: slot 0 near 6.6 $R_\oplus$ and slot 1 at 3.8 $R_\oplus$, which is Luna's accretion radius to one percent. Slots 2 and below are Roche-forbidden and feed the open seats. The machine therefore predicts one large moon and at most one small companion, and it predicts the companion's fate. The two-seat configuration is born packed: $\Delta r = 2.73\,R_\oplus$ against a mutual Hill radius of 0.84 $R_\oplus$ gives $P \approx 3.2$, far below the Lissauer threshold, so relaxation is compulsory by the same criterion that convicts 55 Cancri and HD 69830. Earth's spin-down sets the direction: tidal recession scales as $m/a^{11/2}$, so Luna receded some 600 times faster than its thirty-times-lighter outer sibling and ran it down, caught it into the co-orbital points, released it as the migrating geometry destabilised, and absorbed it at roughly escape velocity, the collision regime that accretes rather than craters. Lunar science recovered both bodies and the outcome independently: Luna at slot 1, and a companion whose slow farside merger explains the highland asymmetry (Jutzi & Asphaug 2011); we name the companion *Endymion*. Tidal recession carried Luna from 3.8 to 60.3 $R_\oplus$ at a rate still measured today: a directly observed post-formation clock.

The configuration was the machine's most extreme interior. The vice stood nearly closed, with the corotation boundary at 2.3 $R_\oplus$ against the dam at 6.6: compression $C = 0.35$, fifty times Sol's, a hair short of inversion, which is why the debris disc was compact. Of the cascade's fifteen slots, eleven lay inside Earth itself and two inside the Roche limit, leaving the two seats. The machine's accounting applies only to the debris captured between the vice jaws; ballistic fallback that never circularised into the disc lies outside it. Of the captured disc, the in-planet slots' allocations are the magnetically forced-down share – the disc-borne component of the 50–70% fallback fraction impact simulations find, not its whole – the Roche-forbidden slots' allocation is the Roche-interior disc that viscously feeds the moon, and the disc's much-noted compactness is the kinetically delivered λ-record itself, capped by the magnetopause dam.

The slot, not the Roche limit, is Luna's address. The conventional account reaches "just outside the Roche limit" and stops; Roche is a floor where accretion becomes possible, not a reason the Moon sits at 3.8 rather than 3.0 or 4.5 $R_\oplus$. The machine supplies the address: slot 1 of the impact disc's cascade. Roche explains why not closer; the slot explains why exactly there, and the Moon becomes an ordinary slot occupant, the same kind of object as Venus at Sol's slot 7 or Callisto at Jupiter's slot 0. The agreement also hides a discriminator. Mainstream accretion simulations recover $\sim 3.8\,R_\oplus$ as ~ 1.3 Roche radii, a density-anchored number; the machine derives the same value as $\rho \times R_{\rm dam}$, a field-and-wind-anchored number. For Earth the two coincide. In any other impact-built satellite system where they separate, Pluto–Charon among the candidates and resolved exomoon systems eventually, the moon's address decides whether tides or the machine set it. The test is entered in the register (§10).

Venus is the control case: no late giant impact, and its three absences (no spin, no magnetic field, no moon) are one absence: the kinetic delivery never arrived. Mars runs the same sequence at one-tenth scale and supplies receipts Luna cannot. Its gas-era dam computes to 260 km, subsurface as the prohibition requires; the Borealis impact, the proposed origin of the hemispheric dichotomy (Andrews-Hanna et al. 2008), delivered the pair, and the dynamo's receipt survives as the crustal magnetization, the field dying ∼4.0 Gyr ago, after the formation epoch it enabled. The impact-era magnetopause dam brackets at 4.5–6.9 $R_{\rm M}$, and the cascade beneath it holds exactly two seats above the rocky Roche limit (2.67 $R_{\rm M}$): slot 0 at 6.92 $R_{\rm M}$, where Deimos sits in situ to this day, and slot 1 at 4.04 $R_{\rm M}$, Phobos's formation seat, from which measured tidal decay carried it to its present 2.77, a gentler start than near-synchronous origins demand of the tidal quality factor. The corotation radius (6.03 $R_{\rm M}$) falls between the moons, and both obey it: Deimos recedes, Phobos falls. Why Mars kept both children while Earth ate one is a single number: the pair packs at $P = 285$ against Luna–Endymion's 3.2; same architecture, tiny occupants, relaxation impossible. The mass ledger reads differently from Luna's, and informatively so: even the engine's minimum disc fraction allocates $\sim 8 \times 10^{19}$ kg per seat, the canonical Borealis disc scale, while the moons total $\sim 10^{16}$ kg; the deficit is the grinding ledger of the ring–moon cycle (Hesselbrock & Minton 2017), which is running now: Phobos sits inside the fluid Roche limit for its own density (2.77 against 3.13 $R_{\rm M}$), strength-held, ring-bound within ∼50 Myr: the Roche zone is permanent and its fill recurrent. Across the terrestrial zone the machine is off by default and switched on only by collision, and every case matches.

7.6. *The recursion run upward: Sol's progenitor*

Nothing restricts the recursion to descent, and Sol's own boundary supplies the key. Its 30.07 AU dam is read not as a free jaw-lock value but as the *tide truncation* of its natal disc at its seat in the parent's system – the Alpha Cen B configuration one generation up – so $D_\odot = 1$ is the truncated-disc record, which resolves why no inherited ambient value reproduces it. The truncation constraint then derives the parent with no calibration: $r_t = 30.07$ AU at mass ratio 2.28 puts Sol's seat at $q = 139.8$ AU; seat = slot 1 gives the parental dam at 239.7 AU, hence $D = 0.0234$; and the geometry closes on itself three ways – the primary's tide at the seat (30.1 AU), the Hill truncation (0.41 R_H of Sol's 73.7 AU Hill sphere, 30.2, at the centre of the standard 0.3–0.4 fraction), and the half-separation to the equal-mass inward neighbour Eos (29.1) all reproduce the observed dam: the disc filled exactly its allotment at the seat, bound by the inward side. (For comparable-mass standing neighbours the gap partitions at the half-separation; the Breslau flyby law, correct where the perturber dominates, over-truncates in this regime.) Then $f = 0.115$

follows by requiring the slot-1 allocation to build Sol *and* its disc ($379{,}558 \times 1.0104$, closing at 1%) – landing inside the multi-star family band 0.07–0.13 unprompted. We name the parent *Hyperion* (after the Titan father of Helios; the name is shared with Saturn VII). The derived family: a $0.20\,M_\odot$ M-dwarf keeper on the dam; Sol at slot 1; *Eos, a G-type solar twin*, at slot 2 (81.7 AU, $376{,}652\,M_\oplus = 0.992$ of Sol's primordial mass); a 1.87-unit star at slot 3, seated on the snow-line pile-up (47.7 against 47.5 AU); brown dwarfs at slots 4–5 and giants at 6–8, all inside the snow line. The nest's history follows from its own spacing. One pair is compulsorily unstable: slots 2–3 pack at $P = 0.70$ mutual Hill radii, the system's tightest, and the mass-ordered kick rule resolves it – the 1.87-unit star holds, Eos takes the ride, ejected gigayears before anything else moved. The remainder of the nest is wide in absolute terms (the planet-chain packing index does not transfer to comparable-mass stellar hierarchies), and Sol kept its seat for Hyperion's full ~ 1.3 Gyr lifetime – the parent's light at 139.9 AU contributing a negligible 0.2% of Sol's own insolation at 1 AU, leaving the luminosity chronometers untouched – until the asymptotic-giant mass loss halved the system's binding mass and released Sol, the keeper, and the A-class star together, at the epoch van Maanen's star's cooling age records. The residence is a statement about Sol's first gigayear, not only Hyperion's: the magma-ocean era, the Theia impact, the Noachian, and the emergence of life all ran inside a multiple-star system – a quintuple at birth, a quadruple after Eos's ejection – whose companions contributed negligibly to the energy budget ($\lesssim 0.5\%$, leaving the luminosity chronometers intact) while dominating the night sky: Hyperion at 140 AU and the A-class star at conjunction each shone near magnitude -20, a thousand full Moons, and the Sun has been a single star for only the latter 80% of its history. The reconstruction also answers the meteoritic short-lived radionuclides (^{26}Al, ^{60}Fe, ^{53}Mn), which require a supernova within parsecs of Sol's formation: Hyperion's birth cluster supplies the neighbour.

The reconstruction ends in a falsifiable package. Hyperion's mass and epoch fix its remnant: at $2.28\,M_\odot$ and a 1.27 Gyr lifetime, a carbon–oxygen white dwarf of $0.67\,\mathrm{M}_\odot$ with cooling age 3.3 Gyr ($T_{\rm eff} \approx 6{,}100$–$6{,}400$ K), on a thin-disc orbit displaced from Sol's by the tens of $\mathrm{km\,s^{-1}}$ that 4.6 Gyr of independent disc heating imposes on any co-natal pair. A candidate already exists in the catalogue, fourteen light-years away: van Maanen's star matches the package in mass ($0.68\,\mathrm{M}_\odot$), temperature (6,100–6,200 K), and cooling age ($\sim$3.2 Gyr), and the literature's own progenitor reconstruction for it ($2.6\,\mathrm{M}_\odot$, 0.9 Gyr main sequence) agrees with the inheritance-law inversion to within 4% in mass. It is moreover the archetypal metal-polluted white dwarf, currently accreting debris from its old planetary system, which supplies the decisive archival test: its photospheric metal ratios sample the planets condensed from Hyperion's disc, the reservoir that made Sol, and should be solar-rock to high precision if the identification holds, and *dry*: the derived geometry places Hyperion's snow line at 47.5 AU, beyond every sub-stellar seat, so the predicted diet is waterless rock, which is what the star's hydrogen-free DZ pollution already shows. The candidate's single status is consistent: Hyperion's asymptotic-giant mass loss shed over half the system's mass and unbound the remaining family.

The derived family then defines three searchable siblings. (i) *Eos*, the slot-2 solar twin: ejected early by the slot-3 instability, it carries a gigayears head start plus ejection velocity over the AGB-released members, so the framework does *not* expect it in the solar neighbourhood – the nearby solar twins (18 Scorpii foremost) are controls rather than candidates, each still decidable by the solar isotopic fingerprint whenever screened. (An earlier placeholder seating identified Eos as a $0.33\,M_\odot$ M dwarf and matched the candidate GJ 4040; that chain is retired with the seating.) (ii) The slot-3 star: $708{,}732\,M_\oplus = 1.87\,M_\odot$, seated on the snow-line pile-up, lifetime $\sim$2.0 Gyr – it outlived the nest by $\sim$0.8 Gyr and died in the field. The framework therefore predicts a *second white dwarf sibling*: $\sim 0.64\,\mathrm{M}_\odot$, $T_{\mathrm{eff}} \approx 7{,}000$ K, cooling age $\sim$2.5 Gyr – younger-cooling than van Maanen's star by the $\sim$0.8 Gyr survival gap. A candidate *pair* exists 18 light-years from van Maanen's star: Stein 2051, whose white dwarf B carries the only model-independent WD mass ever measured ($0.675 \pm 0.051\,\mathrm{M}_\odot$ by gravitational deflection; Sahu et al. 2017) at $T_{\mathrm{eff}} \approx 7{,}100$ K and cooling $\sim$2–2.5 Gyr, *bound to an M dwarf of* $0.24\,M_\odot$ – against the keeper's predicted 0.20. Two members pairing at the low-relative-velocity AGB release is dynamically natural, and both component masses, the cooling age, and the cooling-age *gap* to van Maanen's star match the package. The configuration is also self-consistent: the three members form a near-isoceles triangle (Sol–van Maanen 14.1 ly, Sol–Stein 2051 18.0, van Maanen–Stein 2051 18.3 – the two corpse baselines equal to 1.5%), no member an outlier, as equal-status dispersal from one release point predicts. B is spectrally featureless (DC), so the pollution test is unavailable; the decisive tests are the companion's spectrum – Stein 2051 A's abundance fingerprint against the solar clone – and the corpses' mutual space velocity from Gaia astrometry, a quantity 3.3 Gyr of field heating randomises for unrelated stars but can leave small for a pair that dispersed together. (iii) The slot-0 keeper, a $0.20\,M_\odot$ M dwarf. Kinematic coherence with Sol or the white dwarf is not expected after gigayears of independent field heating (Holmberg et al. 2009); the searches anchor on composition and remnant properties. All three are entered in the register (§10).

7.7. *The root of the recursion*

The tree has a measured top. Across the fitted catalogue, systems at essentially one galactocentric radius span 5.6 decades in D. The parent that sets a system's pair is always the local structure, a disc or a core or a cloud, never the Galaxy at large. At the root, where no parent disc exists, the pair becomes an environmental draw, and this paper measures the draw's statistics twice: as the stochasticity of f_{disc} across the planetary catalogue (§11), and one tier up as the λ-distribution of galaxy formation. They are the same measurement, made at two scales, under two nomenclatures (§9).

8. THE EXTERIOR RESERVOIR: KUIPER BELT, OORT CLOUD, AND THE BOUNDARY CENSUS

Standard theory supplies *mechanisms* for the trans-Neptunian reservoirs (the Oort cloud as scattered planetesimals lifted by galactic tides (Oort 1950; Duncan et al. 1987), the Kuiper belt as the remnant outer planetesimal disc) but their locations, masses, and mutual structure enter as initial conditions, and each carries a standing anomaly (the belt's two-orders-of-magnitude "missing mass," the unexplained Kuiper cliff, the Oort cloud's $\sim 5\%$ trapping efficiency). In the cascade framework both reservoirs are instead *necessary consequences* of the two outer boundaries every star possesses: the magnetic boundary (the Davis Dam) and the gravitational boundary (the star's Hill radius in the galactic potential, Eq. 24 applied one level up the hierarchy). Material pools at a boundary while the boundary has working fluid, and fossilizes in place when it dries. The belt and the cloud are the deposit sequences of the two boundaries.

8.1. *The dam's exterior side*

The dam traps from both directions. In steady accretion $\dot{M} = 3\pi\nu\Sigma$ is constant in radius, so surface density scales as $\Sigma \propto 1/\alpha$: gas flowing inward through the MRI-active exterior ($\alpha \sim 10^{-2}$) meets the dead-zone interior ($\alpha \sim 10^{-3}$–10^{-4}) and queues against the transport-capacity step, a pile-up of order the viscosity contrast (10–100×) over a width of order the scale height. Solids respond aerodynamically: drag delivers pebbles to the resulting pressure maximum, where the concentrated ring crosses the streaming instability threshold (Youdin & Goodman 2005) and collapses directly into $\sim$ 100-km bodies: natively producing the like-coloured binaries observed throughout the cold classical population (Nesvorný et al. 2010). The cascade itself cannot operate exterior to the dam (the slot geometry is anchored to the dam from inside), so growth out there stops at Kuiper-belt-object scale: *the exterior pool makes a belt because it cannot make a planet.* The dam's outer face is the lattice's final zonal flow: the same magnetically organised trapping that builds every interior slot, standing at the boundary with no interior geometry behind it, so everything it builds forms at one radius, wherever the dam stands. The factory also feeds its own advance. As pooled material locks into bodies it stops exerting pressure, the counter-push on the dam's outer face weakens, and the boundary steps outward onto fresh material, carrying the accretion front with it: deposition itself walks the shoreline.

This dissolves the classical mass problem. In-situ coagulation required a 10–30 $M_\oplus$ primordial belt eroded by 99.5% to today's $\sim 0.02\,M_\oplus$ (Pitjeva & Pitjev 2018); streaming-instability assembly at the dam bump requires only ~ 0.1–$1\,M_\oplus$ of exterior solids, comfortably supplied by a percent-level residue of envelope infall that never crossed the dam, and needs no disposal mechanism. Li & Chiang (2025) object that pressure-bump trapping generically over-feeds the belt, and favour late-nebula streaming instability (at 2–5% residual gas) for the cold classicals. Their objection applies to bumps fed by the full disc's drifting dust; the dam-edge trap is fed only by the exterior pool, the interior pebble flux having been consumed by the cascade slots upstream, so the trapped mass is capped by the exterior supply. The late-nebula timing they favour is moreover natural here: the dam persists as the gas wanes, and the exterior pool is the last reservoir still feeding it.

Assembly at boundary interfaces is observed at far larger scales: star formation occurs *within* galactic outflows (Maiolino et al. 2017) and is widespread in them (Gallagher et al. 2019), flow–medium frontiers, environments naively expected only to disperse material, instead pooling, compressing, and collapsing gas into bound objects. This is not an analogy but the same physics: a wind–medium pressure boundary in magnetized flow, pooling what arrives at it until the local collapse threshold is crossed, with only the participants scaled by twelve orders of magnitude; the dam-edge factory pooling exterior solids into Kuiper belt objects and the outflow frontier condensing gas into stars are one boundary class (§8.7), and the product mass in each case is set by the local capture physics, not by the mechanism.

8.2. *Belt architecture from a stationary Neptune*

Neptune is the dam's interior product (cascade slot 0, gas-fed); the dwarf planets are its exterior products (ice-fed, streaming-instability-born). Both sides deliver to the same orbital radius, with the consequences developed in §4.1: exterior products born while the dam lay within Neptune's $\sim 10\,R_H$ reach ($\lesssim 38$ AU) are milled (scattered, eccentricity-pumped, surviving only in protecting resonances) while products born after the dam moved beyond that reach are never disturbed. The observed hot/cold dichotomy follows from a *moving factory and a stationary planet*: the hot population (Plutinos, scattered disc, high inclinations) is early output born inside the reach; the cold classicals (low-e, low-i, binary-rich) are later output born beyond it. The belt's inner edge (~ 39 AU) is Neptune's clearing reach plus the 3:2 resonance at 39.4 AU; the Kuiper cliff coincides with the 2:1 at 47.7 AU, the outermost first-order trap for inward-drifting exterior material (§4.1). The cliff is doubly determined: the dam was transiting the ~ 48 AU station just as the gas-era supply died (§8.3), so the 2:1 trap and the factory's shutdown radius coincide: the belt ends where production ended.

8.3. *The migrating dam: generational deposition*

The dam's position is set by the pressure balance between the wind and the ambient medium, and the stellar spin is that balance's recorded history: compression causes spin through the jaw-lock ($\Omega = D^{2/3}$, §2), so as the ambient pressure relaxes and the dam marches outward, the braking star writes the march into its rotation (Skumanich 1972), and Eq. 4 reads the station back from the record. With Sol's primordial rotation period of 1.55 d and today's 25.4 d, the recorded stations are approximately: 30 AU through the gas era (disc-locked spin); ~ 48 AU by ~ 0.2 Gyr; ~ 80 AU by ~ 1 Gyr; and $30 \cdot (0.061)^{-1/2} = 121.4$ AU today, within 0.2% of the heliopause measured by Voyager 1 at 121.6 AU (Stone et al. 2013). We identify the present-day dam with the heliopause: the surface where the Sun's magnetized outflow meets the interstellar medium is the same boundary, holding nothing because it has no working fluid. The boundary today is a balance of two near-vacuums (wind ram pressure at 121 AU is $\sim 10^{-4}$ of its 1-AU value (~ 0.1 pPa), matched against a comparably tenuous interstellar medium) roughly twelve orders of magnitude below the disc-era working-fluid density at the same boundary. A pressure-balance surface is only as strong as the medium it holds, so the dam's effect has shrunk in proportion: from pebble pile-ups in the gas era to the deflection of sub-0.1 μm charged grains (the only particles whose charge-to-mass ratio still couples to the boundary) today. (The identification carries caveats: the agreement uses the primordial-mass convention and equatorial rotation period, the heliopause breathes with the solar cycle, and a derivation showing the wind–interstellar-medium standoff scales as $\Omega^{-1/2}$ remains outstanding. The formation-era law itself has a candidate derivation: $R \propto \Omega^{-1/2}$ is the angular-momentum-conservation locus of a pressure-confined system (§2), with environmental compression as cause and spin as record. The identification then requires the same causality to persist after the disc: the boundary sits where ambient pressure dictates, while spin, braked by the wind that mediates that very balance, continues to track the boundary's secular expansion. Spin records the secular trend; instantaneous environmental excursions (dense-cloud crossings, which deposit rings) perturb the boundary about the spin-read baseline.)

The present-day dam, though feeble, visibly carries its pile-up: the *hydrogen wall* (the density enhancement of interstellar neutrals stacked against the heliopause, detected in Lyman-α absorption toward nearby stars and crossed by the Voyagers (Wood 2004)) together with the measured filtration of small charged interstellar grains at the boundary. The same absorption technique detects *astrospheres*, hydrogen walls, around other nearby stars (Wood 2004), establishing the dam as a universal, directly observed stellar structure rather than a solar peculiarity. A third, independent manifestation is imaged rather than crossed: Cassini's all-sky energetic-neutral-atom maps revealed an unanticipated belt of enhanced particle pressure wrapping the heliosheath, organised by the draped interstellar field (Krimigis et al. 2009) – the dam's pile-up seen whole, its azimuthal structure set by the external medium that does the confining, as a pressure-balance wall requires. Read as the slot-0 band of the live lattice (§8.4), the heliosheath acquires edges with addresses: the band crest is the heliopause (121.7 predicted, 121.6 crossed) and the band's inner boundary is the comb's outermost trough at $\sqrt{r_0 r_1} = 93.0$ AU – where Voyager 1 met the termination shock at 94.0 AU (+1.1%; Voyager 2's 83.7 reflects the shock's known solar-cycle breathing and asymmetry about the time-averaged station). The shock parks at the standing pressure minimum because that is where wind ram naturally balances sheath pressure, and the band width ($\sigma = 0.3\,R$, 37 AU, the same constant that sets the formation-era edge pile-up) brackets the measured sheath thickness of 28–35 AU. These identifications are temporally post-hoc but accommodation-free: every constant involved (ρ from geometry, the 30.07 AU anchor from Neptune's orbit, the spin-down record, the $0.3\,R$ band width from the formation-era edge allocation) was fixed by formation-scale physics, and no heliosheath quantity entered any calibration – the model could not have absorbed different radii. Under this reading the structure the boundary missions catalogued as anomalous is expected: the thin sheath is the band width, the stagnant interior is a standing structure rather than a flow region, the weak termination shock is a soft trough rather than a ram wall, and the two Voyager crossings sit on the live comb's two outermost stationary points. The slot-0 band is saturated and shock-bounded (nonlinear), so the linear η attenuation gauge does not extrapolate from it.

The live lattice.—If the zonal-flow ladder is anchored to the boundaries rather than to its fossil occupants, it re-anchors as the jaws move: the planets persist at the formation addresses, but the *flows* stand on today's geometry. With the present jaws (dam at 121.7 AU; the Alfvén law extrapolated to the current spin gives 0.041 AU $= 8.7\,R_{\odot}$, within a factor of two of the Parker-Solar-Probe-measured Alfvén surface at 13–20 $R_{\odot}$), the current ladder holds fifteen slots: 121.7, 71.1, 41.5, 24.2, 14.1, 8.25, 4.82, 2.81, 1.64, 0.96, 0.56, 0.33, 0.19, 0.11, and 0.065 AU. Two land on known structure: slot 1 (71.1 AU) at the inner edge of the 70–90 AU ring (Fraser et al. 2024), and slot 2 (41.5 AU) in the heart of the cold classical belt, where a standing band would contribute to the population's noted dynamical coldness. The working fluid today is the supersonic wind rather than a Keplerian disc, so the disc-era band mechanism does not transfer naively and the amplitude must scale with the twelve- order pressure decline; the claim is the boundary-anchored standing-structure grammar, at whatever amplitude survives. The test is archival, and the slot's nature orders the observables: the slot is geometric in origin and magnetic only in operation, so the magnetic perturbation is the *instantaneous* signature, as strong as today's feeble medium, while trapped dust is the *time-integrated* one, accumulating at the geometric address over megayears regardless of the band's present amplitude. Dust tracers therefore lead: radial zodiacal-light and dust-counter profiles binned against the comb (both existing receipts below are dust), with New Horizons' dust counter currently sampling the outer comb – its reported flux excess rising toward 60 AU (Doner et al. 2024) sits on the approach to live slot 1. Magnetic or plasma banding at fixed heliocentric radii is the confirming probe, separable from solar-cycle and transient structure because Pioneer 10/11, Voyager 1/2, Cassini, and New Horizons crossed the same radii at different epochs – a feature standing at fixed r across spacecraft and decades is spatial, not temporal. The two signatures should moreover converge inward: the innermost rung (0.065 AU $= 14\,R_{\odot}$) sits at the silicate sublimation edge, so its aggregate is selected toward conductive iron-rich grains carrying the highest photoelectric charge states in the strongest ambient field – a standing shell of such grains is a current structure, not a passive tracer. Cross-correlating Parker Solar Probe's dust impact rates (which already require an unexplained third impactor population) with its magnetometer, binned against slots 12–14, tests the geometric reading's distinctive joint signature: co-located dust and magnetic banding with a magnetic-to-dust ratio growing inward. Because zonal flows are alternating high- and low-pressure bands, the comb signature is oscillatory in $\log r$ at period $\log(1/\rho)$, permitting a matched filter with frequency *and* phase fixed a priori (crests anchored to the dam) – zero free parameters. A first application to the public Voyager merged daily averages finds a comb-frequency signal in both spacecraft's field magnitudes with cross-craft phase agreement of 6° (epochs years apart, different solar-cycle phases; amplitudes at the ∼90th percentile of the off-comb background) and the field in anti-phase with the ram pressure to 181° – the pressure-trading structure alternating bands require. This is a candidate, not a detection: daily averages, no solar-cycle normalization, no excision of merged interaction regions. The same filter applied to the Helios 1/2 hourly archive (slots 9–11, $\sim 9\times10^4$ records, all solar-cycle phases self-averaging within each radial bin) returns a clean null: standing band contrast $\lesssim 1\%$ in field, thermal, and ram pressure, with no cross-craft phase replication. The null is in fact required by the generative construction: the first band stands at the dam (the system's steepest gradient, observed as the hydrogen wall) and each daughter band organises at its parent's half-amplitude shoulder, so band contrast attenuates geometrically inward, $A_n \sim A_0\,\eta^n$ with $\eta \approx 1/2$. That gauge places slots 1–5 at few-to-tens-of-percent contrast (the Voyager candidate's range) and slots 9–11 below 0.1%, an order of magnitude under the Helios limit – while megayear dust integration keeps the inner rungs' traps visible, which is why their two receipts are dust. The attenuation law sharpens the falsifier: the full pipeline (hourly data normalized by transit-lagged 1-AU conditions, MIR excision, Pioneer 10/11 as third and fourth epochs) must find contrast decreasing monotonically from slot 1 to slot 5; bands at flat or inverted amplitude would falsify the shoulder construction even if the comb itself is present. Two further archives extend it. Cassini's thirteen years in Saturn orbit rode the planet's radial drift from 9.0 to 10.1 AU – a slow scan of slot 5's outer flank, the only dataset where decade-averaging can resolve a middle rung's predicted $\sim 2\%$ contrast (and thereby confirm that the $\pm20\%$ Voyager structures near 8.3 AU were stream noise, as the gauge requires). And Ulysses supplies the axisymmetry control: the bands are disc-plane structures, so its out-of-ecliptic passes should show *no* comb at radii where its in-ecliptic segments do – a latitude-dependence discriminator with a-priori sign. It also yields a dated, prospective prediction (registered June 2026). New Horizons, at ~ 67 AU and moving at 2.95 AU yr^{-1}, is mid-flank on slot 1: its detrended ambient pressure should sit on a *plateau* from the 54.3 AU trough (-25%, crossed ~ 2022) to the 71.1 AU crest – a 17-AU stretch in which the smooth wind's -42% r^{-2} decline is cancelled to $\approx +3\%$ – reaching the crest at +25–35% in late 2027 (the η-law gauge doubled from Voyager 2's slot-2 measurement gives +33%; V2's own 2003 crossing of this rung gave +25%), with a coincident dust-rate maximum (the Doner et al. 2024 excess being the flank entry). A smooth r^{-2} profile through 71 AU kills the rung; a peak displaced by more than a few AU kills the dam anchor; the right peak at the wrong amplitude breaks the $\eta = 1/2$ gauge. The inner slots are better than archival: band contrast scales with the ambient working pressure, which at 0.1–1 AU exceeds the boundary region's by orders of magnitude, and the radii are continuously instrumented – STEREO-A's orbit rides slot 9 (0.958 AU, perihelion 0.956), Helios 1/2 swept slots 9–12 repeatedly (0.29–0.98 AU, 1974–1985), and Parker Solar Probe now crosses slots 9–14 on every orbit. Two candidate receipts are already on record, both filed as curiosities at detection. The Helios dust experiments found the apex-particle spatial density *peaking* at 0.5–0.6 AU (Grün et al. 1980) – a radius with no planet and no proposed trap – against live slot 10 at 0.559 AU. And the circumsolar dust ring near Mercury's orbit (Stenborg et al. 2018), reported as a surprise because Mercury was thought too small to trap one, is eccentric with its brightness maximum aligned to Mercury's *perihelion* (~ 0.31 AU) – the geometry of a planet sweeping through a standing band at live slot 11 (0.326 AU) rather than trapping its own ring. (The Earth and Venus rings, by contrast, sit at their planets' orbits and off the live comb; they are resonant-trapping candidates and do not bear on this test.) A radial superposed-epoch analysis of the assembled datasets, binned by heliocentric distance against the predicted comb, is the register's cheapest test, with slot 9 – unexamined, just inside Earth's orbit, and continuously occupied by STEREO-A – the natural first target.

Cementing requires both working fluid and supply, so deposition is episodic: the gas era builds the main belt, and later cohorts form only when the Sun traverses dense interstellar material, which both re-immerses the dam in a dissipative medium and supplies fresh solids. The radial map of the belt is then a *time log* of the Sun's environmental history read against the spin-down clock. The newest observations show exactly such banded structure: dust fluxes above all model predictions approaching 60 AU (Doner et al. 2024), and a populated ring at 70–90 AU separated from the known belt by a near-empty gap at 55–70 AU (Fraser et al. 2024). In this reading the 70–90 AU ring is a wet episode at the ~ 1 Gyr dam station, and the gap is a quiet transit. The present-day silence is overdetermined: the Sun currently resides inside the Local Bubble, a supernova-blown cavity of hot, $\sim 100\times$ underdense gas (Frisch et al. 2011), which it entered ~ 5 Myr ago (Zucker et al. 2022), the factory today lacks fluid *and* feedstock. Kuiper-belt-object formation is a property of local interstellar conditions, switched on only when a star's environment supplies both. (Interstellar processing of Oort-cloud comet *surfaces* (erosion by interstellar grains, cosmic-ray irradiation) is established (Stern 2003); the framework extends the star–interstellar-medium interaction from surface weathering to episodic accretion and formation.)

The spin-down clock makes deposition radius a birthdate: a cohort at radius R was cemented when the dam stood there, at the epoch when $\Omega(t) = (30\,\mathrm{AU}/R)^2$ in primordial units. This yields a sharp prediction. The youngest possible Kuiper belt objects date to the last wet episode before Local Bubble entry, and because the dam has moved less than 1% over the past ~ 100 Myr, *any recently-formed cohort must lie within a few AU of the present heliopause radius* ($\sim$ 105–121 AU): dynamically cold (born too recently to be stirred) and spectrally fresh (surfaces younger than the space-weathering timescale, in contrast to every known trans-Neptunian population). The clock carries one stated exception, and the exception is itself dated. A sufficiently dense interstellar cloud drives the dam far *inward* for the duration of the transit (the ram pressure of the Sun's ~ 25 km/s motion through the cloud dominating the balance), and a production flash at the compressed stations deposits a thin cohort of small bodies at radii the baseline clock forbids. Independent evidence indicates exactly one recent such event: the live ^{60}Fe and ^{244}Pu layers in deep-sea archives, which Opher et al. (2024) attribute to a cold-cloud passage $\sim$ 2–3 Myr ago that compressed the heliosphere drastically. The framework therefore predicts a sparse population of *young, volatile-fresh, kilometre-class bodies at* $\sim$ 40–150 *AU carrying live* ^{60}*Fe* (half-life 2.6 Myr : still $\sim$50% extant): self-labelling, lab-verifiable, and tied to an independently dated event. The two-sided test is then clean: a fresh-surfaced cohort whose location and radionuclide stamp match a dated transit *confirms* the excursion clause, while a fresh-surfaced population at small radii with *no* corresponding transit record falsifies the clock outright. (Distant comets active beyond conventional sublimation radii, usually attributed to supervolatile ices, are noted as candidate transit-cohort members without being claimed as such.)

The firehose budget.—The primordial-mass convention adopted in §2 fixes the factory's total throughput with no new assumptions: Sol shed $\sim 0.14\,\mathrm{M}_\odot$ across the T Tauri epoch, and every gram of it crossed the Davis Dam. At disc metallicity that is $\sim 660\,M_\oplus$ of solids delivered *through* the dam wall over ~ 10 Myr: of order $65\,M_\oplus$ per Myr at peak. The belt's assembly then requires only that a fractional residue stall and stick at the pressure boundary: a capture efficiency of a few $\times 10^{-3}$ supplies the $\sim$ 0.1–$1\,M_\oplus$ exterior pool the streaming-instability reading needs (§8.1), while even the classical massive-primordial-belt requirement (10–30 $M_\oplus$) would demand only $\sim$4%. The budget is overdetermined in the framework's favour: the factory was supply-rich by two to three orders of magnitude, and the belt's final mass is set by the capture physics at the wall, not by the availability of material. The same convention names the epochs: the *firehose* era (gas-age, $\sim 65\,M_\oplus$/Myr of solids at the wall) built the main belt; everything since is the *trickle*.

Production-rate budget.—The belt's census constrains the shape of the production history. The main belt's $\sim 10^5$ objects larger than 100 km overwhelmingly date to the gas era, when the factory ran at rates of order 10^4 objects per Myr; the post-gas factory is supply-limited and episodic, delivering clutches of order 10^2–10^4 objects per wet episode (streaming instability collapses clumps in bursts, not singly) across a few tens of episodes over solar history. Averaged, that is roughly one new Kuiper belt object per Myr-scale interval: but the delivery must be bursty rather than continuous:

a steady drip would smear deposits along the dam's entire outward track, and the observed emptiness of the 55–70 AU gap (Fraser et al. 2024) is itself direct evidence of the episodic duty cycle.

Compositional variance structure.—Cohort deposition predicts a distinctive variance pattern: *low scatter within a cohort, high scatter between cohorts.* The gas-era population formed from a single well-mixed reservoir over ~ 10 Myr and should be internally consistent in composition, isotopic ratios, and age: as the cold classicals are, constituting the most homogeneous (uniformly very red) colour group in the trans-Neptunian region (Tegler & Romanishin 2000). The same survey's colour *bimodality*, a very-red class and a neutral class, maps onto the framework's two source populations: dam-born cohort material versus milled inner-disc debris mixed into the dynamically hot population. Each later ring instead samples whatever interstellar cloud powered its wet episode, so successive rings should differ from one another (in ice inventory, dust mineralogy, isotopic vintage, and surface age) while remaining uniform internally. The vintage prediction has a thermometer: deuterium enrichment in ice is set by low-temperature grain-surface chemistry (Tielens 1983), and successive dam stations sit at progressively colder radii, so the cohort ladder predicts a *monotonic D/H gradient* with formation radius across the rings: a measurable ordering, not merely a qualitative variance claim. The dynamically hot population, in this picture a mixture of milled dam-born bodies and inner-disc-born swarm debris scattered outward (§4.1, §5.6), should by contrast be compositionally bimodal at any radius. Radius, in short, predicts composition class: discrete rings, discrete vintages.

The age inversion.—This reading inverts the classical understanding of the trans-Neptunian region. Classically, Kuiper belt objects and comets are the Solar System's *oldest* accessible material: pristine relics of the solar nebula, the rationale for cometary sample missions. In this framework the belt and cloud are by definition its *youngest* components: the only parts of the Solar System still (episodically) under construction, with assembly ages spanning from the gas era down to the last wet episode before Local Bubble entry. The planets finished forming within ~ 10 Myr; the belt never stopped. Later cohorts are built from interstellar matter that postdates the Sun: including condensates from stars that died *after* the Solar System formed: *post-solar grains*, the inverse of the presolar grains sought in cometary samples. This yields the framework's most direct sample-return test: a radiometric or isotopic assembly age younger than 4.56 Gyr in any trans-Neptunian solid is inexplicable classically and expected here, with the youngest material residing in the outermost rings.

Existing cometary samples do not yet test this (every comet sampled or closely visited is a Jupiter-family comet, drained from the *old* hot population) but they already test the mixture prediction: comet 81P/Wild 2's classically-surprising content of inner-solar-system igneous material, including high-temperature crystalline silicates and a CAI-like refractory inclusion, alongside an anomalously low presolar-grain abundance (Brownlee et al. 2006) is the expected positive signature of inner-disc-born swarm debris rather than of pristine outer-nebula ice: the refractory content is diagnostic of inner-disc origin, not merely consistent with it. We note a classification bias: anomalous-isotope stardust in samples is classified as *pre*solar by assumption, so post-solar grains would be misfiled; cometary grain inventories have not been screened with the age direction treated as open. Encounters with dynamically-new Oort-cloud comets (e.g., ESA's *Comet Interceptor*) are the first opportunity to sample populations where young material is predicted.

8.4. *The product-mass law, the firstborn, and the size-clock*

The dam is one production line at every scale; what changes between systems is the surface density queued at the boundary, and the product mass follows it quadratically:

$$\Sigma_{\rm dam} = \frac{f_{\rm disc}\,M_*}{R_{\rm disc}^2}, \qquad m_{\rm ext} = m_{\rm Sol}\left(\frac{\Sigma_{\rm dam}}{\Sigma_{\rm Sol}}\right)^2, \tag{35}$$

with $\Sigma_{\rm Sol} = 4.36\,M_\oplus/{\rm AU}^2$. The quadratic is what streaming-instability collapse at a trap supplies (clump mass tracking the square of the local solid loading), and the law spans six decades in the catalogue: Sol's dam mints Pluto-class bodies, while the densest fitted dam (Upsilon Andromedae, $\Sigma \sim 5\times10^3$) mints brown-dwarf-grade products.

The firstborn anchor.—The calibration constant $m_{\rm Sol}$ is pinned by a body this paper has so far carried as an oddity: Triton. The factory's first product is minted while the dam still stands at $R_{\rm disc}$ – *co-orbital with the dam-keeper*, at maximal capture cross-section. A keeper seated on the dam should therefore hold a captured companion of exactly the at-dam product mass, and capture from an adjacent, slowly diverging orbit is precisely the geometry that yields an otherwise regular orbit with reversed sense. Triton is that receipt: retrograde, near-circular, 3.59 milli-$M_\oplus$, and – uniquely among the large trans-Neptunian bodies – carrying zero displacement ($\Delta a = 0.00$: it is still *at* the 30.07 AU gate, because Neptune holds it there). The standard capture account (exchange capture from a binary; Agnor & Hamilton 2006) explains the retrograde sense but leaves the mass coincidence unaddressed; here the mass *is* the calibration: $m_{\rm Sol} = 3.59\times10^{-3}\,M_\oplus$, the firstborn, physically pinned at the dam face.

Size is the clock.—Along the dam's retreat $\Sigma \propto R^{-2}$, so the product mass falls as $m \propto R^{-4}$, and the law inverts: an exterior body's *observed mass* names the stance where it was minted – its original AU – regardless of where scattering later put it. Current orbital radius is *not* an indicator for these bodies; mass is vintage. The dam's chronology has three gears: parked at $R_{\rm disc}$ through the gas era ($t_{\rm disc} = 5.1$ Myr for Sol); the firehose sweep, $R_{\rm disc} \to 1.6\,R_{\rm disc}$ over ~ 3 Myr as the gas-era supply dies; then the long retreat, $R \propto t^{0.138}$, calibrated on a single point – the heliopause at 121 AU after 4,570 Myr (§8.3). Reading Sol's dwarf planets through Eq. 35 then dates the entire population: Triton (3.59 m$M_\oplus$, firehose onset, at the gate); Eris (2.78, born 32.1 AU, ~ 5.4 Myr); Pluto (2.18, born 34.1 AU, ~ 5.8 Myr); Haumea (0.67, 45.7 AU, ~ 7.7 Myr); Makemake (0.52, 48.8 AU, ~ 9 Myr); then the retreat-era cohort – Gonggong (~ 25 Myr), Quaoar (~ 50), Sedna (~ 67, born 64.5 AU and since thrown to 506), Orcus (~ 159), Salacia (~ 253 Myr). The assembly line runs monotonically: heavier is older, with no exceptions in the catalogued population.

The Pluto entry carries an independent check. The size-clock, using nothing but Pluto's mass, places its minting at 34.1 AU – and the classical dynamical reconstruction of the 3:2 resonance capture (Malhotra 1993, 1995) independently requires Pluto to have originated near ~ 33 AU before being carried out to 39.5. Two methods with no shared inputs – a mass-to-density inversion and a resonance-sweeping integration – recover the same origin radius.

The survival law.—Whether a product survives where it was minted depends on one body: the dam-keeper. Its chaotic reach extends $\sim 11\,R_H$ (Neptune: 8.2 AU beyond the dam, $1.27\,R_{\rm disc}$), and each minting stance is therefore SWEPT (inside the reach: survival ~ 0, parking-lot resonance capture excepted), EDGE (within $\sim 10\%$ of the boundary: resonant islands, $\sim 10\%$ survival – the plutinos), or SAFE (beyond: grinding aside, survival of order unity – the cold classicals). The hot/cold dichotomy of §8.2 is this law read radially.

The census and the missing siblings.—The stock available per stance is supply-proportional, $\sim 5 \times 10^{-6} f_{\rm disc}\,M_*$ ($\sim 0.02\,M_\oplus$ per stance for Sol), so dividing by the stance's product mass counts the characteristic-mass members minted there. The count is an audit, vintage bin by vintage bin, and the onset bin fails it informatively: ~ 6 Triton-grade products expected, 3 catalogued (Triton, Eris, Pluto). *Three Pluto-class siblings are predicted undiscovered* – scattered-class objects, high inclination, currently far from perihelion, with search range $a \approx 30$–271 AU and $q \gtrsim 30$ AU. The later bins predict larger cohorts of smaller bodies (~ 27 Haumea-grade, ~ 95 Quaoar-grade, climbing into the hundreds at the outer stations), consistent with the steeply incomplete census beyond 60 AU. These are falsifiable in bulk by LSST-era surveys: the size-clock requires the discovered population to keep its mass-vintage monotonicity as it grows.

The whole apparatus – product law, firstborn rule, size-clock, survival law, census – uses only $(M_*, f_{\rm disc}, \Omega)$ and the keeper's mass, so it runs on every fitted system in the catalogue. The firstborn rule in particular becomes a binary forecast: a dam-keeper found in situ should hold a captured retrograde Triton-analog of the at-dam product mass; a keeper displaced from its dam should have lost it. Sol passes on Neptune–Triton; the catalogue's displaced keepers supply the contrapositive tests.

8.5. *The gravitational boundary: the Oort cloud*

The second boundary is gravitational: the Sun's Hill radius in the galactic potential, $R_{H,\odot} \approx 1.3$ pc $\approx 2.7 \times 10^5$ AU, with long-term-stable orbits confined (as for any satellite system) to roughly half that: matching the outer Oort cloud's observed extent of $\sim 10^5$ AU. This boundary is also epoch-dependent: in the birth cluster the tidal radius was only $\sim 2 \times 10^4$ AU, and the inner Oort cloud and Sednoid region plausibly fossilize that cluster-era station (Brasser et al. 2006), while exchange within the cluster commons stocked the cloud with sibling debris: indeed Levison et al. (2010) conclude the Sun *captured* a large fraction of its Oort cloud from other cluster members, which predicts a compositional split the framework inherits gladly: long-period comets (partly sibling material) should be isotopically heterogeneous relative to the native, dam-pooled Kuiper cohorts. The Kuiper belt and Oort cloud are thus the same phenomenon at the two boundaries: drag pools in the disc plane at the magnetic boundary (a flat belt); gravity pools in three dimensions at the tidal boundary, isotropized by galactic tides (a sphere). The flat-versus-spherical dichotomy, explained in standard theory by two unrelated histories, follows here from the holding mechanism alone.

Table 8. The Solar System's boundary census. Every boundary the system has had either persists today or left a fossil population at its old station.

Boundary	Physics	Primordial	Today
Alfvén Dam	magnetic (stellar)	0.20 AU	$\sim$0.04 AU ($\sim$9 $R_\odot$)
Davis Dam	pressure (interstellar)	30.07 AU	$\sim$121 AU (heliopause)
Resonance brackets	drag fossil	39.4–47.7 AU	frozen
Cluster tidal radius	gravitational	$\sim 2 \times 10^4$ AU	inner Oort fossil
Galactic Hill radius	gravitational	,	2.7×10^5 AU

8.6. *The desert, and exterior slots*

The two-boundary census makes a hard negative prediction: neither boundary ever occupied $\sim$ 150–2,000 AU (the dam will reach only $\sim$ 160–180 AU by the end of the main sequence; the cluster-era tidal radius never compressed below a few thousand AU), so *no in-situ population should exist in that range*: only high-eccentricity lifted bodies passing through, consistent with the known Sednoids (Sedna, $e = 0.84$).

8.7. *The factory class*

The full Minor Planet Center census (6,034 trans-Neptunian objects) shows the dam's exterior carries no standing geometry of its own: structure outside the dam is *temporal*. The interior, a held dissipative disc, writes space-structure: slots. The exterior has one organizing element only, the boundary itself, the lattice's final zonal flow: a single moving shoreline whose deposits record *when* it stood at each radius

(§8.3), post-processed by the nearest giant's resonances. And the shoreline is not a solar peculiarity. It is the local instance of a factory class operating at every wind–medium interface, and every component of that class is already in the literature under its own name: observed most directly at the galactic scale, where external systems' boundaries can be imaged.

At stellar scale the boundary and its stocked shelf are directly observed: the hydrogen wall at our heliopause and the equivalent *astrospheres* of nearby stars (Wood 2004). At galactic scale the boundary is named: the *galactopause*, where Shull & Moss (2020) show that during strong outflows thermal instability precipitates kpc-scale clouds at the wind–IGM interface (30–65 kpc), which then lose pressure support and rain back inward on ~ 200 Myr timescales. Condensation at the pressure boundary, followed by inward delivery of the product: the same two factory steps as the dam's, eleven orders of magnitude up, with the product mass set in each case by the local capture physics, pebble pile-ups cement km-to-100-km ice bodies at a stellar dam; thermally unstable coronal gas condenses cloud-mass objects at a galactic one, which fall inward as the disc's fuel. Star formation at flow boundaries is likewise established componentwise: sequential triggering at shock fronts (Elmegreen & Lada 1977), star formation *within* outflows predicted as positive feedback and then observed (Maiolino et al. 2017; Gallagher et al. 2019), young clusters manufactured at ram-pressure interfaces, and globular clusters condensing from the protogalactic boundary medium (Fall & Rees 1985). The product differs (belts at one scale, disc fuel at the other); the mechanism is identical.

What has not previously been assembled is the identification: heliopause, astrospheres, and galactopause as *one class of production boundary whose output is dated cohorts*. Each boundary's deposit sequence is a time log, radius (or infall phase) maps to epoch through the boundary's own motion, so every boundary population records its system's environmental history, and the records are mutually calibrating: the Solar System supplies the resolved, sample-accessible instance (cohort vintages, the D/H ladder, the transit-cohort radionuclide stamp above), while external systems supply the statistics. At the galactic tier the same clock logic predicts age–radius *stratigraphy* in nuclear stellar populations rather than spatial lattices; Section 9 develops this against the Galactic Center's measured, strongly episodic star-formation history.

The deep desert prediction of the two-boundary census stands as stated: a Rubin-class survey finding a substantial low-eccentricity in-situ population at 150–2,000 AU would falsify the census; sparse late-cohort material near the dam's historical stations ($\lesssim 121$ AU) is expected; the deep desert is not populated in situ.

8.8. *Universality*

Every magnetized accreting star has a dam, and every star has a galactic Hill radius; the framework therefore predicts that exo-Kuiper belts are universal, located at $R \propto M_\star\,\Omega_{\rm prim}^{-1/2}$, with comet clouds filling each star's Hill sphere to its stability edge. Cold debris belts are indeed detected around $\gtrsim 20\%$ of FGK stars (sensitivity-limited, so the true fraction is higher), as rings at tens-of-AU radii, and their measured radii already show a systematic increase with host luminosity (Matrà et al. 2018). Regressing the resolved-belt sample against the dam scaling is an archival test requiring no new observations: if it holds, every resolved debris ring is a measurement of its host's primordial rotation.

β Pictoris already provides a sharper, system-level instance. Fitting the cascade to its two imaged giants alone yields $R_{\rm disc} = 28.4$ AU (spin 2.64), with b and c as inward migrants from slots 1 and 0 and the inner cascade (slots 2–7, $\sim 700\,M_\oplus$ of predicted inventory) shredded by their passage: supplying the archetypal debris disc that in-situ readings leave unexplained. The fit is then checked out-of-sample by the system's resolved planetesimal belts: mid-infrared imaging locates dust-replenishing rings at $\sim$ 6, 16, and 30 AU (Okamoto et al. 2004), coincident with cascade slots 3, 1, and 0 (5.6, 16.5, 28.4 AU) to within 3–6%, three independently observed belts on three slots of a ladder fitted only to the planet positions, reading naturally as collisional debris at the giants' formation slots and at the destroyed slot-3 body.

HR 8799 is a second instance of the same mechanic: its four wide super-Jupiters leave every inner slot obliterated by simultaneous scattering in the fit, and the system indeed hosts a warm inner dust belt while no inner planets are known. Two of two wide multi-super-Jupiter systems in the catalogue thus carry shredded inner cascades, generalizing to a population prediction with a mass threshold: outer giants near Jupiter's mass *preserve* their inner systems, consistent with the observed positive correlation between cold Jupiters and inner super-Earths (Zhu & Wu 2018), while multiple wide super-Jupiters *destroy* them, leaving inner-planet deserts filled with multi-ring debris at slot radii. The known excess of directly-imaged giant planets around debris-disc hosts (Meshkat et al. 2017) reads here not as a shared massive-disc origin but as cause and wreckage; the discriminators are the dust's location and structure (interior to the giants, slot-laddered) and the absence of surviving inner planets specifically above the super-Jupiter threshold.

8.9. *Born-hot chains: the packing index at formation*

The cascade fixes where bodies form and how massive they are; Hill stability (Eq. 24) then determines whether the configuration can persist. For each adjacent pair of slots define the *packing index* $P = \Delta r / R_{H,\rm mut}$, the slot separation in units of the pair's mutual Hill radius, evaluated at the formation positions, with destroyed slots carried at their predicted masses (stability acts on the at-birth configuration, not the surviving one). Numerical integrations of multi-planet chains require $P \gtrsim 7$ for long-term stability (Smith & Lissauer 2009); a system fitted below that limit was born dynamically hot and *had* to relax.

The fitted catalogue splits cleanly at the boundary. Quiet systems sit far above it (Sol's terrestrials at $P = 41$–56, TRAPPIST-1 at 8–10 on its half-step lattice (§2), the Galilean satellites at 15–18) and none carries scattering, migration, or remnant signatures. Sol's one marginal pair is Jupiter–Saturn at $P = 7.2$, grazing the limit: precisely the pair implicated in the framework's single dynamical event (§5.6). Every system fitted below the boundary shows independent evidence of violence, and the outcomes sort into recognizable fates. *Resonance lock*: adjacent slots sit at a period ratio $\rho^{-3/2} = 2.24$, just wide of 2:1, so modest convergent drift captures a packed chain into the neighbouring integer commensurability rather than chaos. We emphasize the distinction: 2.24 is the *birth* ratio, and the observed integer locks are the *relaxed* states, GJ 876 ($P = 3.8$) settled into the canonical Laplace 4:2:1 chain (Rivera et al. 2010), and HR 8799's four giants ($P = 3.7$) into a proposed 8:4:2:1 protection (Goździewski & Migaszewski 2014). The cascade explains why capture lands specifically in first-order 2:1 chains, the resonances convergent disc migration also selects (Pierens & Raymond 2011): the ladder is born adjacent to them. The Luger lattice of the inverted regime (§2) supplies the second observed family by the same logic, $\rho^{-3/4} = 1.497$ sits on 3:2 essentially at birth, which is why compact chains like TRAPPIST-1's barely needed to move to lock (Luger et al. 2017), and why its observed pairs sit within $\sim 1\%$ of exact commensurability while the giant chains had to converge 11% to reach theirs. An occupancy prediction follows directly: HD 7924 (the catalogue's deepest inversion ($C = 241$), whose known planets occupy lattice sites while the ladder's entire lower half (six rungs, $\sim 10\,M_\oplus$ of allocation) formed inside the Hayashi photosphere and was consumed) carries vacant full rungs at 2.5–$6.8\,M_\oplus$ (0.02–0.09 AU; radial-velocity semi-amplitudes 1–2.5 $\mathrm{m\,s^{-1}}$): either detectable with modest precision gains, or absorbed in mergers like its slot 2, which the fit already credits to planet b. The HR 8799 lock model carries a bonus that doubles as a slot test: Goździewski & Migaszewski (2014) predict a fifth, innermost planet (HR 8799f) at ~ 7.5–9.7 AU to extend the chain, and the cascade fitted to the four known giants independently places its next rung, slot 4, at 7.9 AU, inside their predicted window. Two unrelated formalisms, a resonance-dynamics fit and a geometric ladder, point at the same empty orbit. *Outside-in shredding*: β Pictoris ($P = 3.5$) and HR 8799, whose packed outer giants destroyed their inner cascades (above). *Inside-out wrecking*: HD 134987 ($P = 3.3$ at the fitted parameterization) places c in situ at the dam edge (5.8 AU) and b as an inward migrant from slot 2 ($2.0 \rightarrow 0.81$ AU), with slot 1's occupant, the chain's dominant member, now absent despite decades of radial-velocity coverage: ejected during the exchange. The survivor is a lone warm Jupiter over a swept interior, parked where magnetospheric-cavity simulations show inward-migrating giants halt and accumulate (Romanova & Lovelace 2006): the configuration whose population-level signature is the well-documented loneliness of close-in giants

(Steffen et al. 2012), whose differential form (warm Jupiters and hot Neptunes retain companions; hot Jupiters specifically do not) maps the launch-energy threshold of the wrecking fate. 55 Cancri repeats the pattern in compressed form ($P = 5.1$; eleven slots inside 1.3 AU): b swept inward from slot 2 ($0.46 \to 0.11$ AU), capturing c near the observed 3:1 commensurability on the way past, while d was driven outward ($1.3 \to 5.5$ AU) and the innermost survivor e was ground to a bare core. Notably, wreckage *visibility* is age-gated: β Pic (~ 20 Myr) and HR 8799 ($\sim$ 30–40 Myr) still display their debris, while ancient wrecks like 55 Cancri (~ 10 Gyr) retain only dynamical scars , consistent with the absence of any confirmed debris disc there after gigayears of collisional grinding and drag removal. *Marginal rearrangement*: Mu Arae's three giants pack at $P = 6.0$–6.1, just under the limit, and show precisely a marginal outcome, the lightest giant displaced inward 2.1 AU, and a $10.5\,M_\oplus$ body parked deep inside the Alfvén void, reading as the 5–10% Martian-band survivor of a 105–210 $M_\oplus$ casualty of the relaxation.

Across every packed chain the *lightest* member absorbs the relaxation, and the rule continues unbroken to stellar masses. Alpha Centauri B and Proxima formed at adjacent slots 10.4 AU apart with a mutual Hill radius of 13 AU: $P = 0.8$, overlapping Hill spheres at birth (the Hill formalism is only indicative at stellar mass ratios, but an index below unity needs no precision). Coexistence as born was impossible; the $7.5\times$ lighter Proxima took the kick, and its wide, bound, plane-aligned orbit (Kervella et al. 2017) is the relaxation product (§6.1). GJ 667 repeats the pattern at $P = 0.7$ with the kick rule executed exactly: B formed at the dam edge (slot 0, 21.6 AU) and settled inward 9 AU to the observed 12.6 AU A–B separation, while C, the runt of the triple, was flung from slot 1 to its observed ~ 230 AU. The same fit predicts a $0.46\,M_\odot$ K-class sibling at slot 2 and a $17.5\,M_{\rm J}$ brown dwarf at slot 3, mutually evicted and ejected outright, GJ 667 was born a quadruple plus a brown dwarf and fired two runaways. Packed-chain ejecta (stars, brown dwarfs, giants like HD 134987's, and planetesimal debris) are thus a standing output of cascade formation, stocking the interstellar inventory that the exterior factories of §8 later sample.

8.10. *Gravitational stripping: encounter-edited cascades*

Two catalogue systems carry violence their own packing indices forbid. HD 20794's chain packs at $P = 14$–22 with no giant anywhere, yet its ledger shows a destroyed dam-edge body, a half-stripped outer member, and inner planets carrying mass surpluses an order of magnitude beyond Sol-style late delivery; Proxima Centauri's planets defeat every unperturbed anchor family outright. Born-quiet systems do not wreck themselves: the editing was external, and for Proxima the framework independently names the editor (§6.1). We therefore promote encounter editing from forensic narrative to a fitted mechanism. A flagged system gains one parameter (the encounter periapsis q, with the perturber mass $M_{\rm pert}$ supplied ($0.91\,M_\odot$ for Proxima's named assailant; $0.5\,M_\odot$ for an anonymous cluster-era flyby)) from which the *stripping fac-*

tor follows as the tidal truncation radius $r_t = 0.28\, q\, (M_{\rm pert}/M_\star)^{-0.32}$ (Breslau et al. 2014). The transform edits the baseline allocation in three zones: slots exterior to r_t are *destroyed* (predicted planet mass zero; inventory freed), the band $0.5\, r_t$–r_t is *stirred* (graduated loss, up to 60% at r_t), and the interior is *enriched*, 15% of the freed inventory rains inward and is swept up outermost-first. q is then bisected against the mean per-planet $|\log(\mathrm{pred}/\mathrm{obs})|$: precisely the quantity that consensus mass-closure cannot see, and the quantity encounter editing exists to close.

Both systems close. HD 20794 fits at $q = 15.8$ AU ($r_t = 5.2$ AU): the dam-edge slot is destroyed, g sits in the stirred zone at −23%, f's raw +310% surplus collapses to −10% (the late-delivery band) once the inward debris is credited, and b ($2.7\, M_\oplus$ parked beneath the Alfvén dam) is the *encounter fragment*: 8% of the $34\, M_\oplus$ freed inventory, the inward share of the same 5%/95% asymmetric split that delivers Theia and the Mars survivor in Sol (§5.6), here operating on an encounter's spoils rather than a swarm's. Proxima fits at $q = 19.7$ AU, inside the energetically allowed window (§6.1) – and with its disc geometry *inherited* from A's fit rather than bisected (§6.1): d, b, and c close at +0.6%, −0.0%, and −0.0% respectively, the cleanest per-planet closure in the catalogue, from the system that was previously unfittable at any compression. The two fits agree on the transport physics: the inward-capture fractions implied independently are 10–15% (Proxima) and $\sim$ 18% (HD 20794), bracketing the calibrated 15%. One parameter, physically derived from stellar mass and periapsis, closes the framework's two previously resistant systems.

8.11. *The fitting laws: a planet must justify its existence*

The catalogue's fits are not curve-fits; they are adjudicated by a set of rejection laws that grew out of the catalogue's own failure modes, each one a physical statement about what a formation history may and may not contain. We state them here because the two systems that follow – Upsilon Andromedae and the resolved Alpha Centauri seating of §6.1 – were cracked by these laws, not by parameter search.

Existence justification, at two epochs.—A fit is rejected if any observed planet would be scattered or shredded at its observed mass and orbit by the fit's own actors – and the test runs at *both* epochs, formation and arrival, because a fit that destabilises a planet with a migrant's final position is as dead as one that never formed it. (An early 47 UMa solution seated a brown dwarf whose chaotic zone enclosed both observed giants at their formation seats; a later HD 142 solution passed at formation and failed at arrival. Both are now rejected automatically.) The fit must leave every planet formable where it formed and stable where it sits.

Void rejection.—Unfilled interior slots that the fit's own scattering cannot account for are rejection-grade, not cosmetic: the slot count must equal the observed census plus the event ledger.

The overlord rule.—No fit may conjure an unobserved body more massive than every observed member. A hypothetical that outweighs everything actually seen explains nothing; it merely hides the discrepancy in an invisible giant. (This rule alone resolved Alpha Centauri: the seating that survives it is B on the dam seat and Proxima at slot 1, both at 10^{-5}-level closure.)

The mass-supply gate.—A seat must be able to *feed* its occupant: predicted allocation (plus any devoured credit) must reach within a factor 2.5 of the observed mass, with encounter-stripped bodies priced at their iron-fraction floor.

The devoured-mass ledger, with table manners.—A migrant that grew by consuming interior slots must justify its *formation* mass – observed minus meals – at its formation seat, with each meal priced through the retention line (Eq. 27): the devourings were not intact. Victims must not outweigh their eater, and no eater's meals may exceed 60% of its observed mass. (HD 134987 b fits at its formation mass of 395 $M_\oplus$, not its post-meal 505; the difference is two named victims, retention-priced.)

Event closures count as exact.—A deviation fully explained by an identified, budget-closed event (an impact dated by the truncation clock, a devouring with its ledger entries, an eviction with its energy bill paid) is model output, not error: the fit is scored as exact there. Deviation without a receipt remains error.

Under these laws the catalogue is self-governing: all of its systems re-fit from observables alone with no per-system exemptions, and the laws transfer because none of them mentions Sol.

8.12. *Upsilon Andromedae: the eviction ratchet*

Upsilon Andromedae has been a standing dynamical puzzle: three giants whose c–d pair carries a 30° mutual inclination, a hot Jupiter (b) that matches no formation story at its 0.059 AU perch, and an M-dwarf companion (ups And B) at ~ 750 AU. The fit that survives the laws above is a nine-body system, and it identifies the companion as a cascade product.

B was minted on the dam seat.—The energy ledger is the discriminator. Seated at slot 0 (8.99 AU, the dam seat, where the pile-up supplies what no interior seat can), B's predicted mass closes on its observed 66,589 $M_\oplus$ at 0.04% – the second cascade-sourced stellar companion in the catalogue, after Alpha Centauri B. One rung in, the seat supplies only 29,600: the dam seat is the only address that can pay for B.

The ratchet.—B did not leave in one kick. The system's dam ran the catalogue's densest factory ($\Sigma_{\rm dam} \approx 5{,}450$, six hundred times Sol's), minting *giant-grade* exterior products ($m_{\rm ext} \sim 3{,}400\,M_\oplus$ by the product-mass law of §8.4) with a throughput budget of $\sim 74{,}000\,M_\oplus$ – about twenty product-units. B's 11 R_H chaos zone engulfs every exterior rung, so each product was destabilised on minting and relayed *down*, a sunk star-grazer, with B stepping outward on every exchange. The arithmetic closes:

B's full climb costs $\sim$ 196,000 $M_\oplus\sqrt{\mathrm{AU}}$; each sunk product pays $\sim$ 13,000; fifteen installments needed against twenty produced: repeated product ejections over Myr displaced the keeper from its own dam.

The receipts.—B's orbit is peri-anchored – periapsis 17.7 AU, twice the dam seat, planted over the rungs it harvested – with a ratcheted apoapsis near 1,500 AU: the signature of repeated kicks at fixed periapsis, not one ejection. The repeated passes shear the outer slots thoroughly (the 30° c–d inclination and the abrupt edge at e are the stirring receipts), and radial velocity excludes any surviving giant inside 25 AU: the cleanup was total, as fifteen installments require. The fifteenth receipt is positional: a corpse seat at 15.39 AU, which is 8.99×1.713 – exterior rung 1 exactly – the last installment, sunk.

ups And b is factory product, not resident.—The hot Jupiter owns no interior seat: its 218 $M_\oplus$ matches no allocation, but it sits precisely on the exterior product ladder ($m \propto R^{-4}$ along the retreat: 3,413 at the dam, $\sim$ 396 at rung 1), at a minting stance of $\sim 1.28\,R_{\mathrm{disc}}$. It is a late product, destabilised by B like its fifteen sunk siblings and relayed down – but caught at the Alfvén parking lot, the terminal trap, instead of burning. The system's one hot Jupiter is the one installment that did not sink.

The forecast.—The same passes that stirred A's outer slots sheared B's own natal disc, exactly as the A–B encounter sheared Proxima's (the calibration analog). The encounter law gives B a truncation radius $r_t = 2.52$ AU: any survivors compact inside $\sim$ 1.3 AU, anything at 1.3–2.5 AU eccentricity-stirred, nothing beyond 2.5 AU, no exterior cohort. B is unsurveyed, so a clean null is currently consistent; five icy seats at 0.20–1.70 AU (0.2–1.1 $M_\oplus$ at $f_{\mathrm{disc}} = 0.01$) are the live prediction, with $\sim$ 0.2% transit depths.

8.13. *GJ 667: the triple that runs on one disc*

The treatment that resolved Alpha Centauri and Proxima applies wholesale to the next nearby triple, and GJ 667 extends it in one important way: *both* of A's companions carry inherited discs, one at each exponent of the hierarchy law.

The stellar tier.—A's cascade ($\Omega = 3.04$, dam at 12.6 AU) seats both companions at the 10^{-6} level: C – the lighter – minted *on the dam seat* (108,881 predicted vs. observed) and took the big ejection to its observed $\sim$ 230 AU; B at slot 1 (229,748, exact) rose +5.25 AU into the vacated dam radius – today's A–B binary ($a = 12.6$, $e = 0.58$) orbits *at* the old dam. The mass-ordered kick rule executes exactly as in Alpha Centauri: the lightweight takes the ride, the heavyweight the step. The interior paid: slots 2–7 (a $\sim$ 5,000 $M_\oplus$ brown dwarf and a four-giant ladder) were obliterated in the relaxation, so A is predicted planetless – the GJ 667 system was born a quadruple-plus and fired its middle children into the field.

Two inheritances, one disc.—C inherits at $x = 1$ ($D_C = 0.39\,D_A = 2.06$); B inherits at $x = \rho$ ($D_B = 0.39\,\rho^{-2.12} D_A = 6.46$). Every disc quantity of both daughter systems – ladders, rims, snow lines – follows with no fitted parameter, and the formation-era geometry supplies the tides: B sat *inside* C's dam seat at a separation of 5.25 AU, giving C a standing sibling tide ($r_t = 1.16$ AU, Holman–Wiegert cross-check 0.95 AU), while B's system rides the present binary's $q = 5.29$ AU tide ($r_t = 1.46$, $a_{\rm crit} = 1.18$).

C adjudicates its own controversy.—GJ 667 C's claimed six-planet system has been disputed for a decade (activity contamination; Anglada-Escudé et al. 2013; Robertson & Mahadevan 2014). The inherited geometry rules on each signal with no dial to turn. The nine-seat ladder, truncated at 1.16 AU, leaves four safe seats (0.52, 0.31, 0.18, 0.10 AU) allocating 0.03–0.6 $M_\oplus$, plus the Alfvén parking lot below 0.086 AU. The verdict inverts the naive reading, and the discriminating witness is A's *factory*. A's dam ran dense ($\Sigma_{\rm dam} = 439$, a hundred times Sol's), so the product-mass law (§8.4) mints 36.4 $M_\oplus$ at the dam face – and the size-clock prices the two *robust* detections exactly: b ($m \sin i = 5.6$) is the product of stance 1.60 $R_{\rm disc}$ – *the firehose-cliff product*, vintage $\sim$ 24 Myr – and c (3.8) the early-retreat product of stance 1.76 ($\sim$ 49 Myr), both priced by A's fitted Σ with zero C-system parameters and no inclination correction. C, the formation-era keeper co-orbital at the gate, captured them – the Neptune–Triton grammar at planet mass. Sequential capture from a retreating dam imposes a hard ordering law: products mint lighter and farther out as the dam withdraws ($m \propto R^{-4}$), so a captured chain must be *mass-descending and vintage-ascending with radius*. The claimed roster splits cleanly against it. b $\to$ c $\to$ f is a perfect chain ($5.6 \to 3.8 \to 2.7\,M_\oplus$; $24 \to 49 \to 90$ Myr). d violates it outright (5.1 $M_\oplus$ at 0.276 AU, heavier than both interior neighbours, its 28-Myr vintage demanding a berth between b and c): artifact – independently the literature's rotation-alias suspect (Robertson & Mahadevan 2014). g violates identically (4.6 at 0.549 AU, vintage 34 Myr): artifact. And e claims a mass *identical* to f's (2.7 $M_\oplus$ – identical stance, identical vintage): the two are mutual aliases of one real product. The surviving chain also dates the kick twice over: C's eviction lies in the 90–391 Myr window – after the youngest captured product, before the first uncaptured cohort (minted into B's 90-AU chaos reach and swept). Two consequences follow. First, GJ 667 C c, long advertised as a habitable-zone super-Earth, is a *captured icy factory product residing in the habitable zone* (0.084–0.160 AU at $L = 0.0137$) – a water world by construction, decidable by bulk density the moment it transits or its true mass is measured. Second: GJ 667 C's actual planets have never been detected, and most no longer exist. Every catalogued body is an adopted factory product, RV-visible only because factory masses dwarf anything a metal-poor disc ([Fe/H] $= -0.55$) can allocate. The adoption also destroyed the natives: the settled chain's 11 R_H chaos zones engulf the native seats at 0.104 AU (removed by c, which now occupies the habitable zone in its place) and 0.178 (removed by f), and the captures' circularization sweep crossed everything inside $\sim$ 0.6 AU – $\sim$ 0.3 $M_\oplus$ of natives devoured into the chain, the rest ejected or sunk. The surviving natives are two or three light seats (0.4–1.1 $M_\oplus$ at 0.3–0.9 AU, $K = 11$–$18\ \mathrm{cm\,s^{-1}}$), invisible to current and next-generation instruments alike: the detected and the formed populations are disjoint. The capture grammar adds its signature: co-orbital capture's stable attractor is retrograde – it is why Triton circles Neptune backwards – and a captured chain must be uniform-sense (mixed senses collide), so the entire adopted system should orbit *retrograde* to C's spin ($P_{\mathrm{rot}} \approx 105$ d is known, so a Rossiter–McLaughlin measurement of $\lambda \approx 180°$ decides it at the first transit), with residual eccentricity increasing outward through the chain as tidal circularization weakens (checkable in archival radial velocities now). With the ice-rock bulk density and atmospheres that should match A's photosphere rather than C's formation budget, the captured reading carries four independent observational discriminators.

B's quiet forecast: a habitability window.—B's inherited ladder under the mutual tide keeps one safe seat: 0.12 $M_\oplus$ at 0.441 AU, inside B's habitable zone ($T_{\mathrm{eq}} \approx 289$ K), watered at 4–16 oceans by the icy conveyor seat at 1.29 AU – the Alpha Cen B slot-6 mechanism at one-sixth the mass. But the seat fails the audit's third axis, and the framework prices the failure from its own inputs: *the spin parameter is also the XUV dial.* The inherited $\Omega = 3.47$ holds a K dwarf in or near activity saturation ($L_X/L_{\mathrm{bol}} \sim 10^{-3}$) far longer than a Sol-spun star – the same nebula density that built the dense disc built the harsh star – and the seat sits at 0.441 AU, five times the dose of an Earth-distance orbit. At Mars's escape velocity ($5.2\ \mathrm{km\,s^{-1}}$) with an early dynamo death, escape runs at $\sim$ 5–10 oceans/Gyr against a conveyor drip of 2–8: the window is marginal at best – transient wet episodes when delivery spikes, never a standing ocean – and today, the system being kinematically old, a desert with extreme atmospheric D/H (the escape signature observed at Venus, here at Mars-scale gravity). The coupling passes its differential check: Alpha Cen B's inherited spin is 1.016, Sol-quiet, so its drowned slot-6 verdict stands untouched – one law separating the catalogue's two delivered habitable-zone seats in opposite directions. The audit is thereby a triad: *zone* (where the seat sits), *delivery* (what arrives, on what duty cycle), and *retention* (what the seat's gravity can keep). Earth scores all three; Alpha Cen B's seat over-scores delivery; this one fails retention on a finite timescale. B's radial-velocity silence is consistent: one spent Mars-class world ($K \sim 0.25\ \mathrm{m\,s^{-1}}$) and a dying conveyor.

Within 24 light-years the framework now carries two triples, and in both, a single fitted disc – the primary's – prices every member: seats, evictions, truncations, planet rosters, and water budgets; the daughter systems are derived, not fitted.

9. THE GALACTIC TIER

A galaxy forms the way a planetary system does, one rung up. This section states the generation sequence, identifies its fossils in the present-day Galaxy, and gives the tests that are running now. The law driving the sequence is the same inheritance pair measured at 0.9% three rungs down (§7.1); at this tier the pair already has names.

9.1. *The generation sequence*

A seed condenses in a collapsing protogalactic cloud. Its outflow against the ambient medium sets a pressure boundary, the galactopause, which is the Davis Dam of the seed's natal disc (§9.4), and the core is then constructed exactly the way Sol's planets were. While the collapse holds the dam deep, the natal disc runs the machine of §2 unchanged: cascade seats at the fixed ratio ρ beneath the dam, each allocating by the same rule, with only the products rescaled by local capture physics. At galactic seat allocations of 10^5–$10^6\,M_\odot$ no single body can hold together as a star, so the seats fill with direct-collapse objects and massive clusters where Sol's seats filled with planets. The interior seats are consumed by the seed, as in protostellar consumption of a compact system's inner slots: occupants dragged across the inner boundary contribute to seed growth. That inner boundary is now directly observed. Five years of deep ALMA imaging resolve a parsec-scale evacuated cone in the cold molecular gas around Sgr A*, carved and actively maintained by the seed's own wind, with the region's stellar winds too weak to have done the work (Gorski & Murchikova 2026), and horizon-scale polarimetry shows the dynamically dominant ordered fields of a magnetically arrested flow (EHT Collaboration 2024), a configuration whose accumulated flux halts infall exactly as a stellar magnetosphere truncates its disc. The inner dam is the same engine at every rung: rotation powers a magnetic rotor, the rotor launches the outflow, and the outflow delivers the boundary, its rim standing where wind pressure balances the ambient gas, far outside the machinery that pays for it, precisely as the heliopause stands far outside the corona that drives it. The surviving deposit of this deep-dam epoch is the bulge and nuclear star cluster, and black-hole mass tracks classical-bulge mass ever after because seed and bulge are the same epoch's co-products: one cascade, two ledger columns. As ambient pressure falls, the boundary sweeps outward and lays the disc down inside-out at constant mass per radius, with successive cohorts deposited at golden-angle offsets along the slot ladder. Once the deposited mass outweighs the seed, dynamics pass from Keplerian to self-gravitating, the sweep ends at the galactopause, and the templating epoch survives only in what it built. A generative integration of this sequence, the inheritance pair driven forward 13.6 Gyr under a pressure-driven factory law with one tuned normalisation, reproduces the Milky Way's gross statistics eight for eight: stellar mass by construction, and star count, white-dwarf census, star-formation rate, half-mass epoch, old-mass fraction, unconverted-baryon fraction, bulge fraction, and the inside-out age gradient emergent, with the bulge entering through the compression dial below.

9.2. *The inheritance pair under its ΛCDM names*

ΛCDM galaxy formation parameterises a galaxy by halo mass and spin (Peebles 1969), and the correspondence to the framework's pair is identity of content, not analogy. The budget is the same number outright: $B = f_b\, M_{\text{halo}}$. The spin draw is the same physical variable read at opposite ends of the system. λ measures the system-wide angular momentum acquired by tidal torques; the framework's D records the central compression the collapse achieved; a shallower angular-momentum barrier means deeper collapse and higher central compression, so the two are anticorrelated readings of one underlying draw, collapse depth. ΛCDM discovered the pair top-down, by simulation, at the one scale where the parent is the environment itself, and within ΛCDM the pair is phenomenology. The recursion supplies the missing why: budget and boundary are what any parent hands any child, and the same two-number law fits Saturn's moons, which no semi-analytic model can attempt with $(M_{\text{halo}}, \lambda)$. The theories tile rather than compete. This paper imports ΛCDM's cosmological scaffold wholesale, and ΛCDM's galaxy recipe acquires lower rungs and a derivation. Each becomes evidence for the other: the stochasticity of f_{disc} across the planetary catalogue (§11) and the λ-distribution of haloes are the same measurement, made at two scales, under two nomenclatures.

9.3. *Fossils of the templating epoch*

The seed's present dynamical irrelevance does not erase what it templated. Sgr A* rests at the dynamical centre to less than 1 $\text{km}\,\text{s}^{-1}$ (Reid & Brunthaler 2004), the founder still at the barycentre of everything it organised, and the templating epoch survives in four fossils.

The deposition profile.—The allocation rule lays down constant mass per radius, which is the Mestel profile, the one surface density whose rotation curve is flat (Mestel 1963). The shape of the Galaxy's baryonic deposition is thereby a direct print of the allocation law. The dark component that dominates the full rotation curve is ΛCDM's, imported with the rest of the scaffold (§9.5).

The spiral pitch.—Cohorts deposited at the most-irrational (golden) angular offset along the slot ladder trace a logarithmic spiral of pitch $\arctan[\ln(1/\rho)/2.400] = 12.65°$, with a doubled-step harmonic at 24.2° and a forced ratio $\tan\varphi_2/\tan\varphi_1 = 2$. The measured pitch distribution across 4,378 disk galaxies is bimodal at $12.0° \pm 3.4$ and $23° \pm 4.3$ (Yu & Ho 2020), and $\tan 23°/\tan 12° = 1.997$. One constant, imported unchanged from the planetary cascade, places both peaks; the mainstream alternatives require two unrelated mechanisms to do the same. The theory further predicts a third harmonic at 33.9°, suppressed density near 18°, and pitch independent of shear rate within each peak (§10).

The founder coupling.—The machine makes the organised inventory linear in the seed ($\sigma_{\rm AAF} \propto M_{\rm seed}$), so a templated structure stays mass-correlated with its founder long after the founder is dynamically negligible. This is the black-hole–bulge relation and its famous asymmetry: near-linear ($M_{\rm BH} \propto M_{\rm bulge}^{1.17\pm0.08}$), tight for classical bulges, absent for pseudobulges and discs (Kormendy & Ho 2013). The asymmetry is structural. Classical bulges are regime-1 deposits, co-eval with seed feeding; pseudobulges are later secular rearrangement, causally disconnected from the templating epoch, and therefore uncorrelated. The zero-zero corner exists: M33, bulgeless, carries a nuclear black-hole limit of order $10^3\,{\rm M}_\odot$ (Gebhardt et al. 2001). The relation's ~ 0.3 dex scatter is the environmental draw again, the same stochasticity the planetary catalogue measures in $f_{\rm disc}$.

The bulge family.—The generative sequence has one structural dial, the early-compression history $\gamma = d\ln R_{\rm dam}/d\ln(1+z)$, and the dial is the spin draw under another name. Setting $\gamma = 2.1$, which corresponds to protogalactic overdensity growth modest beside virialisation, reproduces the Milky Way's 25% bulge and ends central deposition at $z \approx 1.7$, matching both the measured ages of bulge stars and the observed "blue nugget" compaction epoch, the compact star-forming phase through which massive galaxies are seen to pass at $z \approx 2$–4 before growing inside-out (Zolotov et al. 2015). Bulge-to-total ratio is therefore a one-parameter family in collapse depth, and the family's corollary has been in the data for decades: the morphology–density relation (Dressler 1980). Ambient density is the dial, so morphology tracks environment because the dial is the address.

9.4. *The boundary factory at galactic scale*

The galactopause *is* the Davis Dam, one rung up. The same wind–ambient pressure boundary that stands at 121 AU around Sol, holding the hydrogen wall, stands at tens of kiloparsecs around the Galaxy, holding the precipitating halo clouds: one object, two addresses, and the galactic instance was named and its boundary condensation modelled before this paper asked for it (Shull & Moss 2020). The rest of the factory class (§8.7) is likewise already published piecewise, star formation at flow boundaries established from triggered shells (Elmegreen & Lada 1977) to outflow interiors (Maiolino et al. 2017; Gallagher et al. 2019). What the framework adds is the identification across scales, and the clock. Boundary deposits are dated cohorts, so nuclear populations carry age–radius stratigraphy rather than spatial structure, and the Galactic Centre already reads this way. The nuclear stellar disc's star-formation history is strongly episodic: $\sim$ 80% of its mass is older than 8 Gyr, gigayears of near-quiescence follow, and a discrete burst arrives $\sim$ 1 Gyr ago (Nogueras-Lara et al. 2020). That is the duty cycle in fossil form. The current boundary components are individually observed: the quiescent inner rim at the circumnuclear disc, the wind that sweeps it detected directly as the parsec-scale evacuated cone in the cold molecular gas (Gorski & Murchikova 2026), front jitter in X-ray light echoes, and the escaped polar front of the last major episode in the Fermi bubbles (Su et al. 2010). The "paradox of youth," stars at 0.04–0.5 pc with ages of 4–6 Myr (Paumard et al. 2006) where standard star formation is forbidden, coincides in epoch with that last episode. It is the factory's most recent firing, deposited exactly where and when the front last stood. The stratigraphic tests are running: the clockwise disc's stellar age will equal the bubble-launch epoch as both datings tighten; the distinct disc features will show measurably different ages, where a single fragmentation event predicts one; and the ~ 1 Gyr nuclear burst will prove radially annular. Externally the programme is statistical. Astrospheres establish the boundary as universal (Wood 2004); cavity-edge star-forming rings in face-on hosts test cohort age against independently dated activity fossils, with the bar-resonance alternative discriminated by unbarred hosts and by radial rather than azimuthal age structure; and ram-pressure fronts manufacturing young clusters supply the running existence proof.

9.5. *Scope: the dissipation gate*

Everything above concerns the dissipative fraction. Dark matter cannot cool, therefore cannot pool, therefore cannot enter any dam, slot, or shoreline. It is the permanently unorganisable remainder: present everywhere, including the Solar System, where some five Jupiter masses of it stream unbound through the Sun's Oort volume at any instant, and organised nowhere. The theory makes no claim on its dynamics and imports ΛCDM's cosmological core intact. The claim of this paper is the baryonic layer: the visible fraction, organised by one inheritance law from the Galaxy down to the moons.

10. PREDICTIONS AND OBSERVATIONAL TESTS

The model makes several falsifiable predictions:

Asymmetric scattering split (5%/95%).—N-body simulations of Jupiter's gravitational scattering of the slot 4-5 embryo swarm should reproduce the framework's predicted asymmetric inner/outer mass partition: roughly 5% of the swarm mass delivered inward (yielding the Mars survivor and the Earth-impacting body) and roughly 95% delivered outward (producing cumulative impacts on Saturn and Uranus).

Cascade-slot signatures in exoplanet populations.—Surveys of compact exoplanet systems should show the geometric ratio $\rho \approx 0.584$ in their orbital spacings, with deviations attributable to scattering events or migration through the chaotic zones of dominant perturbers. The $\sim$ 5–10% surviving-mass-fraction predicted for partial swarm dispersal (§5.6) is testable against the observed mass-deficit-at-position patterns in multi-planet systems.

The lattice in living discs.—Resolved ring systems should stand on the ρ ladder anchored at the disc's outer edge, with bands at every slot *including vacant ones*: the adjacent-ring ratio comb (1.31, 1.71, multiples) and the planet-free status of most rings are joint discriminators against planet carving, which requires an occupant per gap and predicts no universal ratio. Continued null results from deep planet searches inside DSHARP gaps support the lattice reading; a survey-complete census finding planets in most gaps would falsify it. In evolved systems the same test runs on debris: vacated seats identified by the catalogue's fits predict warm debris rings at computable radii (the asteroid belt at Sol's dispersed slots 4–5; the eroding 2.76 AU seat of Alpha Cen B). And in Sol itself the lattice should be *alive* on the present jaws (§8.3): fifteen slots from the heliopause down to the Alfvén surface, with faint axisymmetric magnetic or plasma banding at the predicted radii detectable in cross-epoch archival magnetometer data, and slots 1–2 already coincident with the 70–90 AU ring and the cold classical belt.

Packed-chain ejecta.—Chains fitted below the Lissauer limit shed their lightest members (§8.9). The framework therefore predicts a standing galactic population of runaway stars, brown dwarfs, and free-floating giants as routine cascade output, and that wide, eccentric, plane-aligned low-mass companions (Proxima, GJ 667 C) are relaxation products of their own systems rather than captures: testable through composition, since a scattered sibling shares its system's abundances while a capture need not.

Multi-star formation thresholds.—The framework predicts multi-star formation outcomes wherever the slot-0 mass allocation exceeds the deuterium-burning threshold ($\sim 25{,}400\, M_\oplus = 0.08\, M_\odot$). The calibrated $f_{\rm disc}$ catalog (§11.2) provides direct test cases. Future ALMA-era disc-mass surveys combined with companion-search programs around the catalogued systems can validate the slot-0-threshold criterion.

Resonant Kuiper belt census.—The co-orbital survivorship channel (§4.1) predicts resonance-island populations loaded by survival statistics from a source at Neptune's own orbit: strong 3:2, weak 2:1, strong 5:2, with deep-libration, Kozai-rich microstates after 4.5 Gyr of erosion. Migration sweeping loads the 2:1 most heavily. The observed census (3:2 strong, 2:1 weak, 5:2 anomalously strong; Gladman et al. 2012) already leans toward survivorship; a dedicated N-body comparison (Pluto-mass bodies injected co-orbital with a stationary Neptune, integrated over the age of the Solar System, counted against the observed resonant census) is a decisive and currently feasible test. Two further discriminators separate the channels: migration sweeping ties a captured body's eccentricity to the migration distance (Malhotra 1995), predicting a tight e–resonance relation, whereas scatter-then-survive predicts a broad survival distribution with *no* such correlation; and drag-driven trapping at a stationary Neptune (Weidenschilling & Davis 1985) sets population ratios by drift speed alone, with no migration timescale anywhere in the prediction.

Exterior-reservoir tests.—Three follow from §8: (i) *exo-belt scaling*, resolved debris-belt radii should regress onto $R \propto M_\star \, \Omega_{\rm prim}^{-1/2}$ (archival ALMA data). The existing empirical relation is $R \propto L_\star^{0.19}$ (Matrà et al. 2018) (roughly $M_\star^{0.8}$ on the main sequence, shallower than the dam law's linear mass term) with scatter that dominates the trend; the framework's distinctive content is precisely that the scatter is not noise but the $\Omega_{\rm prim}^{-1/2}$ term, so the test is whether independently-inferred spin orders the residuals; (ii) *the desert* (no in-situ (low-eccentricity) population at 150–2,000 AU under the two-boundary census, versus low-e clustering at the exterior-slot radii (151, 259, 444 AU) under the outward ladder) a clean three-way verdict for Rubin-class surveys; (iii) *cohort stratification* (members of the 70–90 AU ring (Fraser et al. 2024) should be compositionally and surface-age distinct from the cold classicals if they are a later, dam-station generation; (iv) *the youngest cohort*) any objects formed in the last wet episode before Local Bubble entry must sit at $\sim$ 105–121 AU, just inside the modern heliopause, dynamically cold and spectrally fresh; the largest such members fall within the reach of Rubin-class coadded photometry, and the New Horizons dust counter should see sustained or rising fluxes if its trajectory approaches such a ring.

The impact-forensics class.—The truncation clock and polarity law (§4.6) run on masses and orbits alone, so every fitted system now carries a falsifiable impact history. In Sol the standing entries are: Uranus's mass is a *bare core* to -2.9% (no impact-stripping scenario should be needed at any future mass refinement); the asteroid belt's parent population should carry slot-4 isotopic kinship (one reservoir, 34% ice); and improved giant-planet gas inventories must keep Saturn and Uranus on the same $t^* = 4.61$ Myr clock.

The missing Pluto-class siblings.—The vintage-bin census (§8.4) predicts *three undiscovered Pluto-class bodies* (the onset bin's deficit: $\sim$ 6 expected, 3 catalogued), scattered-class orbits with $a \approx$ 30–271 AU, $q \gtrsim 30$ AU, high inclination, currently far from perihelion – LSST-grade targets. The size-clock further requires the growing trans-Neptunian census to preserve mass–vintage monotonicity (heavier is older), and the firstborn rule makes a binary forecast for every fitted system: an in-situ dam-keeper holds a captured retrograde Triton-analog of the at-dam product mass; a displaced keeper lost its firstborn.

The Alpha Centauri B system.—The framework's most consequential near-term register entry (§6.2): six seats at 0.19–2.76 AU with $f_{\rm disc} \lesssim 0.02$ measured by the RV nulls; an Earth-analogue at 0.94 AU (0.77 $M_\oplus$, $P \approx 349$ d, $K \approx 7.5$ cm s^{-1}, just below current floors); a volatile forecast of a rock-floored global ocean ($\sim$ 1.5–5 surface oceans, no continents) with the sealed ice-floor variant requiring sequestration to fail tenfold – the two endmembers separated by a single transit-radius measurement; and a live debris discriminator (an eroding icy seat at $\sim$ 2.5–2.8 AU; a warm-debris signature there is the standing-conveyor's receipt). The companion forecast for Upsilon Andromedae B (§8.12): five icy sub-Earth seats inside 1.7 AU, nothing beyond 2.5 AU, $\sim$ 0.2% transit depths. GJ 667 (§8.13) adds five decidable entries: d and g are activity artifacts (capture-law violators; an activity-robust reanalysis should remove them), while e and f are mutual aliases of one real $2.7\,M_\oplus$ product; the surviving chain b–c–f must hold its mass-descending, vintage-ascending order under any re-analysis; C c is a water world, not a rocky super-Earth (bulk density at first transit or true-mass measurement); C's actual native planets are undetected (0.027–$1.1\,M_\oplus$, $K = 1.3$–18 cm s^{-1}); and B hosts one Mars-class habitable-zone world at 0.441 AU, watered by its conveyor.

11. DISCUSSION AND FUTURE WORK

11.1. *Novelty statement: which claims are new*

The framework presented here synthesises established disc-physics and impact-dynamics results into a coherent allocation rule for planetary mass and orbital position. Many individual ingredients are well-established in the literature; others are proposed here for the first time. We list each major claim with its provenance, distinguishing rigorously-derived from synthesised from hypothesised content. We hedge speculative claims and emphasise that several core proposals are testable predictions rather than proven results.

Existing-literature foundations (well-established).—The viscous-disc surface density profile $\Sigma \propto r^{-1}$ (Lynden-Bell & Pringle 1974; Pringle 1981), the magnetospheric Alfvén truncation (Alfvén 1942; Königl 1991), ALMA-era observations of disc sizes (Andrews 2020; Long et al. 2018; Manara et al. 2023), the canonical pebble-accretion framework (Lambrechts & Johansen 2012; Bitsch et al. 2015), the Tanigawa-Ikoma runaway gas accretion ($M_{\rm gas} \propto M_{\rm core}^2$; Tanigawa & Ikoma 2007), and the Chatterjee-Tan inside-out planet formation reservoir (Chatterjee & Tan 2014) are adopted directly from the published literature.

Novel framework claims.—The following claims, to our knowledge, do not appear in the existing literature and are proposed here:

1. *The geometric cascade ratio* $\rho = 1 - \frac{\sqrt{\ln 2}}{2} \approx 0.584$ (§2). Derived from the half-amplitude-at-45° projection of the Gaussian accretion-zone HWHM; a pure geometric constant with no empirical fit parameter. The cascade slot positions $r_n = R_{\rm disc} \cdot \rho^n$ fit Sol's planets and the 22 further catalogued systems. Unlike the Titius–Bode tradition and its exoplanet generalizations (per-system ladders $a_n = a_0 e^{bn}$; Bovaird & Lineweaver 2013), the cascade ratio is a single derived universal constant anchored to a physical boundary ($R_{\rm disc}$), not a per-system fit.

2. *Jupiter and Saturn essentially in-situ at slots 3 and 2* (§5); the slot 4-5 embryo-swarm dispersal as the Solar System's single dynamical event beyond cascade formation. Contradicts the Grand Tack and Nice Model migration sequences; testable via N-body verification of the asymmetric $\sim 5\%/95\%$ inner/outer scattering split.

3. *The Borealis impactor and Theia share a common Jupiter-driven slot 4-5 swarm origin.* Hypothesis; the literature establishes both impacts independently but does not link them dynamically. Testable via Borealis-ejecta cosmochemistry vs. Theia-snow-line-origin signatures.

4. *Alfvén Dam and Davis Dam* as paired terms for the inner (magnetospheric) and outer (pressure-confinement) disc truncations (§2). Terminological proposal honouring the discoverers of each boundary's physics (Alfvén for the magnetospheric truncation concept, Davis for the pressure-confined cavity (Davis 1955)) rather than claiming coinage: the underlying physics is established; what is new is the dam-and-reservoir framing, the identification of the outer boundary as the planet-forming edge, and its present-day identification with the heliopause.

5. *The Annulus Allocation Factor* σ (Eq. 7) as a named intensive quantity. The mathematical quantity ($dM/dr = 2\pi r\Sigma$) follows directly from Lynden-Bell & Pringle (1974) but was not previously named.

6. *Mercury as a slot-8/slot-9 merger remnant* with iron enrichment from the high-energy encounter retention function (§3.1). The slot-9 Vulcan body is a framework prediction; the merger mechanism resolves Mercury's mantle anomaly without invoking post-formation impact stripping, hit-and-run sequences, or extreme nebular processing. The retention formula (Eq. 27) is calibrated against both Mercury and Tau Ceti e.

7. *Venus and Neptune as the ISU primordial-plane anchors of the Solar System* (§4.1). Identification via four convergent diagnostics (most-circular orbits, one-way-integral eccentricity, inverse-capture diagnostics, precise mass-cascade match). The component facts are long catalogued (Souami & Souchay 2012; Brasser et al. 2013; Jacobson et al. 2017); Neptune's in-situ status requires the non-migratory reading of the resonant Kuiper belt (Weidenschilling & Davis 1985 rather than Malhotra 1995). The joint use of the pair as a primordial-plane reference for reading impact history is, to our knowledge, new (§4.1).

8. *Scale-invariant cascade ratio* across stellar, sub-stellar, and multi-star scales (§6). Demonstrated for Jupiter's Galilean moon system (all four moons in situ on the inherited pair, closing at 0.9%; §7.1) and Alpha Centauri's triple-star architecture. The Crab Nebula progenitor reconstruction (§6.3) is presented as an illustrative thought experiment only; Sol's own progenitor is treated as a calibrated reconstruction with named falsifiers (§7.6).

9. *The trans-Neptunian reservoirs as the Solar System's youngest components* (§8). The Kuiper belt and Oort cloud are derived as necessary deposits at the system's two outer boundaries, assembled episodically across the entire age of the Solar System rather than only at its formation: inverting the pristine-relic

paradigm and predicting cohort-stratified ages, compositions, and *post-solar* grain content in the outer rings.

10. *Stochastic $f_{\rm disc}$ across stellar class, and its identifiability structure* (§11.2). Across the eight cleanly machine-calibrated systems, $R^2 \leq 0.03$ vs $M_\star$ with $f_{\rm disc}$ spanning a factor ~ 21 over three decades of host mass; and the model itself delimits where $f_{\rm disc}$ is measurable at all: all-giant, no-ISU architectures leave it formally unconstrained. Consistent with the cross-cluster ALMA disc-mass compilation of Manara et al. (2023).

Mathematically derivable vs. empirically calibrated.—The local-capacity allocation function $M(r)$ is mathematically defined by the closed-form expressions of §2. Given the fundamental system inputs $(M_\star, D, f_{\rm disc})$ – with the nebula density D and the recorded spin locked through $\Omega = D^{2/3}$ (§2), two readings of one dial – and the universal constants of Table 2, the planet masses are uniquely determined, this is a derived (not fitted) result. D is the causally primary jaw: it sets the dam radius ($R_{\rm disc} \propto D^{-1/3}$), the factory surface density and hence every exterior product mass (§8.4), the quantity the inheritance law propagates to daughter systems, and – through the locked spin – the XUV dose that prices volatile retention (§8.13). Encounter-edited and bound-companion systems add a perturber block: $M_{\rm pert}$ and, for bound companions, the orbit $a_{\rm bin}$ – observational data, not fitted parameters – with the encounter periapsis q either supplied by the geometry or bisected against the fit (§8.10). The Jupiter-driven slot 4-5 swarm dispersal hypothesis and the inner/outer asymmetric scattering split are physically motivated but *not* mathematically derivable from the allocation function alone. We present these as hypotheses consistent with the observational record, testable by the specific predictions of §10, rather than as theorems flowing from the underlying disc-physics framework.

What our framework cannot prove.—We explicitly do not claim to have proven: (i) that the asymmetric scattering split has the precise 5%/95% ratio predicted (requires N-body verification with disc-gas torques); (ii) that the Borealis impactor was a slot 4-5 outer-swarm sibling of Theia (requires ejecta cosmochemistry to test); (iii) that the Vulcan slot-9 body existed as a distinct embryo or remains as buried iron mass within Mercury (requires high-precision gravity-gradient or interior-structure measurements distinguishing merger-remnant from primordially-mantle-stripped chondritic body); Each of these is a candidate explanation consistent with observation, not a proof. Additionally, the Crab-progenitor reconstruction (§6.3) is *not* a claim but an illustrative thought experiment of how the framework behaves under a speculative scenario; it should not be cited as a testable hypothesis, prediction, or evidence for the framework. Sol's progenitor reconstruction (§7.6) is, by contrast, calibrated, and carries its falsifiers with it.

11.2. *Empirical finding: disc-to-star mass ratio is stochastic across stellar classes*

A non-trivial empirical finding emerges from machine calibration of the catalogue. Calibration is restricted to systems whose architectures actually constrain the disc mass: $f_{\rm disc}$ has leverage only through sub-threshold rocky planets (whose masses scale directly with the disc budget), mantle-stripped remnants, or ISU anchors. In a system composed entirely of gas-eligible giants with no ISU anchor, the per-planet formation-time fit absorbs any disc mass, and $f_{\rm disc}$ is *formally unconstrained*: a structural identifiability property of the model, not a fitting deficiency. Planets observed deep inside the Alfvén Dam ($r < 0.5\,R_A$; e.g. hot Jupiters) occupy no slot at all: they are delivered-inward bodies whose formation slot is indeterminate, and they are likewise excluded from calibration targets. Fifteen of the 22 catalogued systems fall in the unconstrained class (viable parameterizations exist for all of them, but their disc masses are consistency conditions, not measurements). Table 9 reports the eight systems with clean, identifiable machine calibrations.

Table 9. Machine-calibrated primordial spin and $f_{\rm disc}$ for the ten systems whose architectures constrain the disc mass (sub-threshold rocky planets, stripped remnants, or ISU anchors among the calibration targets), ordered by host mass. Systems composed entirely of gas-eligible giants are formally unconstrained in $f_{\rm disc}$ (see text) and excluded from statistics. Stellar masses are in primordial-solar units ($1 \equiv 1.14$ present-day $M_\odot$). †Fitted with one encounter parameter (gravitational stripping, §8.10): HD 20794's innermost planet is the encounter fragment, the inward share of the 5%/95% split, parked beneath the Alfvén dam, and Proxima closes only under its Cen B encounter ($q = 19.7$ AU, disc geometry inherited from A; §6.1); both encounter fits close to $\lesssim 0.2\%$ per planet. inhNot fitted: inherited from Alpha Cen A via the hierarchy law at $x = \rho$ ($\Omega = (0.39\,\rho^{-2.12} D_A)^{2/3}$).

System	$M_\star$ ($M_\odot$)	Ω	$f_{\rm disc}$
Jupiter (sub-cascade)	0.00095	5.18	0.031
TRAPPIST-1 (M8V)	0.089	1861	0.048
Proxima Centauri (M5.5V)†	0.122	$2.17^{\rm inh}$	0.025
Tau Ceti (G8V)	0.78	308	0.022
HD 20794 (G8V)†	0.81	10.3	0.010
HD 219134 (K3V)	0.81	487	0.045
55 Cancri (G8V)	0.95	454	0.157
Mu Arae (G3V)	1.10	165	0.090
Kepler-90 (G0V)	1.13	1127	0.032
Sol (G2V)	1.00	0.995	0.010

Across the eight cleanly-calibrated systems the disc-to-star mass ratio is *stochastic* with respect to stellar class: $f_{\rm disc}$ spans a factor of ~ 16 (0.010–0.16); systems of nearly identical host mass differ by a factor of sixteen (HD 20794 at 0.010 versus 55 Cancri at 0.16, both late-G hosts); and a regression of $\log f_{\rm disc}$ on $\log M_\star$ yields $R^2 < 0.001$ for the nine stellar hosts ($R^2 < 0.001$ including the circumplanetary sub-cascade, which extends the host-mass baseline by three decades). The Jupiter sub-cascade calibrates to $f_{\rm disc} = 0.032$: a planetary-scale disc drawing its ratio from within the stellar-host range, consistent with scale-invariant boundary physics.

Multi-star formation outcomes require the slot-0 allocation to cross the deuterium- or hydrogen-burning thresholds, which demands a high-$f_{\rm disc}$ disc; the catalogued multi-star systems (Alpha Centauri, GJ 667) admit viable parameterizations at $f_{\rm disc} \sim 0.2$–0.3, but their all-stellar architectures place them in the unconstrained class, so those disc masses are consistency conditions rather than measurements.

This empirical finding is independently supported by ALMA-era protoplanetary disc surveys, which measure $M_{\rm disc}/M_\star$ directly via millimetre continuum emission. Surveys of the Lupus (Ansdell et al. 2016), Chamaeleon (Pascucci et al. 2016), Ophiuchus (Williams et al. 2019), and ODISEA (Cieza et al. 2019) star-forming regions consistently find ratios spanning 0.1%–30% at fixed stellar mass. The mean trend is real and in fact super-linear (Pascucci et al. (2016) report $M_{\rm disc} \propto M_\star^{1.7\text{–}1.9}$, steepening with cluster age) but the stochasticity claim rests on the *scatter*, not the slope: 0.5–0.9 dex at fixed stellar mass (Ansdell et al. 2016), dwarfing any deterministic prediction for an individual system (Manara et al. 2023). (A super-linear mean is itself congenial to the cloud-density conjecture of §11: denser cores yield both more massive stars and proportionally heavier, faster-spinning discs.)

The combined model + observational finding is: $f_{\rm disc}$ *is set primarily by stochastic local conditions during cloud collapse and disc assembly* (turbulent fragmentation, infall asymmetries, magnetic flux trapping), *not by the deterministic properties of the resulting star*. This is consistent with the conclusion of Manara et al. (2023) in the Protostars and Planets VII (PPVII) review that disc-mass diversity reflects formation-environment variability more than stellar-mass scaling.

Theoretical expectation.—The stochasticity is not only observed but predicted. The angular momentum that seeds a disc is set by the turbulent velocity field of its natal core: effectively a random draw with a large intrinsic spread at fixed core mass (Burkert & Bodenheimer 2000). Zoom-in simulations following collapse from giant-molecular-cloud scales find the accretion process heterogeneous “in time, in space, and among protostars of otherwise similar mass” (Kuffmeier et al. 2017), and disc population synthesis from radiation-hydrodynamical cluster simulations produces an enormous diversity of disc masses and sizes from the chaotic dynamics of star formation itself (Bate 2018).

The cascade's structural explanation.—The framework supplies its own reason for the same decoupling. When star formation occurs within discs (§6), a star of a given mass can emerge from progenitor discs of almost any mass: the same stellar mass can be allocated at slot 0 of a modest disc, at an inner slot of a massive one, or assembled by independent cloud collapse. The new star's own disc budget is in turn inherited from its local share of the *progenitor's* disc, not set by the star's final mass. Stellar mass and $f_{\rm disc}$ are therefore decoupled by construction: $f_{\rm disc}$ records a system's formation lineage rather than any property of the star it orbits. The catalogue's $R^2 < 0.1$ is the planet-architecture-level imprint of that decoupling.

A single-variable hypothesis (unproven).—The two per-system inputs may not be independent of each other. Spin, on the pressure-confinement reading of §2, is the recorded *compression* of the birth site; $f_{\rm disc}$, on the structural reading above, is the recorded *mass supply* of the birth site. Both would then be functions of a single underlying variable, the density of the formation environment, and planet formation would reduce to a two-parameter theory: stellar mass and birth-site density, plus universal constants. We emphasize that this is a hypothesis with no proof at present. Across the ten cleanly machine-calibrated systems the rank correlation between fitted spin and $f_{\rm disc}$ is $\rho \approx 0.60$ at $n = 10$: positive, as a common density driver would produce, but not significant ($p \approx 0.07$). We report it with its full volatility history: development revisions of the catalogue moved the value between $\rho \approx 0.6$ and 0.81 as calibrations improved and the constrained sample grew from eight to ten, rank statistics this volatile should be treated as motivation for a larger sample, not as an established correlation. Density also enters the two quantities through different moments (confining pressure versus local surface density). Establishing or refuting the hypothesis requires a larger identifiable sample, birth-environment tagging of calibrated systems, and rotation–density statistics in young clusters. The other piece of internal evidence already pointing this way is the quasi-universality of the specific-angular-momentum constant (30 AU at solar values) across the cleanly-calibrated set.

Implications.—This stochasticity has direct consequences for the framework's generalisability:

- The cascade slot mass budget at each radius scales with $f_{\rm disc}$; systems with high $f_{\rm disc}$ have more massive cascade bodies at every slot, and the slot-0 mass allocation crosses the stellar formation threshold at lower $f_{\rm disc}$ for massive primaries (driving the multi-star outcome statistics).
- Two systems with identical $M_\star$ but different $f_{\rm disc}$ produce qualitatively different cascade architectures (planet-only vs. multi-star), even at fixed stellar class.
- Population-level statistical inferences about planetary architectures cannot rely on stellar class alone; the system-specific $f_{\rm disc}$ is an independent, observable, dominant input.

11.3. *Model limitations*

The cascade framework is closed-form and deterministic given its fundamental system inputs ($M_\star$, D, $f_{\rm disc}$; spin $\Omega = D^{2/3}$ under the jaw-lock), the perturber block ($M_{\rm pert}$, q, $a_{\rm bin}$) where a system is encounter-edited or companion-bound, plus per-planet $(r, t_{\rm form})$ via the AAF cascade rule (Eq. 22).

The grain-opacity constant is calibrated only on Sol.—All catalogue fits use $g = 0.82$ (§2); present mass-position data cannot constrain it per system. Resolved snow-line imaging or compositional measurements of individual systems would lift the degeneracy and test whether g is genuinely universal: or merely uniform across the solar neighbourhood that the calibration sample occupies.

Observational anchors carry tracer and calibration systematics.—Two soft spots are inherited from the disc-observation literature. ALMA "disc sizes" are tracer-dependent, CO radii are chemically depleted relative to the true gas extent (Trapman et al. 2019), so identifying a measured radius with the physical dam edge carries a tracer correction. And absolute gas-disc masses remain systematically uncertain (the disc mass-budget problem; Manara et al. 2023), so the catalogue's *absolute* $f_{\rm disc}$ values are reliable only to that systematic floor; the stochasticity conclusion and all cross-system comparisons are differential and robust to it.

Linear rocky allocation neglects pebble-drift physics.—The linear $\sigma\, r$ rocky allocation neglects detailed pebble-drift dynamics in favor of an aggregate empirical scaling. A more physics-rich treatment would couple Birnstiel-style two-population dust evolution (Birnstiel et al. 2012) to the local-capacity framework, particularly for the pebble accumulation at outer cascade slots that contributes to gas-giant core masses.

Empirical A_0 accretion coefficient.—The Tanigawa-Ikoma H/He accretion coefficient ($A_0 = 4.39\, M_\oplus^{-1}$, $k = 0.691$ Myr^{-1}) appears as an empirically-tuned constant. Its order of magnitude is reproduced from MMSN disc parameters at Jupiter's slot-3 position (Eq. 21), but a fully first-principles derivation from disc viscous evolution and gas-accretion timescales remains outstanding.

δM post-formation modifications.—The δM post-formation modification term is a structural feature of the model that absorbs collisional and dynamical events following primary cascade assembly. We emphasize that most δM assignments have independent physical interpretation: Jupiter-induced dispersal of the slot 4-5 embryo swarm is constrained by dynamical and isotopic evidence; impact-retention energetics for Mercury follow from the geometric cascade and $\Delta v/v_{\rm esc}$ encounter physics (§3.1) without ad hoc mass-stripping mechanisms. The remaining flexibility is bounded by the cascade's slot-position predictions and the four-line ISU criterion (§4.1); arbitrary δM assignments would violate the inclination/eccentricity baseline established by the Venus-Neptune primordial plane.

Sub-cascade scale-invariance is only partially validated.—The cascade ratio $\rho = 0.5837$ generalises successfully from stellar to sub-stellar scales (Jupiter's circumplanetary cascade, §6). However, the framework's $R_{\rm disc}$ formula uses stellar primordial-spin scaling (Eq. 4); for circumplanetary discs around gas giants, Hill-sphere truncation by the host star may impose additional boundary conditions not captured by the spin-only formula. Saturn's moon system, where Titan dominates the mass budget by 75× over the next-largest body, was identified as a non-cascade "runaway-growth + fragments" regime where the geometric cascade does not apply.

f_{disc} is an input only at the recursion's root.—For a system whose parent is identified, the framework now *computes* the pair rather than fitting it: the budget is the parent's seat allocation and the boundary follows from the inheritance law (§7), as measured at 0.9% for Jupiter's disc and run upward to calibrate Hyperion's. The limitation that remains is confined to root systems, those with no parent disc, where f_{disc} is a genuine environmental draw taken as input: its distribution across stellar classes is documented but not predicted, and deriving it from cloud-collapse physics (turbulent fragmentation, infall asymmetries, magnetic flux trapping) would require coupling the cascade framework to larger-scale star-formation theory.

11.4. *Extensions*

Future work should: (i) implement N-body simulations of Jupiter-driven slot 4-5 swarm dispersal to verify the predicted $\sim 5\%/95\%$ inner/outer scattering asymmetry and the cumulative impact-energy delivery to Saturn and Uranus; (ii) extend the embryo-swarm dispersal criterion to other multi-planet systems to predict scattering signatures across the exoplanet catalog; (iii) derive the A_0 and k envelope-accretion constants from disc physics rather than from their present calibration against Jupiter and Neptune; the slot's physical embodiment as a zonal-flow pressure bump (§2) names the route, since A_0 is physically the bump-fed accretion efficiency, accessible to the magnetized-disc simulations already cited, with Pollack–Bitsch–Mordasini-class disc-evolution models constraining the k depletion clock; (iv) test the prediction that gas-disc Type-I and Type-II migration are minor or absent dynamical effects, with observed planet positions explained by cascade-default formation plus scattering events alone; and (v) test the single-variable hypothesis that Ω and f_{disc} both derive from birth-site density (§11.2), beginning with per-system spin refits and rotation–density statistics in young clusters.

11.5. *Relation to existing frameworks: the LBP track that was disregarded*

The present model adopts viscous-disc surface-density scaling from the gas-dynamics tradition (Lynden-Bell & Pringle 1974; Shakura & Sunyaev 1973; Pringle 1981) and applies it to planet mass allocation, a perspective rarely combined in the planet-formation literature, which has predominantly used Hayashi (1981)'s MMSN $\Sigma \propto r^{-1.5}$ scaling. Modern ALMA observations (Andrews 2020; Andrews et al. 2010; Long et al. 2018; Trapman et al. 2019; Manara et al. 2023) robustly favour the LBP $\Sigma \propto r^{-1}$ profile, with measured power-law indices $\gamma \approx 0.9$–1.1. The MMSN framework, which underweights the disc's outer mass reservoir, can be considered superseded.

The Annulus Allocation Factor was implicit in the 1970s viscous-disc literature.—A historical observation deserves explicit acknowledgement. For the LBP $\nu \propto r$ self-similar solution, the mass per unit radius $dM/dR = 2\pi R\Sigma(R) =$ constant emerges as a *trivial mathematical consequence*, the quantity we name the Annulus Allocation Factor sits one line below their derivation. Lynden-Bell & Pringle (1974) proved it; Pringle (1981) consolidated it; Shakura & Sunyaev (1973) established the α-disc viscosity prescription that makes $\nu \propto r$ natural. The framework was set aside not because the math was wrong, but because:

1. Hayashi (1981) adopted a steeper $\Sigma \propto r^{-1.5}$ for solid material in the MMSN, partly because it produced the correct rough orders of magnitude for Solar System planetary masses under contemporary observational constraints;

2. the planet-formation community subsequently built on the MMSN framework for four decades, treating viscous-disc gas-dynamics work as relevant to stars (accretion onto compact objects, AGN, X-ray binaries) rather than to planet allocation;

3. the closed-form constant dM/dR that emerges in the LBP framework was never crystallised into a named planet-formation quantity, because no one was looking for it within the MMSN-shaped planet-formation landscape.

Safronov (1972), Cameron (1978), and Weidenschilling (1977) all worked the relevant territory in the 1970s, Safronov with his "feeding zone" concept (planet mass = integrated annular mass), and Weidenschilling with the inverse reconstruction of disc surface density from observed planetary masses (the empirical demonstration that planet masses read out a smooth annular mass law, of which our forward allocation is the mirror) but none crystallised the intensive radial constant. The load-bearing departure is the profile: Weidenschilling's reconstruction assumed the MMSN's $r^{-3/2}$, where the LBP r^{-1} adopted here is what renders the per-annulus allocation a radial constant at all.

11.6. *Component provenance: who found the pieces*

The same acknowledgement extends beyond the allocation constant. Nearly every component mechanism of this framework was first identified (as a phenomenon, if not in its final mechanistic role) by earlier workers, each within the context where their methods encountered it. We record that provenance explicitly, because the framework's claim is the *assembly* and its universality, not the parts:

- *The allocation constant*: Lynden-Bell & Pringle (1974) proved it, Pringle (1981) consolidated it, Shakura & Sunyaev (1973) supplied the viscosity that makes it natural; Weidenschilling (1977) demonstrated empirically that planet masses invert to a smooth annular mass law, and Safronov (1972) supplied the feeding-zone calculus and the $\Delta v/v_{\rm esc}$ accrete-or-erode parameter.

- *The inner dam*: magnetospheric truncation by Ghosh & Lamb (1979) (the magnetic-pressure concept tracing to Alfvén 1942); its coupling to stellar spin by disc-locking, Königl (1991) and Bouvier et al. (2007); the trapping of solids against it as a planet-forming reservoir by Chatterjee & Tan (2014); the parking of inward-migrating giants at it by Romanova & Lovelace (2006).

- *The outer dam*: the pressure-confined cavity (the boundary itself, in its present-day form) by Davis (1955), inflated by Parker's wind (Parker 1958); the ionization-bounded viscosity transition (the dam's disc-era masonry) by Gammie (1996), sharpened by the demonstration that the disc bulk is magnetically dead (Bai & Stone 2013); solid trapping at pressure maxima by Pinilla et al. (2012); the transport-set character of observed disc edges by Najita & Bergin (2018).

- *The packing physics*: the Hill-spacing stability threshold by Smith & Lissauer (2009); chain relaxation at gas dispersal by Liu, Ormel & Lin (2017); the observation that real chains assemble at marginal spacing by Hansen & Murray (2013).

- *Rung inference itself*: geometric spacing regressions in exoplanet systems by Bovaird & Lineweaver (2013); the prediction of an unseen member from the occupied rungs of a chain by Goździewski & Migaszewski (2014) for HR 8799f, and its confirmed precedent, TRAPPIST-1h, predicted from the inner chain's resonance relations and found at the predicted period (Luger et al. 2017); the half-step occupancy state of §2 is named the *Luger lattice* accordingly.

- *The exterior factory's parts*: collapse at pressure maxima by Youdin & Goodman (2005); its binary signature by Nesvorný et al. (2010); drag-trapping into resonances at a stationary planet by Weidenschilling & Davis (1985); the cluster-commons exchange by Levison et al. (2010); boundary mass assembly at galactic scale by Maiolino et al. (2017).

- *The standing barrier in cosmochemistry*: the sustained NC/CC reservoir separation by Kruijer et al. (2017).

Each of these works contains a piece of the present mechanism, observed where its discoverers' instruments and questions happened to point. What this paper adds is the identification: that they are one structure (two dams, a geometric ladder between them, and the dynamics of its occupants) operating identically from circumplanetary discs to star-forming clouds.

12. CONCLUSION

ΛCDM galaxy formation describes a galaxy with two numbers and does not say why two suffice. This paper has proposed the answer and calibrated it where the hierarchy can be resolved: (M, λ) is the inheritance pair of dissipative baryonic organisation, the mass budget and the collapse-depth record that every parent system hands every child, processed by one universal machine whose formation seats, the natal slots, stand at the fixed geometric ratio $\rho = 1 - \sqrt{\ln 2}/2 \approx 0.584$ between two pressure boundaries.

At planetary scale the machine is calibrated, not conjectured. In primordial solar units Sol sits at exactly $(1, 1, 1)$ with its dam at Neptune; ten clean calibrations and 23 fitted systems follow across spectral classes A through M and both dynamical regimes of the disc. The Solar System is reproduced parsimoniously: Jupiter and Saturn essentially in situ; one dynamical event, the slot 4–5 swarm dispersal, accounting through its asymmetric split for the Moon-forming impactor, the Mars survivor, and the gas deficits and tilts of Saturn and Uranus; and the Mercury–Vulcan merger as the only other invocation. Migration throughout the catalogue is gravitational, angular-momentum exchange with heavier bodies, with the gas disc trapping rather than torquing: migrators park on seats and rims, and the parking position dates the move.

The hierarchy closes in both directions. Jupiter's satellite disc prefers Jupiter's formation seat over its present orbit by a factor of 35 and closes at 0.9% on the inherited pair; Saturn's refusal to fit dates its rings; Uranus's rebuilt system records the tilt event; Triton arrives from Sol's own dam cohort; and the kinetic regime extends the same machine to impact-born discs, where Earth's two seats packed at $P = 3.2$ and merged into one moon and a farside scar, while Mars's two seats packed at $P = 285$ and kept both children, one of which is being ground back into the Roche zone now. Run upward, the recursion reconstructs Hyperion, the A-class star whose disc produced Sol, predicts its remnant, and identifies a candidate fourteen light-years away whose measured debris pollution matches the predicted dry-rock composition; the same seat arithmetic computes stellar metallicity offsets, reproducing Sol's protosolar bulk composition unprompted and the measured metallicity excess of Alpha Centauri B over A that co-natal formation cannot explain.

Run outward, the dam is a production line whose deposits are dated cohorts: the Kuiper belt is the residue of the T Tauri firehose, the cold classicals its mid-era deposition band, the Kuiper cliff its shutdown radius, and the boundary still operates, its stocked shelf observed as the hydrogen wall, its dense-cloud excursions predicting young radionuclide-bearing bodies at accessible radii. The same boundary, under the name the literature independently gave it, is the galactopause, and the galactic core is constructed by the same machine that built Sol's planets, with seats filled by direct-collapse objects and massive clusters where Sol's seats held planets. The templating epoch survives in fossils the paper enumerates, the deposition profile, the spiral pitch lattice, the founder coupling, and the compression family of bulges, and a generative integration of the sequence with one tuned normalisation reproduces all eight of the Milky Way's gross statistics.

One empirical finding stands beside the framework rather than inside it: across the cleanly calibrated catalogue the disc-to-star mass ratio is stochastic with respect to stellar class, exactly as the inheritance structure requires, since the draw is environmental at the recursion's root and seat-determined everywhere below it. One reconstruction remains fenced as a thought experiment, the Crab progenitor cluster; Sol's progenitor, by contrast, is calibrated and carries its falsifiers.

Two numbers suffice to describe a galaxy because two numbers are what any parent hands any child; at the top of the recursion, where the parent is the environment itself, the pair carries the names cosmology gave it. Every mechanism above is listed in the register (§10) with the observation that would falsify it.

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DATA AVAILABILITY

The complete model implementation, including the TypeScript/JavaScript engine, calibration data for the 22 catalogued exoplanetary systems, and a browser-based interactive tool, is available at:

- **Interactive simulator:** https://jamesgdahl.github.io/HYDROS-Planet-Formation-Model/
- **Source repository:** https://github.com/jamesgdahl/HYDROS-Planet-Formation-Model

The interactive simulator allows direct exploration of the fundamental system inputs ($M_\star$, D, $f_{\rm disc}$, with Ω jaw-locked to D), the perturber block ($M_{\rm pert}$, q, $a_{\rm bin}$) for encounter-edited and companion-bound systems, and per-planet $(r, t_{\rm form})$ adjustments, displaying the resulting mass allocation and predicted vs observed masses in real time. Built-in presets reproduce the Solar System fit (Table 3) and all 22 catalogued systems (§11.2), plus the thought-experiment reconstructions of §6.

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